

Characterization of two novel lytic bacteriophages for *Klebsiella pneumoniae*: molecular studies

Bruna R. Guerrero¹, Stephen C. Emencheta^{1,2}, Carlos A. Quinde³, Basilio Cieza⁴, Aakash Basu⁵, Safia Samir⁶,
Anthony A. Attama⁷, Ebele B. Onuigbo², Marta M. D. C. Vila¹, Victor M. Balcão^{1,8,*}

Academic Editor: Agostinho Antunes

Abstract

Introduction: Multidrug-resistant *Klebsiella pneumoniae* (Kpn) poses a growing global health threat due to its role in severe hospital-acquired infections and increasing resistance to antibiotics. The objective of this study was to isolate, characterize, and evaluate the therapeutic potential of two novel lytic bacteriophages targeting multidrug-resistant *K. pneumoniae*, aiming for the development of alternatives to conventional antibiotics.

Materials and methods: Phages KpnSo1BRG and KpnSo2SCE were analyzed using transmission electron microscopy (TEM) and whole-genome sequencing (WGS). Replication kinetics were determined through single-step growth curves (latency period and burst size). Antibacterial efficacy was tested in vitro using a phage cocktail at a multiplicity of infection (MOI) of 10,000 over 12 h. Additionally, a genome mechanics analysis was conducted to evaluate viral DNA cyclizability and flexibility.

Results: TEM revealed that both phages belong to the class *Caudoviricetes* with a siphovirus-like morphology, and WGS classified them within the genus *Webevirus* (family *Drexelviriidae*). Phage KpnSo1BRG exhibited a latency period of 60 min and a burst size of 17.7 virions/cell, while KpnSo2SCE showed a latency of 5.8 min and a burst size of 18.6 virions/cell. The phage cocktail reduced the bacterial load by approximately 97.2% after 12 h, relative to the untreated control. Genomic analysis indicated that the higher DNA flexibility in KpnSo2SCE correlated with a slightly higher virion production, corroborating previous findings on genome mechanics.

Conclusions: The novel phages demonstrated potent antibacterial activity (i.e., strongly reduced bacterial counts after 6 h) and favorable genomic characteristics, establishing themselves as promising candidates for phage-based therapeutic strategies against multidrug-resistant *K. pneumoniae* infections.

Keywords: bacteriophage, siphovirus, *Klebsiella pneumoniae*, in vitro phage inactivation, molecular studies, *Drexelviriidae* family, genus *Webevirus*

Citation: Guerrero BR, Emencheta SC, Quinde CA, Cieza B, Basu A, Samir S, et al. Characterization of two novel lytic bacteriophages for *Klebsiella pneumoniae*: molecular studies. *Academia Molecular Biology and Genomics* 2026;3. <https://doi.org/10.20935/AcadMolBioGen8364>

1. Introduction

The global public health danger posed by multidrug-resistant *Klebsiella pneumoniae* (Kpn) is growing [1]. Gram-negative, encapsulated, rod-shaped, and non-motile, *K. pneumoniae* is a significant opportunistic pathogen belonging to the *Enterobacteriaceae* family that is responsible for a significant number of nosocomial infections, especially in poor nations [2, 3], such as urinary tract infections, wound and soft tissue infections, pneumonia, life-threatening sepsis, among other conditions [4–6]. Up to 10% of nosocomial infections caused by *K. pneumoniae* have been reported [7]. Not only does it raise patient rates of morbidity and mortality, but it also extends hospital stays and raises treatment costs [3]. Some *K. pneumoniae* infections are challenging to treat

effectively. Alternative approaches have gained attention as a potential treatment option for bacterial infections, particularly phage therapy [8–12].

Phage therapy is a rapidly evolving field, offering a potential solution or a complementary approach to counteract antibiotic-resistant bacteria [9, 10, 13, 14]. The vast advance of phage therapy in recent years is due to many reasons, such as its specificity to their target bacteria while leaving beneficial bacteria unharmed, and its potentiality to be personalized/tailored for individual patients based on their effectiveness against the specific bacterial strains causing an infection in a particular patient. This personalized approach aligns with the increasing focus on precision medi-

¹VBlab—Laboratory of Bacterial Viruses, University of Sorocaba, Sorocaba, Brazil.

²Department of Pharmaceutical Microbiology and Biotechnology, University of Nigeria, Nsukka, Nigeria.

³Department of Biological Sciences, University of South Carolina, Columbia, SC, USA.

⁴Department of Biophysics and Biophysical Chemistry, Johns Hopkins University, Baltimore, MD, USA.

⁵Department of Biosciences, Durham University, Durham, UK.

⁶Department of Biochemistry and Molecular Biology, Theodor Bilharz Research Institute, Imbaba Giza, Egypt.

⁷Department of Pharmaceutics, University of Nigeria, Nsukka, Nigeria.

⁸Department of Biology and Center for Environmental and Marine Studies (CESAM), University of Aveiro, Aveiro, Portugal.

*email: victor.balcao@prof.uniso.br

cine and tailored treatments [15]. Moreover, phage therapy is versatile: phages can be relatively easily isolated against a wide range of bacterial species and combat various bacterial infections [16].

Identifying a new phage typically involves the following steps, though specific protocols may vary based on study goals and resources [17]. For isolation and purification, phages are extracted from samples and purified to determine their titer and host range [17]. For the genomic characterization, sequencing identifies genes driving host specificity, lysis mechanisms, and potential virulence [18, 19]. For studying replication dynamics, one-step growth curves are used to measure the latent period and virion morphogenesis yield [20]. For assessing environmental stability, testing resistance to pH, UV, and temperature ensures practical viability [21]. The phage's safety profile and ability to overcome bacterial resistance must be validated prior to clinical application.

Interestingly, studying the characteristics of a phage, also known as stability to abiotic factors that can influence its survival, replication, and interactions in its environment, is of great importance [22]. Phages can show the most lytic activity at specific temperature and pH preferences for optimal growth, stability, and replication. Studying the abiotic factors that influence phages' lytic viability/activity is crucial for learning about their behavior, ecology, and uses, helping to figure out the ideal circumstances for phage growth, survival, and application—such as in phage therapy or environmental applications [23].

K. pneumoniae reference strain NCTC-13439 was used for isolating two lytic phages. The well-characterized reference strain *K. pneumoniae* NCTC-13439 was deliberately chosen as the isolation host because its robust capsular polysaccharide serves as a stringent physical barrier against viral entry, thereby ensuring the selective isolation of highly virulent phages capable of overcoming formidable bacterial defenses. Isolating phages that can effectively infect and kill this particular strain can be challenging due to the capsular polysaccharide formed by the bacteria that acts as a physical barrier, preventing the phages from reaching the bacterial surface [24–26]. Additionally, there is a genetic diversity within *K. pneumoniae* strains [27], which means that the isolated phages that infect strain NCTC-13439 may not infect other *K. pneumoniae* strains. Moreover, bacteria may develop resistance mechanisms to phage infection, via changes in surface receptors or the production of enzymes that degrade phage DNA [28], hence limiting the effectiveness of phages in infecting and killing the bacteria.

K. pneumoniae relies heavily on its robust exopolysaccharide (EPS) capsule as a primary virulence factor. This thick capsular layer promotes biofilm formation and effectively shields the bacterium from environmental stressors, host immune responses, and antibacterial penetration. To successfully infect such heavily encapsulated strains, *K. pneumoniae* phages have evolved specialized structural enzymes known as capsular depolymerases, which are typically localized on their tail fibers or tail spikes. These depolymerases play a critical role in the initial stages of viral infection. They specifically recognize, bind, and enzymatically cleave the capsular polysaccharides (CPSs), effectively “stripping” the protective bacterial shield. This degradation allows the phage to reach its secondary outer membrane receptors and inject/translocate its viral DNA. Several researchers have demonstrated that phage-derived depolymerases can be deployed as

highly specific anti-virulence agents. By digesting the bacterial capsule, depolymerases expose the underlying pathogen, thereby re-sensitizing it to both host immune clearance and conventional antibiotics [28, 29]. Consequently, the identification of a putative capsular depolymerase gene within the genome of the newly isolated lytic phage KpnSo2SCE strongly underscores its therapeutic relevance. It indicates that the phage inherently possesses the enzymatic arsenal required to disrupt *Klebsiella* biofilms and encapsulated matrices, positioning it as a highly promising candidate for targeted biocontrol and advanced phage therapy applications.

In the present research work, we report a leap forward in the search for lytic phages against *K. pneumoniae*, fully characterizing and testing in vitro two novel lytic phages together with their complete molecular characterization.

2. Materials and methods

2.1. Biological material

The bacterial strain used as the phage host in all assays, *K. pneumoniae* NCTC-13439, was acquired from the NCTC (National Collection of Type Cultures, UK Health Security Agency (UKHSA), Salisbury, Wiltshire, UK). For phage isolation, two types of samples were employed: the first being a pooled environmental sample consisting of central sewage, fishery, abattoir, piggery, chicken rearing, hospital, and residential wastewater, which was used in the isolation of one of the phages (phage KpnSo2SCE). This sample was sourced in and around Nsukka, the host community of the University of Nigeria, a town and local government area in South-East Nigeria in the Enugu State (GPS coordinates of Nsukka: latitude, 6°51'28.19" N; longitude, 7°23'44.77" E); the second environmental sample was sewage collected from the Veterinary Hospital at the University of Sorocaba (UNISO) (Sorocaba, Brazil) on August 08th 2022 (GPS coordinates of the sampling spot: latitude, 23°29'56.4" S; longitude, 47°23'42.5" W) and was used in the isolation of the second phage (phage KpnSo1BRG).

Bacterial cultures were kept in tryptic soy agar (TSA) at 4 °C and, before starting an assay, a single isolated colony was transferred to 25 ml tryptic soy broth (TSB) and incubated overnight for bacterial growth at the optimum temperature. A sample of the bacterial cultures (100 µl) was transferred to 10 ml of fresh TSB and incubated overnight at the optimum temperature for bacterial growth until attaining an optical density (OD_{600 nm}) of 1.0 (ca. 10⁹ cells per ml).

2.2. Phage enrichment, isolation, and enumeration

The two bacteriophages characterized in this study were recovered from environmental matrices using an enrichment strategy adapted from previously established methods [30–34]. Briefly, 50 ml of TSB in a Schott flask was inoculated with the environmental sample and 50 µl of an exponential-phase Kpn NCTC-13439 culture (OD_{610 nm} ≈ 0.5). Following 24–48 h of incubation at 37 °C, the enriched mixture was clarified via centrifugation (9000 × *g*, 10 min, 4 °C). To eliminate residual host cells, the supernatant was treated with an equal volume of chloroform (1:1, *v/v*). The resulting aqueous fraction was passed through a 0.22 µm polyethersulfone membrane (Sartolab®-RF 250, Sartorius,

Göttingen, Germany) and screened for viral activity using the double-layer agar method on Kpn NCTC-13439 lawns [30, 35]. After 24 h of incubation at 37 °C, plates were inspected for clear or turbid lysis zones. Isolated plaques were harvested into 1 ml of sterile 0.9% (*w/v*) NaCl solution, and their lytic potential was confirmed through spot assays (10 µl) on the host bacterial lawn [30–32].

Isolation of a phage plaque was then carried out following the procedure described in detail elsewhere [30–32]. The titer of the phage suspensions (PFU/ml, plaque forming units/ml) was then determined [30, 33] by plating serial dilutions of the phage suspensions via the double-layer agar method.

2.3. Polyethylene glycol precipitation of phage virions

Viral particles were concentrated by combining the initial phage suspensions (10^{11} PFU/ml for KpnSo1BRG and 10^{13} PFU/ml for KpnSo2SCE) with a sterile precipitating solution containing 10% (*w/v*) polyethylene glycol (PEG) 8000 and 1 M NaCl (Sigma-Aldrich, St. Louis, MO, USA). The mixture was prepared using a 2:1 volumetric ratio of the PEG/NaCl solution to the phage lysate. After overnight incubation at 4 °C, the suspension was centrifuged at 11,000 rpm for 45 min at 4 °C. The resulting phage pellet was harvested and carefully resuspended in 5 mM MgSO₄ until homogenized.

2.4. Transmission electron microscopy (TEM) analyses

To visualize the structural morphology of the phages, PEG-concentrated viral particles were subjected to negative staining with 2% (*w/v*) uranyl acetate, following a protocol adapted from previous studies [30]. High-resolution imaging was performed using a JEOL JEM 2100 transmission electron microscope (Tokyo, Japan) equipped with a LaB6 filament and operated at 200 kV, achieving a resolution of 0.23 nm. Digital micrographs were captured via a GATAN ORIUS™ 832.J4850 SC1000B CCD camera (11 MP, GATAN Inc., Pleasanton, CA, USA) and managed using the Gatan Microscopy Suite (DigitalMicrograph, version 2.11.1404.0). The physical dimensions of the capsids and tails (**Table 1**) were calculated as the average of seven independent virions ($n = 7$) using the ImageJ software package (version 1.52a).

Table 1 • Estimated dimensions of the virions for phages KpnSo1BRG and KpnSo2SCE (based on average measurements of 7 virions).

Structural feature	Phage virion dimensions	
	Phage KpnSo1BRG	Phage KpnSo2SCE
Capsid length (nm)	56.48 ± 3.20	53.17 ± 2.70
Capsid width (nm)	54.47 ± 4.63	47.92 ± 2.36
Tail length (nm)	164.77 ± 4.12	163.58 ± 1.96
Tail thickness (nm)	10.20 ± 1.87	9.22 ± 1.54

2.5. Host range of the isolated phages

The lytic breadth of the two phages was evaluated against the bacterial panel listed in **Table 2**, utilizing a spot-testing protocol adapted from established methodologies [30–33, 36, 37]. Briefly, 100 µl of each bacterial culture was mixed with 5 ml of semi-solid molten top agar-TSB (MTA-TSB) and overlaid onto TSA base plates. Once the overlay solidified, 5 µl of the phage suspension was applied to the surface. After 24 h of incubation at the optimal growth temperature, the plates were inspected for lysis. Strains were classified as susceptible (+) if a clear lysis zone appeared or non-susceptible (–) in the absence of lysis. For all susceptible strains, the efficiency of plating (EOP) was determined via the double-layer agar method. The EOP was calculated relative to the isolation host, Kpn NCTC-13439 (EOP = 100%), using the formula $EOP = \{(\text{Average PFU}_{\text{target bacteria}})/(\text{Average PFU}_{\text{host bacteria}})\} \times 100$ [30, 33, 37, 38]. The reported EOP values represent the mean of three independent experimental replicates [30, 33, 37, 38].

The host range of the two phages was determined via spot testing using the bacterial strains listed in Table S1. These strains included both clinical isolates of the target species and representatives of other enteric genera (e.g., *Escherichia* and *Salmonella*) to evaluate the taxonomic specificity of the phages and their potential impact on non-target commensal bacteria, a critical factor for ensuring the preservation of the host microbiota during therapeutic application.

2.6. Phage virion adsorption rate onto bacterial host cells

To evaluate phage adsorption kinetics, a 25 ml culture of Kpn NCTC-13439 in TSB was grown at 37 °C until reaching the exponential phase. The bacterial concentration was standardized to an OD_{610 nm} of ≈ 0.5 ($\approx 0.5 \times 10^9$ CFU/ml), and the phage suspension was diluted to 8×10^5 PFU/ml (total volume of 15 ml) to achieve a multiplicity of infection (MOI) of 0.001, ensuring sufficient host availability for viral attachment [39]. Following a 30 s stabilization period after mixing, 50 µl aliquots were harvested at 5 min intervals for the first half-hour, and every 10 min thereafter until 150 min. Each sample was immediately transferred to 425 µL of TSB containing 25 µl of chloroform to arrest bacterial activity; notably, chloroform was only applied to the 10^{-1} dilution. Subsequent serial dilutions (10^{-2} to 10^{-5}) were performed in TSB alone. For titration, 5 µl droplets from each dilution were spotted onto pre-labeled TSA plates containing a Kpn NCTC-13439 lawn. After overnight incubation at 37 °C, lysis plaques were quantified across three independent replicates. Adsorption was modeled as the temporal decline in normalized phage concentration. Based on the mechanistic interaction involving reversible phage–bacteria complex formation [40–42] and subsequent modifications [30, 43, 44], the following mathematical model was utilized: $(P_t/P_0) = (\phi/(\delta \times X_0 + \phi)) \times \{1 + (\delta \times X_0/\phi) \times \exp(-(\delta \times X_0 + \phi) \times t)\}$. In this model, P_t/P_0 represents the normalized phage density, δ is the first-order adsorption rate constant (CFU⁻¹ ml min⁻¹), ϕ is the first-order desorption rate (min⁻¹), X_0 is the initial susceptible host concentration (CFU ml⁻¹), and t denotes time (min). Parameter estimation was conducted via non-linear regression using the Solver tool in Microsoft Excel (Microsoft, Redmond, WA, USA).

Table 2 • Frequency of emergence of *K. pneumoniae* NCTC-13439 spontaneous resistant mutants to phages KpnSo1BRG and KpnSo2SCE.

Phage	Control sample (CFU/ml)	Sample treated with phage (CFU/ml)	Frequency of phage-resistant bacterial mutants
KpnSo1BRG	$(0.995 \pm 0.286) \times 10^{10}$	$(1.158 \pm 0.390) \times 10^7$	1.164×10^{-3} (0.116%)
KpnSo2SCE	$(1.040 \pm 0.489) \times 10^{10}$	$(1.793 \pm 0.549) \times 10^7$	1.724×10^{-3} (0.172%)

CFU: colony forming units.

A reversible adsorption model is superior to an irreversible one because it more accurately reflects the multi-step physical reality of how a virus interacts with a bacterial cell. The key justifications for this rationale are as follows: (i) It captures the “docking” phase (In reality, phage adsorption is not an instantaneous “lock-in”. The phage virion first undergoes a reversible “docking” or “scanning” phase where it attaches to primary receptors (like capsular polysaccharides or lipopolysaccharides (LPS)). A reversible model accounts for the fact that a phage virion can detach if it fails to find a secondary receptor for DNA injection/translocation.). (ii) It quantifies binding stability (By using a reversible adsorption model, one can calculate the phage virion desorption rate. This parameter serves as a critical biomarker for “binding strength”. An irreversible model assumes every collision results in a permanent infection, which overestimates efficacy and ignores differences in how “sticky” different phages are.). (iii) It reflects capsular dynamics (For encapsulated bacteria like *K. pneumoniae*, the phage virions must navigate a thick sugar layer. The ability to measure the rate at which phages fall off the capsule (desorption) versus staying attached is vital for predicting how the phage will perform in complex environments like biofilms.). (iv) Its mechanistic nuance (It differentiates between a phage that is highly efficient at finding the host but binds weakly (high adsorption rate constant, high desorption rate constant) and one that binds almost permanently once it hits the target (lower desorption rate constant). This distinction is completely lost in a simpler irreversible model.). In summary, while an irreversible adsorption model provides a single “speed” constant, a reversible model provides a “stability” profile, which is essential for understanding the actual success rate of the infection cycle.

2.7. Phage virion growth curves in a single synchronous cycle

Single-step growth curves were constructed to determine the replication kinetics of phages KpnSo1BRG and KpnSo2SCE. All assays were performed in triplicate employing a synchronized infection model with the host Kpn NCTC-13439 at an MOI ≤ 0.001 , adapted from prior protocols [30, 43]. Briefly, a 10 ml aliquot of logarithmic-phase bacterial culture ($OD_{610\text{ nm}} \approx 1.0$, yielding $\approx 1 \times 10^9$ CFU/ml) was harvested via centrifugation (2500 rpm, 20 min, 4 °C). The supernatant was decanted, and the cell pellet was resuspended in 5 ml of fresh TSB. To initiate infection, 5 ml of a standardized viral suspension (1×10^5 PFU/ml) was introduced. This mixture was incubated at 37 °C under mild agitation (100 rpm) for 5 min to facilitate viral attachment. Unadsorbed phages were subsequently removed by a second centrifugation step (2500 rpm, 20 min, 4 °C). The infected bacterial pellet was gently re-constituted in 10 ml of TSB and returned to the 37 °C incubator with orbital shaking (100 rpm). Starting at $t = 0$, 50 μ l samples were collected at 5 or 10 min intervals over a 150 min tracking period. These aliquots were promptly subjected to serial dilution and titrated using the double-layer agar technique, spotting 5

μ l per dilution onto host-seeded lawns. Plates were incubated overnight at 37 °C for plaque enumeration. To extract key kinetic parameters—specifically the latent period, intracellular accumulation time, and virion morphogenesis yield—the resulting time-series data (PFU/ml) were modeled using four-parameter logistic (4-PL) non-linear regression: $\text{Log}(P_t) = P_\infty + \{(P_0 - P_\infty)/(1 + (t/\psi)^\xi)\}$. Within this 4-PL framework, P_t , P_0 , and P_∞ define the viral titers at time t , the initial baseline ($t = 0$), and the maximum plateau ($t = \infty$), respectively. The variable ψ corresponds to the curve’s inflection point, ξ defines the Hill slope (steepness), and t is the elapsed time in minutes. Optimization of this non-linear fit was executed utilizing the Microsoft Excel Solver add-in.

2.8. In vitro phage–bacteria inactivation assays

Inactivation of planktonic Kpn NCTC-13439 cells (adjusted to a concentration of 10^5 CFU/ml during the exponential growth phase) by phages KpnSo1BRG, KpnSo2SCE, and their cocktail was studied at MOI values 0.0001, 0.01, 10, and 10,000 (a very high MOI was required to ensure the bacterial population is overwhelmed before resistant phenotypes can dominate). For each MOI experiment, a bacterial control (BC) was also prepared, encompassing only planktonic host cells. Both BC and treatments (BP-B, bacteria and phage—bacterial concentration) were incubated with the same time/temperature parameters, viz. 37 °C and 12 h. Two ml aliquots of BC and treatment samples (BP-B) were withdrawn at predetermined intervals of time up to a total treatment timeframe of 12 h, viz. 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 80, 100, 120, 150, 180, 210, 240, 270, 300, 360, 420, 480, 540, 600, 660, and 720 min, and their absorbance was measured at 610 nm.

2.9. Evaluation of the frequency of emergence of bacterial mutant cells resistant to the phages

The spontaneous mutation rate conferring resistance to phages KpnSo1BRG and KpnSo2SCE in Kpn NCTC-13439 was evaluated utilizing established methodologies [30, 36, 45]. Initially, ten independent susceptible colonies were grown in 5 ml of TSB at 37 °C with an orbital shaking speed of 150 rpm for 24 h. Following incubation, the bacterial suspensions were serially diluted to 10^{-7} by mixing 150 μ l of the culture with 1350 μ l of PBS. To isolate phage-insensitive cells, 100 μ l samples from the intermediate bacterial dilutions (10^{-1} to 10^{-4}) were mixed with 200 μ l of a 10^8 PFU/ml phage stock and soft agar (MTA-TSB), which was then overlaid onto solid TSA plates. For the enumeration of the total initial bacterial load, 100 μ l aliquots from the higher dilutions (10^{-5} to 10^{-7}) were subjected to pour-plating on standard TSA medium in the absence of viral particles. Both sets of plates were maintained at 37 °C for 48 h. The frequency of resistance emergence was subsequently calculated by dividing the titer of resistant survivors by the total viable bacterial titer of the phage-free controls [45]. This protocol was independently repeated three times.

2.10. Assessment of the impact of abiotic factors upon phage lytic viability

The impact of T, pH, and solar radiation upon phages KpnSo1BRG and KpnSo2SCE ($10^{8.1-9.5}$ PFU/ml) viability was studied in phosphate-buffered saline (10 ml). For the pH and T experiments, aliquots were withdrawn every 2 h during the first 6 h, every 6 h up to 12 h, every 12 h up to 48 h, and then every 24 h up to 72 h of incubation. For the solar radiation experiments, aliquots were withdrawn every 1 h up to 7 h of exposure to direct sunlight. Phage virion concentrations were determined in triplicate in three independent experiments as described before.

2.10.1. pH

The impact of pH on phages KpnSo1BRG and KpnSo2SCE lytic viability was assessed by adding phage suspensions to sterile phosphate-buffered saline with different pH values (viz. 3.0, 5.0, 7.0, 9.0, and 11.0), under constant temperature (25 °C).

2.10.2. Temperature

The impact of T on phages KpnSo1BRG and KpnSo2SCE lytic viability was assessed by adding phage suspensions to sterile phosphate-buffered saline (pH 7.0), followed by incubation at a constant temperature (4, 25, 37, and 41 °C).

2.10.3. Solar radiation

The sensitivity of phages KpnSo1BRG and KpnSo2SCE to solar exposure was evaluated by suspending the viruses in sterile phosphate-buffered saline (pH 7.0) and subjecting them to natural sunlight (SR group). A corresponding control group (SR-C) was maintained outdoors under identical thermal conditions but shielded from direct light. These experiments were conducted on a clear day with a documented direct solar irradiance of approximately 1636.4 kWh/m², localized at coordinates -23.499789, -47.400936 (Global Solar Atlas). It is well-established that solar radiation, particularly the ultraviolet (UV) spectrum, is a major driver of viral inactivation in environmental settings [36, 44]. Specifically, short-wavelength UV-B radiation induces irreversible genomic damage, protein modifications, and the creation of lethal photoproducts that compromise virion infectivity. To circumvent the UV-shielding properties of borosilicate or mineral glass [44], this study utilized small, colorless, non-mineral glass flasks. This choice of labware ensured that UV-A and UV-B wavelengths could effectively penetrate the vessel to interact with the phage suspensions.

2.11. Whole phage virion genome extraction and sequencing

Genomic sequencing of the isolated phages KpnSo1BRG and KpnSo2SCE was conducted at NGS SOLUÇÕES GENÔMICAS (Piracicaba, Brazil) using the Illumina MiSeq platform. To remove host nucleic acid contaminants, 500 µl of the PEG-concentrated phage suspension was incubated with 1.25 µl each of DNase-I and RNase for 1 h at 37 °C. Viral lysis and protein digestion were subsequently initiated by adding Proteinase K (1.25 µl), sodium dodecyl sulfate (SDS) (final concentration 0.5% w/w), and ethylenediamine tetraacetic acid (EDTA) (final concentration 20 mM, pH 8.0), followed by a 1 h incubation at 60 °C. After cooling, genomic DNA was recovered via organic extraction using a phenol:chloroform (1:1 v/v) ratio. The aqueous phase was

isolated after centrifugation (6000 rpm, 5 min), and the chloroform wash was performed twice. DNA was then precipitated overnight at -20 °C using 3 M NaOAc (1/10 volume) and cold absolute ethanol (2.5 volumes), followed by 30 min at -86 °C. The DNA pellet was harvested by centrifugation (14,000 rpm, 20 min), air-dried, and reconstituted in 50 µl of nuclease-free water. Final DNA purification included an RNase treatment before storage at -20 °C. Concentration and purity (A_{260nm}/A_{280nm} and A_{260nm}/A_{230nm} ratios) were evaluated using a DeNovix DS-11FX spectrophotometer (Wilmington, DE, USA) while further quantification and integrity were verified via the Quant-iT Picogreen assay and an Agilent 2100 Bioanalyzer (DNA 7500 chip, Santa Clara, CA, USA). Shotgun libraries were prepared from 2 to 20 ng of DNA using the Illumina Nextera kit (San Diego, CA, USA). Following purification with Agencourt AMPure XP beads (Beckman Coulter, Brea, CA, USA), library fragment sizes (400–700 bp) were confirmed on a High Sensitivity DNA chip (Agilent). Final sequencing was performed on the Illumina MiSeq (San Diego, CA, USA) using the v3 Reagent kit (600-cycle, paired-end) according to standard manufacturer protocols.

2.12. Phage genome assembly, taxonomy, annotation, and phylogeny

2.12.1. Assessment of sequencing reads

The sequencing data generated using the NextSeq 2000 platform (Illumina, Inc., San Diego, CA, USA) were processed using the BCL Convert software (version 3.8.4). This program performs base calling, transforming raw sequencing images into sequence data in fastq format, with each base assigned a corresponding Phred quality score [46]. Sequencing quality was evaluated using FastQC version 0.12.1 [47], and the results were consolidated into a comprehensive report using MultiQC version 1.14 [48].

2.12.2. Assembly of phage virion genomes

Genomic reconstruction for phage KpnSo1BRG was performed using Shovill v.1.1.0, an optimized pipeline for prokaryotic de novo assembly. This workflow achieves high-fidelity results through a coordinated sequence of operations: (i) genome size estimation via KMC v.3.2.1 [49]; (ii) read subsampling to a target coverage of 250x; (iii) adapter trimming with Trimmomatic v.0.39 [50]; (iv) error correction using Lighter v.1.1.2 [51]; (v) merging of overlapping pairs via FLASH v.1.2.11 [52]; (vi) primary assembly using SPAdes v.3.15.5 [53]; (vii) read mapping with BWA MEM v.0.7.17 [54]; and (viii) assembly polishing with Pilon v.1.24 [55]. Final refinements included the removal of contigs shorter than 500 bp, those with insufficient coverage, or those dominated by homopolymers. In contrast, KpnSo2SCE required a multi-stage iterative approach to maximize contiguity. Initial processing involved BBduk v.38.94 for quality filtering, followed by a “careful” SPAdes (version 3.15.5) assembly [53]. The resulting contigs were taxonomically screened using Kraken2 v.2.1.2 [56] against the PlusPF database version 20230605. Contigs identified as belonging to the *Webevirus* genus were isolated using KrakenTools v.1.2 [57]. These viral sequences were further partitioned by k-mer coverage; high-coverage contigs (>2000) served as templates to “bait” and select specific reads for a secondary Shovill-based assembly. For both phages, final per-contig sequencing depth was determined by aligning reads to the final assembly with Minimap2 v.2.25 [58].

2.12.3. Taxonomic assessment

Contigs were taxonomically classified using Kraken2 [56], a k-mer-based taxonomic classifier, utilizing the PlusPF database (version 20230605). Viral contigs were categorized based on the genus identified by Kraken2 and KrakenTools [57]. Subsequently, ccfind version 1.4.5 [59] was employed to detect circular contigs and trim their terminal repeats.

2.12.4. Phage virion genome annotation

We performed a comprehensive annotation of the viral genomes using the Pharokka 1.4.1 [60] framework (run with the default options) and database 1.4.0 from zenodo.org (record 8267900). This dedicated phage-centric pipeline coordinates multiple bioinformatics utilities to ensure high-fidelity functional predictions. Specifically, structural RNA elements (tRNAs/tmRNAs) were detected using tRNAscanSE (version 2.0.12) [61] and ARAGORN (version 1.2.41) [62], while potential CRISPR clusters were mapped using MinCED (version 0.4.2) [63]. For the protein-coding regions, gene models were established via PHANOTATE (version 1.5.1) [64]. The resulting protein sequences were then functionally categorized by MMseqs2 (version 13.45111) [65] through similarity profiling against the PHROG [66], CARD [67], and VFDB [68] datasets. Beyond individual gene annotation, Pharokka determined the closest genomic neighbors by executing Mash (version 2.3)-based [69] searches within the INPHARED collection [70]. When high-confidence hits were absent from this specialized database, the NCBI NT library was queried using BLAST+ (version 2.14.1) [71] to establish the taxonomic and evolutionary context of the newly recovered contigs [72].

2.12.5. Phylogeny of lineages

To pinpoint the most closely related *Webervirus* genomes, we utilized ReferenceSeeker (v.1.8.0) [73]. This tool quantifies genomic similarity by integrating Mash-based sketches (v.2.3) [69] with average nucleotide identity (ANI) calculations, the latter derived from nucmer alignments (MUMmer package v.4.0.0-RC1) [74]. ReferenceSeeker's searches were performed against a comprehensive database of RefSeq/NCBI bacterial and viral genomes [75]. The genomic dataset for phylogenetic analysis was composed of the top hits from ReferenceSeeker and the two most similar entries from the INPHARED database. We estimated evolutionary distances using Mashtree (v.1.2.2) [76], which generates approximate local alignments to determine pairwise distance metrics. The final phylogenetic hypothesis was reconstructed using the neighbor-joining method in QuickTree (v.2.5) [77]. To ensure the statistical robustness of the branching patterns, 999 bootstrap replicates were performed.

2.12.6. Viral proteomic trees

To investigate the proteomic similarities and evolutionary relationships between phages KpnSo1BRG and KpnSo2SCE and other prokaryotic dsDNA viruses, we utilized the ViPTree web server available from genome.jp [78]. The computational processing required for these genomic comparisons was executed using the Supercomputer System at Kyoto University's Institute for Chemical Research (Kyoto, Japan).

2.13. Mechanical features of the KpnSo1BRG and KpnSo2SCE phage genomes

2.13.1. DNA structural features

To gain deeper insights into the physical properties of the isolated phages, we modeled four primary DNA structural features: ProT (Propeller twist is the rotation of the two nitrogenous bases within a single DNA base pair relative to each other, where they twist along an axis perpendicular to the main helical axis. This deformation makes base pairs non-planar (resembling a propeller) and helps hide hydrophobic base surfaces from water), HelT (Helical twist of the DNA double helix is the angle of rotation between consecutive base pairs, which results in a complete 360° turn of the helix every 10 base pairs. This "twisting" of the two sugar-phosphate backbones provides stability and compacts the molecule.), roll (Parameter describing the conformational rotation of adjacent base pairs around their long axis. It represents a "bending" motion where consecutive base pairs open or close towards the major or minor grooves, essential for determining DNA curvature and flexibility when interacting with proteins.), and MGW (minor groove width). These parameters were derived through the DNASHape prediction tool available from rohslab.cmb.usc.edu [79], utilizing the pentamer-based methodology described in the literature [18, 33, 43, 80]. This predictive approach calculates structural values for each nucleotide by incorporating its specific sequence context. Visualization and downstream analysis were performed using a custom-developed Python (v.3.9.12) workflow executed in Jupyter Notebook (v.6.4.8) via Anaconda Navigator (v.2.1.4). This computational setup allowed for the generation of high-resolution heatmaps, which served as the basis for a thorough characterization of the genomes' structural motifs.

2.13.2. Correlation analysis of the DNA structural features between the two phage genomes

The dependency between different DNA structural attributes was quantified through Pearson correlation testing for both phage genomes. This allowed for a systematic identification of potential co-variations within the structural landscape of the viral DNA. The resulting correlations were visualized via high-resolution heatmaps, which displayed the structural features in relation to the total number of nucleotides. These plots provided a detailed summary of the correlation values, their distribution patterns, and their statistical weight. In these heatmaps, a dual-color gradient was applied to represent the strength of the relationships, where red signifies high correlation and blue indicates low or inverse correlation values.

2.13.3. Dinucleotide distant correlation patterns for the KpnSo1BRG and KpnSo2SCE phage genomes

To investigate periodicities in the viral DNA architecture, we calculated the pairwise distance distribution for both isolated phages following the methodology of Basu et al. [81]. This function measures how often two specific dinucleotides appear at defined separation distances, with results normalized against expected frequencies from a random sequence background. The study provided a comprehensive mapping of all 256 potential dinucleotide pairings. For comparative purposes, the distribution profiles for KpnSo1BRG and KpnSo2SCE were overlaid in the same graphical output. These

plots illustrate the correlation and distribution frequency for each dinucleotide transition within a 100-step window along the genomic sequence.

2.13.4. Dinucleotide frequency in the KpnSo1BRG and KpnSo2SCE phage genomes

The distribution of the 16 primary dinucleotide pairs within the KpnSo1BRG and KpnSo2SCE sequences was assessed via a customized computational workflow. Each dinucleotide's occurrence was enumerated across the entire genome and subsequently expressed as a proportion of the total base count. By calculating these relative percentage frequencies, we were able to characterize the sequence-specific motifs of the two phages. This data was ultimately visualized through bar-chart illustrations to highlight the compositional differences and similarities between the genomes.

2.13.5. Differential dinucleotide frequency between KpnSo1BRG and KpnSo2SCE phage genomes

The divergence in the dinucleotide landscape of KpnSo1BRG and KpnSo2SCE was evaluated using a custom-developed computational workflow. A custom Python script was designed to find the differential dinucleotide frequency between the genomes. We quantified the frequency of all 256 possible dinucleotide combinations in each genome and subsequently calculated the relative difference by subtracting the KpnSo1BRG values from the KpnSo2SCE dataset. The resulting comparative profile was represented through high-resolution heatmaps. In these plots, red and blue gradients were utilized to denote increases and decreases in frequency, respectively, providing a visual summary of the compositional distance between the two viral genomes.

2.13.6. Phage genome cyclizability

The intrinsic looping potential, or cyclizability, of the KpnSo1BRG and KpnSo2SCE genomes was determined using the computational framework established by Basu et al. [81, 82] and Zhang et al. [83], as previously applied in related studies [18, 33, 80]. These values were calculated across the viral genomes using a 50-nucleotide window with a 7 bp sliding step. To investigate the spatial distribution of DNA bendability, the resulting datasets were mapped to specific genomic coordinates. We utilized a bespoke Python script (v.3.9.12) (executed within a Jupyter Notebook environment (v.6.4.8)) to generate comprehensive heatmaps and histograms. Statistical distributions were further characterized using box plots, where the internal black line represents the mean and the whiskers indicate the full range (minimum to maximum) of cyclizability; extreme outliers are highlighted as green triangles. This high-resolution mapping facilitated the calculation of average bendability across the entire sequence, allowing for a standardized structural comparison between the two phages.

2.13.7. Phage genome circular maps

The structural and functional architecture of the isolated phages was visualized using the Proksee genomic server. By applying this specialized visualization suite, we constructed circular genomic profiles for KpnSo1BRG and KpnSo2SCE at individual-base resolution. These maps provide a comprehensive overview of the viral genomes, displaying the distribution of CDSs alongside core

compositional metrics like GC skew and content. To offer a deeper biophysical context, the circular diagrams also map local DNA shape parameters (comprising roll, ProT, MGW, HelT, and cyclizability), thereby facilitating a side-by-side comparison of the physical and genetic traits of both phages.

2.14. Statistical analyses

We performed a lack of fit analysis to evaluate the quality of the non-linear fits for the adsorption and 4-PL growth models [43, 44, 84–86]. The *F*-ratio was derived by dividing the lack of fit Mean Square by the replication Mean Square, using the respective sums of squares and degrees of freedom. This calculated value was then weighed against the theoretical *F*-table critical value ($\alpha = 5\%$). Prior to further testing, the data were screened for normal distribution (Kolmogorov–Smirnov test) and equality of variances (Levene's test). The efficacy of the newly isolated lytic phages in reducing bacterial loads during in vitro inactivation was analyzed via two-way ANOVA, focusing on the interaction between sampling time and treatment. Differences were pinpointed by comparing the phage-treated samples (BP) with the corresponding control cultures (BC) at different time points and MOIs. Statistical significance across all tests was set at $p < 0.05$. All data processing was managed within the Microsoft Excel environment.

3. Results

In this study, two newly isolated lytic phages targeting *K. pneumoniae* NCTC-13439 (Kpn) were thoroughly investigated for their physicochemical, biological, and genomic properties. These phages, which form clear lytic plaques of varying sizes on their host lawn (a hallmark characteristic of lytic phages) were analyzed with the goal of exploring their potential future application in controlling infections caused by Kpn.

3.1. Kpn growth curve

A laboratory strain of *K. pneumoniae* NCTC-13439 was employed as the host for isolating lytic bacteriophages. To determine the optimal time for phage isolation, a bacterial growth curve was generated (**Figure 1**), which was essential for identifying when the cells entered the exponential growth phase. During the initial 3 h of growth, the bacteria remained in a lag phase. After this period, growth gradually began, with the exponential phase commencing around the 5 h mark and continuing until approximately 8 h of growth (**Figure 1**). Following this, the transition into the stationary phase became evident. These findings were critical for the subsequent research, as phage infection is most effective when the bacterial culture is in the exponential growth phase. The experimental bacterial growth data were fitted to the Gompertz function using non-linear regression, which enabled the estimation of key growth parameters: the maximum biomass concentration at $t = \infty$ was calculated to be 1.280×10^9 CFU/ml (equivalent to a maximum absorbance of 1.280), the lag phase duration was determined to be 180 min (approximately 3.0 h), and the mid-exponential growth phase was estimated to occur at around 354 min (approximately 5.9 h).

3.2. Phage plaque isolation

The virions of both phages formed clear plaques on the Kpn lawn, with diameters ranging from ca. 1.5 mm (phage KpnSo1BRG; **Figure 2a**) to ca. 2 mm (phage KpnSo2SCE; **Figure 2c**). A secondary halo could be observed in the frontier of phage KpnSo2SCE lysis plaques (**Figure 2c**), indicative of the production of phage depolymerases by this phage [87–89], but definitive depolymerase activity should be confirmed via genomic identification.

Based on the morphological examination of the phage particles using transmission electron microscopy (TEM) (**Figure 2b,d**), both phages were classified under the class *Caudoviricetes* and exhibited a siphovirus-like morphotype with perfect icosahedral capsids (**Figure 2b,d**) and long, apparently flexible, non-contractile tails.

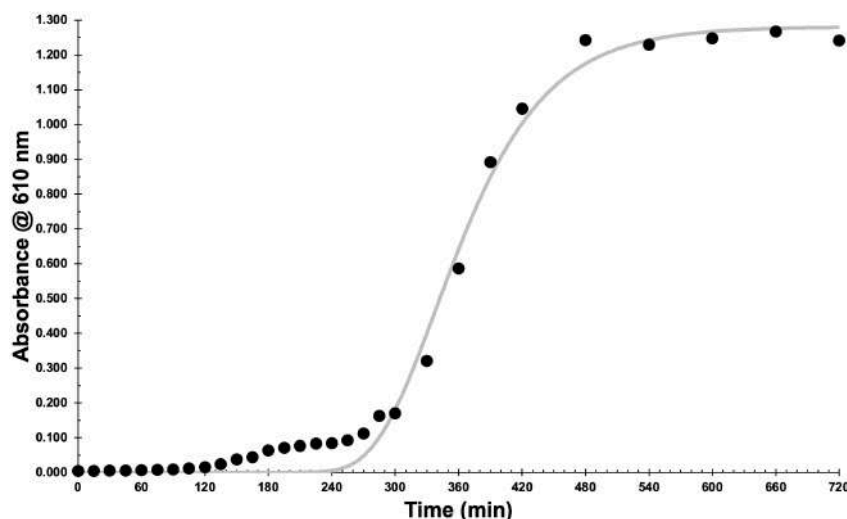


Figure 1 • Growth curve of the host bacteria (*K. pneumoniae* NCTC-13439). The solid gray line illustrates the non-linear fitting of the Gompertz function to the experimental data, enabling the estimation of the maximum biomass concentration at $t = \infty$ of 1.280×10^9 CFU/ml (corresponding to a maximum absorbance of 1.280), a lag period (180 min, ca. 3.0 h), and a mid-exponential growth phase at ca. 354 min (ca. 5.9 h). The values shown are the average of three independent experiments, with (occluded, due to being very small) error bars indicating the standard deviation. CFU: colony forming units.

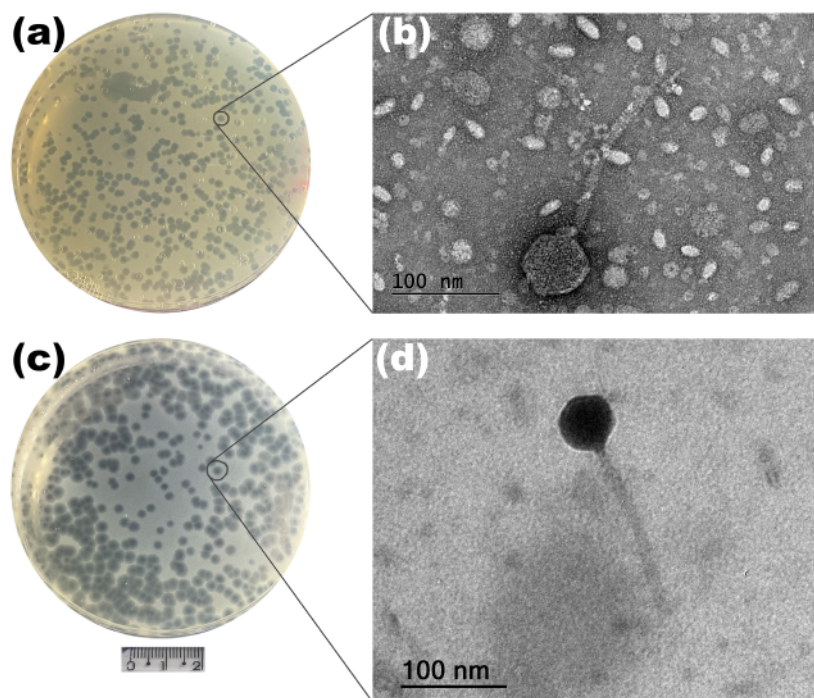


Figure 2 • Morphology of phages KpnSo1BRG (a) and KpnSo2SCE (c) plaques on their bacterial host (*K. pneumoniae* NCTC-13439). The phage virion morphotypes were also determined using transmission electron microscopy (TEM) following negative staining ((b) KpnSo1BRG; (d) KpnSo2SCE). The negative-staining TEM photomicrographs of the KpnSo1BRG ((b) $\times 250k$) and KpnSo2SCE ((d) $\times 200k$) phage virions were obtained at different times, with different magnifications, to observe the intact heads (containing the dsDNA) and long, non-contractile tails.

3.3. Host range of the isolated phages

A total of 66 strains/isolates of nine bacterial species (Table S1) were tested for infection with phages KpnSo1BRG and KpnSo2SCE. Both phages only infected the host strain (*K. pneumoniae* NCTC-13439) with 100% efficiency but failed to infect any other strains of *K. pneumoniae* (one collection strain, 13 isolates from animal samples, and 30 isolates from human samples). The narrow host range (1.5% of tested strains) is a well-documented characteristic of the genus *Webervirus* of the family *Drexelviriidae*, rather than a specific limitation of these two phage isolates. The literature on *Webervirus* phages consistently reports high host specificity, often limited to a single or very few *K. pneumoniae* capsular (K) types. This is primarily because *Webervirus* phages rely on highly specific tail fiber-associated depolymerases to penetrate the diverse capsular polysaccharide layer of *Klebsiella*. Since there are several (at least 77) recognized K-types, a phage targeting one specific capsule will naturally fail to infect the majority of a diverse clinical panel, which underscores the significance of the results obtained. Both phages KpnSo1BRG and KpnSo2SCE exhibited specificity, binding exclusively to the target bacterial strain (Kpn NCTC-13439, their host; Table S1) and killing it, most likely via either phage progeny-induced burst or from extensive piercing of the cell membrane due to high-multiplicity virion adsorption. Hence, phages KpnSo1BRG and KpnSo2SCE displayed a narrow host range. It is widely recognized that the morphogenesis of phage progeny virions within target bacterial cells relies heavily on the metabolic machinery and precursor molecules derived from the infected host cells. This resource utilization during virion assembly significantly disrupts the intricate network of metabolic pathways within the target bacterial cells [90], which, adding to the absence of bacterial cell surface receptor recognition by the phage virions, further imparts a putative incompatibility of the target host metabolic machinery for the phage virion morphogenesis.

3.4. Phage virion adsorption rate onto bacterial host cells

Adsorption assays for phage KpnSo1BRG revealed that approximately 50% of the virions adsorb to Kpn cells within 5 min, with

over 95% of them being adsorbed after 25 min (Figure 3a). Similarly, for phage KpnSo2SCE, around 55% of the virions adsorbed to Kpn cells within 5 min, and more than 98% were adsorbed after 25 min (Figure 3b).

Non-linear fitting of the experimental phage adsorption data to the adsorption model allowed to determine both the adsorption rate (δ) of the phage virions onto their bacterial host cells and desorption rate (ϕ) of the phage virions from reversible virion-bacteria complexes: $\delta_{\text{KpnSo1BRG}} = 1.371 \times 10^{-9} \text{ CFU}^{-1} \text{ ml min}^{-1}$ ($\phi_{\text{KpnSo1BRG}} = 1.200 \times 10^{-7} \text{ min}^{-1}$; $X_0 = 1.0 \times 10^8 \text{ CFU ml}^{-1}$), $\delta_{\text{KpnSo2SCE}} = 1.585 \times 10^{-9} \text{ CFU}^{-1} \text{ ml min}^{-1}$ ($\phi_{\text{KpnSo2SCE}} = 6.005 \times 10^{-11} \text{ min}^{-1}$; $X_0 = 1.0 \times 10^8 \text{ CFU ml}^{-1}$).

A statistical analysis of lack of fit of the adsorption decay model [40, 42] to the experimental data allowed it to be concluded that at $\alpha = 0.05$ (95% confidence), there was no lack of fit to the adsorption data of the two phages (phage KpnSo1BRG: calculated $F_{\text{ratio}} = 0.1557$, standard $F_{\text{ratio}} = 1.9877$, p -value = 0.9997; phage KpnSo2SCE: calculated $F_{\text{ratio}} = 0.1427$, standard $F_{\text{ratio}} = 2.0909$, p -value = 0.9987). A p -value greater than 0.10 (indicating statistically insignificant lack of fit) suggests that the model accurately predicts the experimental response data within a 95% confidence interval (Figure 3).

3.5. Phage virion growth curves in a single synchronous cycle

Non-linear regression analysis of the phage virion growth data in a single synchronous cycle to the 4-PL model showed that phage KpnSo1BRG is characterized by latent and intracellular accumulation periods that last 60 and 30 min, respectively (Figure 4a), whereas phage KpnSo2SCE is characterized by latent and intracellular accumulation periods that last 5.8 and 1.8 min, respectively (Figure 4b). The virion morphogenesis yield of phage KpnSo1BRG was 17.7 virions/host cell (r^2 (coefficient of determination) = 0.9641), whereas the virion morphogenesis yield of phage KpnSo2SCE was 18.6 virions/host cell (r^2 (coefficient of determination) = 0.4242).

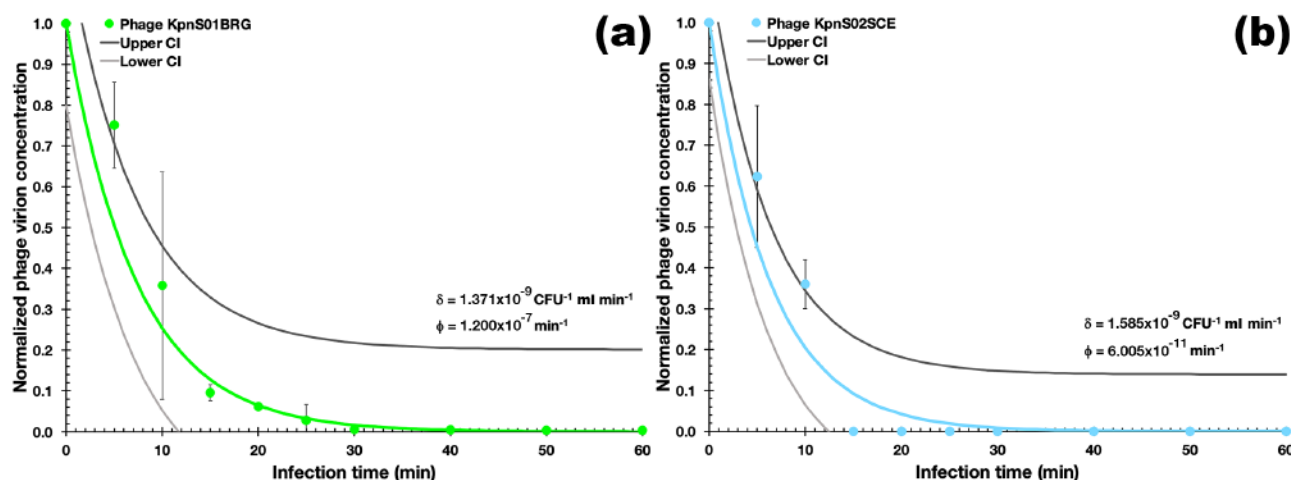


Figure 3 • Adsorption curves of phages KpnSo1BRG (a) and KpnSo2SCE (b) in the presence of Kpn NCTC-13439 as the host. The green (phage KpnSo1BRG, (a)) and blue (phage KpnSo2SCE, (b)) solid lines represent the non-linear fittings of the adsorption decay model to the experimental phage adsorption data. Values are the means of three independent assays; error bars represent the standard deviations. The dark gray and light gray lines represent, respectively, the upper and lower 95% confidence intervals (CIs) for the non-linear fittings performed.

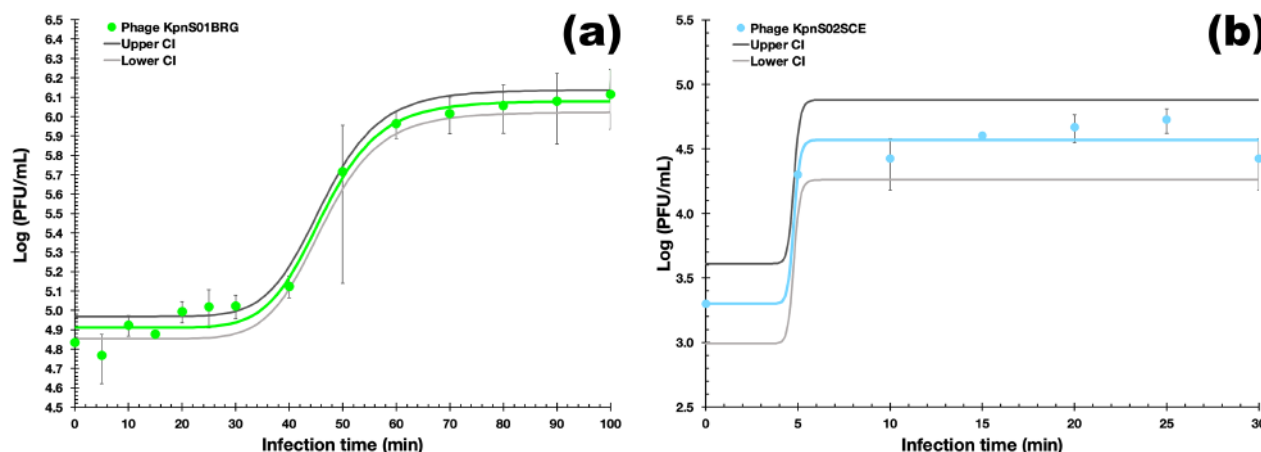


Figure 4 • One-step growth curve of phages KpnSo1BRG (a) and KpnSo2SCE (b) in the presence of Kpn NCTC-13439 as the host. The green (phage KpnSo1BRG, (a)) and blue (phage KpnSo2SCE, (b)) solid lines depict the non-linear fitting of the 4-PL model to the experimental phage growth data. Values represent the averages of three independent experiments, with error bars indicating the standard deviations. The dark gray and light gray lines represent, respectively, the upper and lower 95% confidence intervals (CIs) for the non-linear fittings performed.

A statistical assessment of the lack of fit for the 4-PL model [30, 32, 43] applied to the experimental phage growth data was carried out and allowed it to be concluded that at $\alpha = 0.05$ (95% confidence), no lack of fit was observed for the phage growth data (phage KpnSo1BRG: calculated $F_{ratio} = 0.0622$, standard $F_{ratio} = 2.1425$, p -value = 0.9999; phage KpnSo2SCE: calculated $F_{ratio} = 0.2526$, standard $F_{ratio} = 2.6143$, p -value = 0.9642). Since lack of fit is the variation in the experimental data around the theoretical fitted model, a p -value > 0.10 (lack of fit statistically insignificant) indicates that the model accurately predicts the experimental response data within a 95% confidence interval (Figure 4).

3.6. In vitro phage–bacteria inactivation assays

The bacterial concentration in the control (BC) showed a significant increase (ANOVA, $p < 0.05$) over the 12 h incubation period (Figure 5a,b). When the phages were applied individually at different MOI levels, all tested values effectively controlled bacterial growth by the end of the assay timeframe (Figure 5a,b). At MOI 0.0001, phages KpnSo1BRG and KpnSo2SCE managed to reduce the bacterial load by ca. 94.2% and 84.7%, respectively, after 8 h of incubation with the host, which slightly reduced to ca. 82.3% and 82.6% (ANOVA, $p > 0.05$), respectively, up to 12 h of assay (Figure 5a,b). However, at this MOI, both phages failed at preventing bacterial regrowth after 6 h of contact with their host (Figure 5a). At MOI 0.01, phage KpnSo2SCE was quite successful in achieving high bacterial inactivation, with ca. 94.1% of bacterial reduction being attained after 8 h, which more or less was maintained up to the end of the assay timeframe (Figure 5a,b). At this MOI, however, phage KpnSo1BRG failed at preventing bacterial regrowth after 6 h of contact with its host (Figure 5a). At MOI 10, both phages were quite successful in achieving high bacterial inactivation, with ca. 98.8% of bacterial reduction being attained after 8 h, with only a slight reduction in bacterial inactivation up to the end of the assay timeframe (Figure 5a,b; ANOVA, $p > 0.05$). At MOI 10,000, however, the story was quite different. At this MOI, both phages succeeded in achieving high bacterial inactivation, with ca. 98.8% and 99.9% of bacterial reduction being attained after 8 h for phages KpnSo1BRG and KpnSo2SCE, respectively, which slightly reduced to 95.9% and 99.8%, respectively, by the end

of the experiment (Figure 5a,b). Hence, MOI 10,000 for either independent phage proved to be quite successful in controlling the bacterial levels during a 12 h timeframe. Therefore, MOI 10,000 was selected for preparing a cocktail with both phages and as expected, the lytic phage cocktail was very effective in promoting a high bacterial inactivation, viz. 98.8% after 8 h, which only slightly reduced to 97.2% after 12 h (Figure 5a,b). The bacterial inactivation with both independent phages at MOI 10,000 was much better than that at MOI 0.0001 for both independent phages and than that at MOI 0.01 for phage KpnSo1BRG after 6 h, 8 h, 10 h, and 12 h of incubation (ANOVA, $p < 0.05$). After 8 h of phage–bacteria contact, the difference in bacterial reduction using both independent phages at MOI 10,000 was not statistically different from that using the phage cocktail at MOI 10,000 (ANOVA, $p > 0.05$). Increasing the MOI from 0.0001 g 0.01 g 10 g 10,000 did significantly increase the inactivation factor after 12 h of incubation with both independent phages (ANOVA, $p < 0.05$) (Figure 5a,b). When phages KpnSo1BRG and KpnSo2SCE were independently introduced to planktonic host cells, a notable bacterial regrowth was detected after 6 h of incubation (ANOVA, $p < 0.05$) at a MOI of 0.0001 (Figure 5a). However, by the conclusion of the incubation period, the bacterial densities in all treatments using either phage were significantly lower at every MOI tested compared to the bacterial control (BC; Figure 5a).

While a MOI of 10,000 was extremely high and not physiologically relevant, it was required to ensure that the bacterial population is overwhelmed before resistant phenotypes can dominate. As demonstrated by the data presented, (very) high MOI applications were necessary to evaluate the threshold required for effective short-term control and to prevent rapid bacterial regrowth. Our decision to test (very) high MOI values (up to 10,000) in vitro was not intended to represent a standard clinical application, but rather to establish the biological threshold required to prevent bacterial evasion and regrowth. As observed in the data in Figure 5a, much lower MOI values (0.0001, 0.01, and 10) resulted in significant bacterial regrowth after 5 to 6 h of treatment. A very high MOI value was strategically necessary to overwhelm the bacterial population, potentially utilizing “lysis from without”, and to accurately assess the maximum suppressive capacity of the phages before resistance could establish.

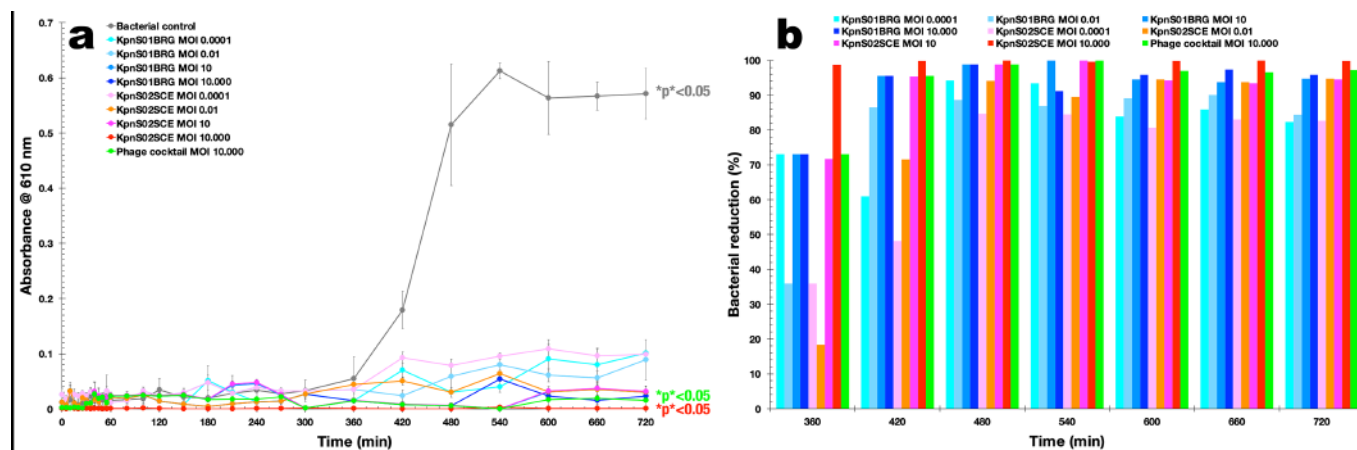


Figure 5 • In vitro inactivation of free-floating (planktonic) Kpn NCTC-13439 cells by independent phages KpnSo1BRG and KpnSo2SCE at a multiplicity of infection (MOI) of 0.0001, 0.01, 10, and 10,000, and their cocktail at MOI 10,000 (a), and the percentage reductions in planktonic bacterial concentrations achieved at all MOI levels for both individual phages and their cocktail (b) over a 12 h treatment period. Values represent the average of three independent experiments. **p* < 0.05 means statistical *p*-value. BC: bacterial control; BP-B: bacteria treated with phage.

3.7. Assessment of the frequency at which bacterial mutant cells resistant to the phages emerge

The frequency of Kpn NCTC-13439 mutants resistant to phages KpnSo1BRG and KpnSo2SCE was 1.164×10^{-3} (0.116%) and 1.724×10^{-3} (0.172%), respectively (Table 2).

As shown in Figure 6, after 3 h of incubation with phages KpnSo1BRG and KpnSo2SCE spotted on a lawn of Kpn NCTC-13439, clear zones of lysis caused by the phages become visible.

These zones gradually become more translucent up to 6 h of incubation. However, after 6 h of incubation in the presence of the host, resistant colonies begin to emerge within the lysis zones for both phages (with nearly identical frequencies; Table 2). These resistant colonies become highly pronounced after 24 h of incubation and remain relatively stable in number up to 72 h. These findings align with the phage–bacteria inactivation assays, where significant bacterial regrowth was observed after 6 h of treatment with both phages at a MOI of 0.0001 (Figure 5a).

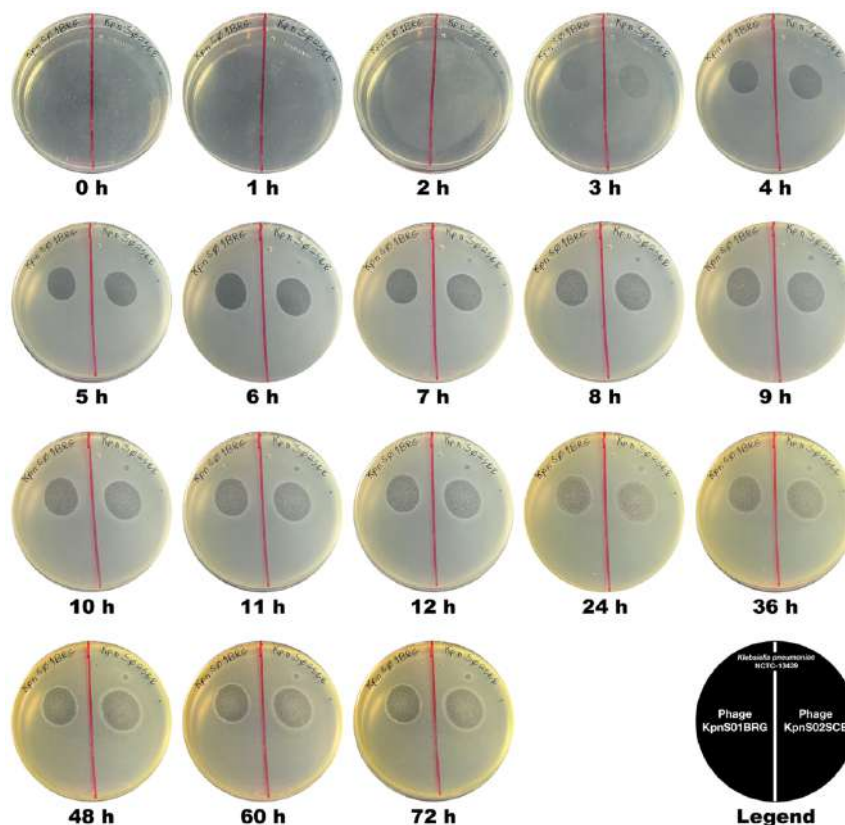


Figure 6 • Emergence of Kpn NCTC-13439 bacterial mutant cells resistant to phages KpnSo1BRG and KpnSo2SCE during a 72 h timeframe assay of in vitro phage spotting on a lawn of the host.

The use of phages to inactivate bacteria inevitably leads to the emergence of phage-resistant bacterial mutants, which represents a major concern [45, 91–96]. Phages KpnSo1BRG and KpnSo2SCE did not fully prevent bacterial regrowth during treatment, especially at very low MOI (viz. 0.0001) (**Figure 5a**). During the phage–bacteria inactivation assays (**Figure 5a**), it is difficult to definitively determine whether the surviving bacteria are truly resistant. This is because bacterial cells are collected from liquid TSB samples where phage virions and planktonic cells coexist, making it impossible to confirm that all tested bacterial cells had actually encountered the virions through random Brownian motion.

A frequent bacterial defense mechanism against phage attacks is the development of resistance by altering cell surface structures recognized by phages, thereby preventing phage adsorption [24, 97].

When applied alone, neither of the two characterized KpnSo1BRG and KpnSo2SCE phages was able to completely suppress the growth of Kpn NCTC-13439 for more than 6 h, especially at very low MOI, most likely due to the inherent capability of *K. pneumoniae* to generate spontaneous phage-resistant mutants (**Figures 5a** and **6**). This is a clear indication that for successful phage treatment of *K. pneumoniae* infections, a cocktail of multiple lytic phages would be absolutely necessary [98–102].

The frequency of emergence of spontaneous resistant mutants of Kpn NCTC-13439 to phages KpnSo1BRG and KpnSo2SCE, specifically for cells that interacted with the phage virions, was relatively low (1.386×10^{-3} for phage KpnSo1BRG and 1.577×10^{-3} for phage KpnSo2SCE) (**Table 2**). The development of resistant mutants of Kpn NCTC-13439 cells for phages KpnSo1BRG and KpnSo2SCE are higher than those obtained by other researchers for *K. pneumoniae* of sequence type 101 and phage vB_Kpn_F48 (1.5×10^{-7}) [103], but are much lower than those obtained by other researchers for *K. pneumoniae* K3574 and phage vB_KpnS-VAC35 (ca. 0.4) or *K. pneumoniae* K3325 and phage vB_KpnS-VAC35 (ca. 0.25) [4].

3.8. Stability of phages KpnSo1BRG and KpnSo2SCE upon exposure to abiotic factors

3.8.1. Temperature experiments

Testing the effects of temperature upon the lytic viability of phages KpnSo1BRG (**Figure 7a**) and KpnSo2SCE (**Figure 7b**) revealed that these phages were very stable at 4 °C and 25 °C, without loss of activity at these temperatures during a timeframe of 72 h of incubation (**Figure 7a,b**; ANOVA, $p > 0.05$). At 37 °C, phage KpnSo1BRG lost 2.15 log PFU/ml, whereas phage KpnSo2SCE lost only 0.18 log PFU/ml of lytic activity over a 72 h period of incubation at that temperature. At 41 °C, however, phage KpnSo1BRG lost 3.53 log PFU/ml, whereas phage KpnSo2SCE lost 1.13 log PFU/ml of lytic activity at this temperature. Hence, both phages revealed to be quite resistant to the effects of high(er) temperatures upon their lytic viability, with the reduction in the concentration of viable phage virions being much higher at 41 °C for phage KpnSo1BRG than that for phage KpnSo2SCE (**Figure 7a,b**; ANOVA, $p < 0.05$).

3.8.2. pH experiments

The impact of different pH values (3.0, 5.0, 7.0, 9.0, and 11.0) upon the lytic viability of phages KpnSo1BRG (**Figure 7c**) and KpnSo2SCE (**Figure 7d**) unveiled an unusual resistance of these phages to alkaline conditions. While at acidic pH values (viz. 3 and 5) both phages lost a few log PFU/ml of lytic activity over a timeframe of 72 h, with phage KpnSo1BRG (reduction of 5.27 log PFU/ml at pH 3.0 and reduction of 2.45 log PFU/ml at pH 5.0) revealing to be more sensitive than phage KpnSo2SCE (reduction of 2.39 log PFU/ml at pH 3.0 and reduction of 1.29 log PFU/ml at pH 5.0) to lower pH values, both phages revealed to be relatively insensitive to higher pH values (viz. pH 9 and 11), with both phages losing only 0.72 log PFU/ml of lytic activity over a 72 h period. At neutral pH, both phages remained with nearly all their lytic activity, with phage KpnSo1BRG losing only 0.08 log PFU/ml and phage KpnSo2SCE losing only 0.49 log PFU/ml of lytic activity.

3.8.3. Solar radiation experiments

When phage KpnSo1BRG was subjected to solar radiation, the number of viable phage particles remained relatively stable over the exposure period, with only a slight difference observed between the control and test samples (**Figure 7e**; ANOVA, $p > 0.05$). However, the concentration of infectious virions (SR) decreased by approximately 2.06 log PFU/ml (**Figure 7e**; ANOVA, $p < 0.05$) after 7 h of direct sunlight exposure compared to the phage control (SR-C). These results indicate that phage KpnSo1BRG exhibits considerable resilience when exposed to direct sunlight. Regarding phage KpnSo2SCE (**Figure 7f**), no decrease in the concentration of viable lytic virions (SR) could be observed during the whole timeframe when compared to the phage control (SR-C), hence revealing a high resilience of phage KpnSo2SCE to the nefarious effects of solar radiation.

3.9. Genomic characterization

The genome sequence of bacteriophage KpnSo1BRG resulted in a contig of 51,716 bases, whereas the genome sequence of bacteriophage KpnSo2SCE resulted in a contig of 49,460 bases, with both being identified as belonging to the *Drexelviriidae* family, genus *Webevirus*. Circularization of the two *Webevirus* contigs was not possible. The GC content of the KpnSo1BRG phage genome is 51.69%, whereas that of the KpnSo2SCE phage genome is 51.08%. The overall features of the genome assemblies of phages KpnSo1BRG and KpnSo2SCE and their annotation are summarized in **Table 3**.

The genomes of phages KpnSo1BRG and KpnSo2SCE did not encode any tRNAs but encoded 87 protein coding sequences (CDSs) (phage KpnSo1BRG) and 85 CDSs, respectively (**Table S2**). A comparison of the annotated coding sequences (CDSs) in the genomes of phages KpnSo1BRG and KpnSo2SCE with various databases unveiled that 50 (phage KpnSo1BRG) and 49 (phage KpnSo2SCE) of these CDSs are predicted to encode hypothetical proteins or proteins with unknown functions. Additionally, typical structural proteins, including capsid (head), neck, tail, tail fiber, and tail length tape measure proteins, were identified and annotated in the genomes of both phages, along with DNA repair and metabolism-related proteins and host lysis proteins (holin, endolysin, spanin). No genes related to depolymerases, toxins,

virulence factors, antibiotic resistance, or integrase enzymes could be detected among the CDSs with predicted functions in the genomes of both phages (Table S2). A little more than half of the CDSs identified in the genomes of phages KpnSo1BRG and KpnSo2SCE were annotated as hypothetical proteins (unknown function) (Table S2). Circular maps of the annotated genomes of phages KpnSo1BRG and KpnSo2SCE are displayed in **Figure 8**. Mechanical features of the genomes of both phages are also displayed (cyclizability, helical twist, minor groove width, propeller twist, roll) in the circular maps.

In the network analysis, phage KpnSo1BRG was grouped with 30 phages at the family level and 13 phages within the same cluster at the genus level. Similarly, phage KpnSo2SCE was grouped with 30

phages at the family level and seven phages within the same cluster at the genus level (**Figure 9**). The phylogeny was constructed using core proteins identified in the analysis. Notably, phages KpnSo1BRG and KpnSo2SCE occupy distinct branches, indicating lower phylogenetic similarity to other phages in the network (**Figure 9**). The phage closest to the KpnSo1BRG phage is *Klebsiella* phage PWKp17 (GCA_021800495.1), with an ANI value of 97.7%, whereas the phage closest to the KpnSo2SCE phage is *Klebsiella* phage pK8 (GCF_021870645.1), with an ANI value of 96.8% (**Figure 9**). These genomes exhibit the closest genomic proximity to the KpnSo1BRG and KpnSo2SCE phage genomes, as determined by a genomic distance metric computed using Mash, a tool that estimates distances between genomes based on k-mer sketches.

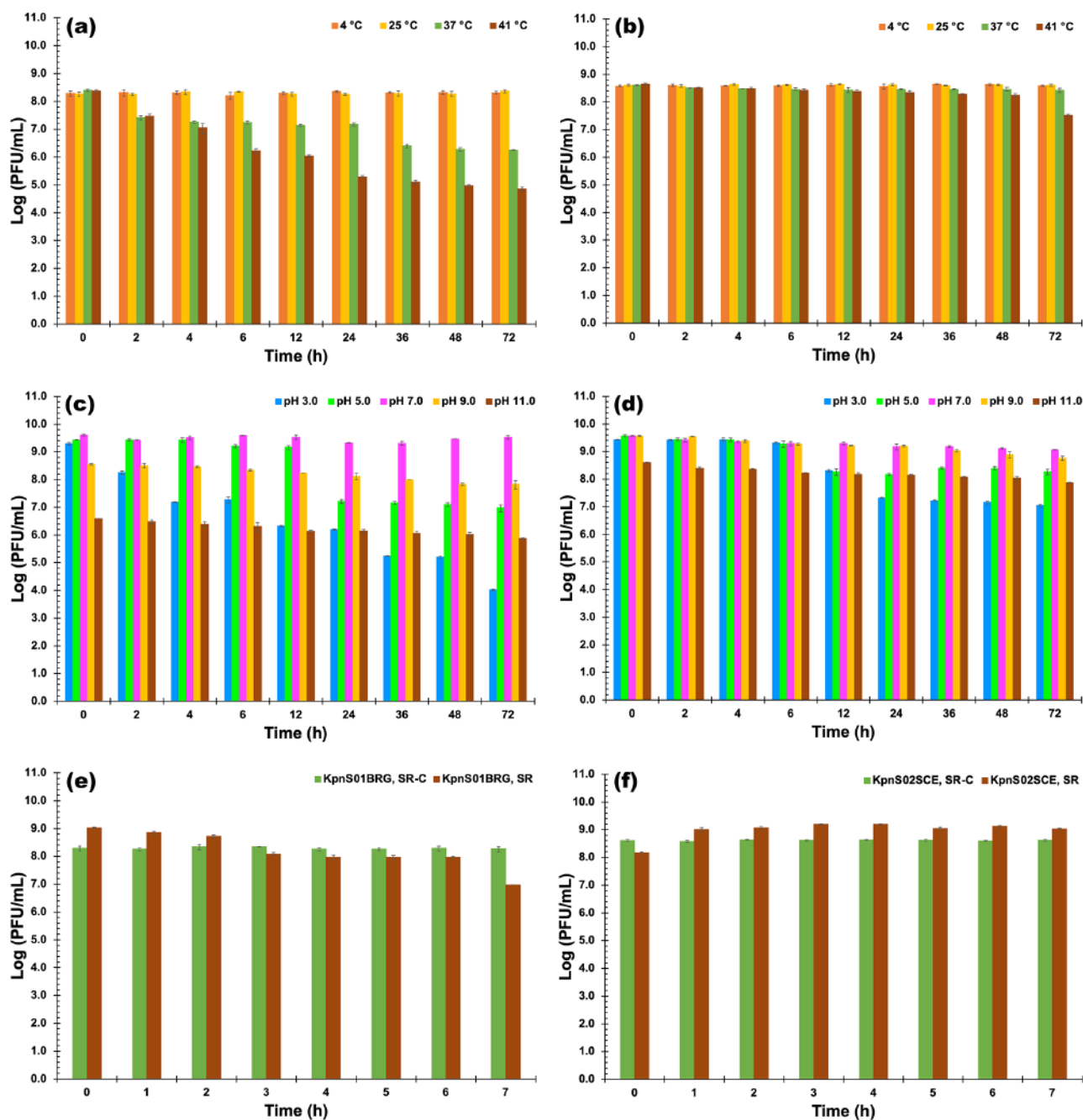


Figure 7 • Survival of phages KpnSo1BRG and KpnSo2SCE following exposure to different temperature values (KpnSo1BRG (a); KpnSo2SCE (b)), different pH values (KpnSo1BRG (c); KpnSo2SCE (d)), and solar radiation (KpnSo1BRG (e); KpnSo2SCE (f)). Values represent the mean of three independent experiments; error bars represent the standard deviation. SR: phage exposed to solar radiation; SR-C: phage control—phage not exposed to solar radiation.

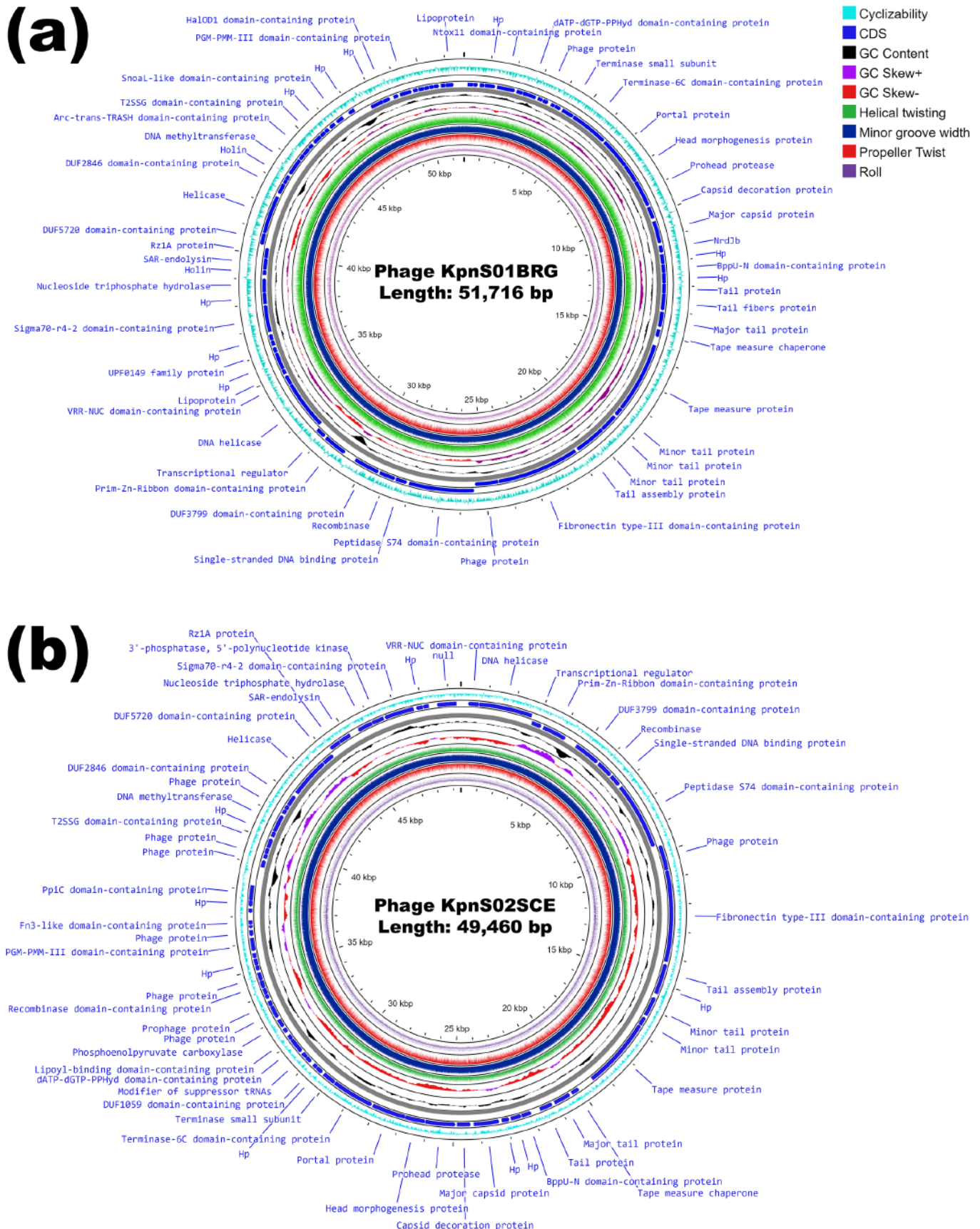


Figure 8 • Annotated genome maps of phages KpnS01BRG (a) and KpnS02SCE (b), illustrating GC skew, G+C content, and predicted coding sequences (CDSs). The outer ring features colored (dark blue) arrows representing the annotated CDSs, as detailed in Table S2. The arrows indicate the direction of transcription (strand + or -). The mechanical features of the genomes of both phages are also displayed (cyclizability, helical twist, minor groove width, propeller twist, roll). Genome maps were prepared using Proksee [104].

Table 3 • Genomic features of phages KpnSo1BRG and KpnSo2SCE genomes.

Feature		Phage KpnSo1BRG	Phage KpnSo2SCE
NCBI/GenBank accession number		PV081216	PV081217
Genome size		51,716 bp	49,460 bp
Number of PE reads mapping in the final assembly		38,600,019 (97.84%)	19,389,318 (36.54%)
Average sequencing coverage (calculated as (length of reads × number of reads mapping)/genome size)		224,523.1×	117,606.1×
GC content		51.69% (coding density: 92.66%)	51.08% (coding density: 93.28%)
tRNA genes		0	0
Protein-coding genes (CDS) predicted		87	85
• With function assigned		37 (42.5%)	36 (42.4%)
• Hypothetical/unknown function		50 (57.5%)	49 (57.6%)
Annotated terminase	terminase large subunit	Putative function: head and packaging	Putative function: head and packaging
	terminase small subunit	Putative function: head and packaging	Putative function: head and packaging
Similar phage genome sequences NCBI/RefSeq accession number of similar phage		Klebsiella phage PWKp17 MZ634348	Klebsiella phage pK8 OL702938
Average nucleotide identity (ANI) when compared to the most similar phage sequence		97.7%	96.8%
Morphotype		<i>Siphovirus</i>	<i>Siphovirus</i>
Family/Genus		<i>Drexlerviridae/Webevirus</i>	<i>Drexlerviridae/Webevirus</i>

NCBI: National Center for Biotechnology Information; PE reads: paired end reads; GC: guanine-cytosine; CDS: coding DNA sequence; ANI: average nucleotide identity.

Utilizing computational resources from the SuperComputer System at the Institute for Chemical Research of Kyoto University (Kyoto, Japan), the web-based ViPTree program [78] was employed to analyze similarities and relationships between phages KpnSo1BRG and KpnSo2SCE and other prokaryotic ds-DNA viruses, following established methodologies [18, 43, 105]. **Figure 10** presents the viral proteomic trees generated by ViPTree, illustrating the relationships between phages KpnSo1BRG and KpnSo2SCE and related phages.

Results showed that phage KpnSo1BRG was grouped into one small group with nine other phages (**Figure 10a,b**), whereas phage KpnSo2SCE was grouped into one small group with 14 other phages (**Figure 10c,d**), all belonging to the *Drexlerviridae* family according to the updated International Committee on Taxonomy of Viruses (ICTV) taxonomic classification. Here we can see that phage KpnSo1BRG and phages KOX1 (NC_047825) and vB_KpnS_IMGroot (NC_049834) (the phages closest to the KpnSo1BRG phage) occupy distinct branches, indicating high phylogenetic similarity (**Figure 10b**). Based on the analyses conducted using the ViPTree program and tBLASTx genomic sequence comparisons of phages KpnSo1BRG and KpnSo2SCE with other phage genomes available in the Virus-Host DB [106] (**Figure 11**), the phages closest to the KpnSo1BRG phage are phages KOX1 (NC_047825) and vB_KpnS_IMGroot (NC_049834), with 95.0% and 94.8% similarity, respectively, whereas the phages closest to phage KpnSo2SCE are phages NJS1 (NC_048024) and vB_KpnS_FZ10 (NC_049840), with 93.1% and 93.0% similarity, respectively.

3.10. DNA structural features of phages KpnSo1BRG and KpnSo2SCE

Studying the mechanical properties of the genomes of these two siphoviruses was crucial for gaining a more comprehensive and detailed understanding of the variations in infection rates of their target bacterial cells, despite the nearly identical yields of virion morphogenesis observed within the infected cells (viz. 17.7 virions per host cell (phage KpnSo1BRG) versus 18.6 virions per host cell (phage KpnSo2SCE)).

Figure 12 displays the results from DNA shape (propeller twist (ProT), minor groove width (MGW), roll, and helical twist (HelT)) calculations for the assembled genomes of the two phages.

The interaction between proteins and DNA is fundamental to numerous biological regulatory processes. These interactions depend on the inherent mechanical properties of DNA, which enable deformation, bending, and twisting [107]. The shape of DNA, including its physical and geometrical characteristics such as minor groove width (MGW), helical twist (HelT), propeller twist (ProT), and roll angle, plays a critical role in this process [108]. HelT refers to the angular rotation needed to transition from one base pair to the next [109, 110]. The minor groove of DNA is particularly significant for recognition by various proteins [111, 112]. ProT involves a rotation along the long axis, resulting in non-coplanar alignment of the two bases within a pair [113], while the roll angle is a local parameter that indicates bending into the DNA grooves [114]. These structural features contribute to DNA bendability and provide valuable insights into the dynamics

of protein–DNA interactions. Advances in computational methods and molecular simulations have enhanced our comprehension of how DNA shape depends on its sequence. Consequently, mechanical properties of DNA can now be predicted from its sequence using high-throughput methods, uncovering inherent conformational preferences [115, 116]. In this study, the DNA shape characteristics of the KpnSo1BRG and KpnSo2SCE phage genomes were predicted using DNAShape, a high-throughput, web-based tool. The results were visualized as heatmaps for clarity and analysis (**Figure 12**). A close examination of the heatmaps displayed in **Figure 12** allowed us to identify similarities and differences between the two phage genomes.

ProT (propeller twist) measures the angle between the planes of two DNA bases. A positive angle occurs when the nearer base rotates clockwise, while a counterclockwise rotation results in a negative angle. This parameter reflects DNA flexibility, with higher ProT values indicating a more rigid DNA structure [113]. The presence of propeller twists in base pairs makes their stacking into dinucleotide steps more challenging compared to in-plane base pairs. This effect introduces a chemical locking mechanism, enhancing the rigidity of dinucleotide steps with high ProT angles. Consequently, smaller ProT values allow base pairs to

slide more easily over one another within the structural limits of the DNA backbone [113]. In the ProT heatmaps (**Figure 12**), regions with higher and lower ProT values are evident within the phage genome sequences. These regions differ between the KpnSo1BRG and KpnSo2SCE phage genomes, with higher values depicted in yellow and lower values in dark red (**Figure 12**, ProT: a, b). Notably, the KpnSo1BRG phage genome exhibits higher ProT angles (**Figure 12a**, ProT), with a mean of -6.92° , compared to the KpnSo2SCE phage genome, which has a mean of -6.96° (**Figure 12b**, ProT). This suggests that, on average, the KpnSo2SCE genome contains more flexible regions than the KpnSo1BRG genome. However, a Mann–Whitney U test, applied to the non-normally distributed data, showed no statistically significant difference in ProT between the two genomes, with a *p*-value of 0.12707. Thus, while the average ProT values are similar across both phage genomes, local variations may occur due to differences in the distribution of ProT within each genome. In both heatmaps, a yellow pattern signifies a higher ProT angle, while a red color indicates a lower ProT angle (**Figure 12a,b**: ProT). Despite the mean values, the distribution of yellow and red patterns varies across both genomes, highlighting differences between them.

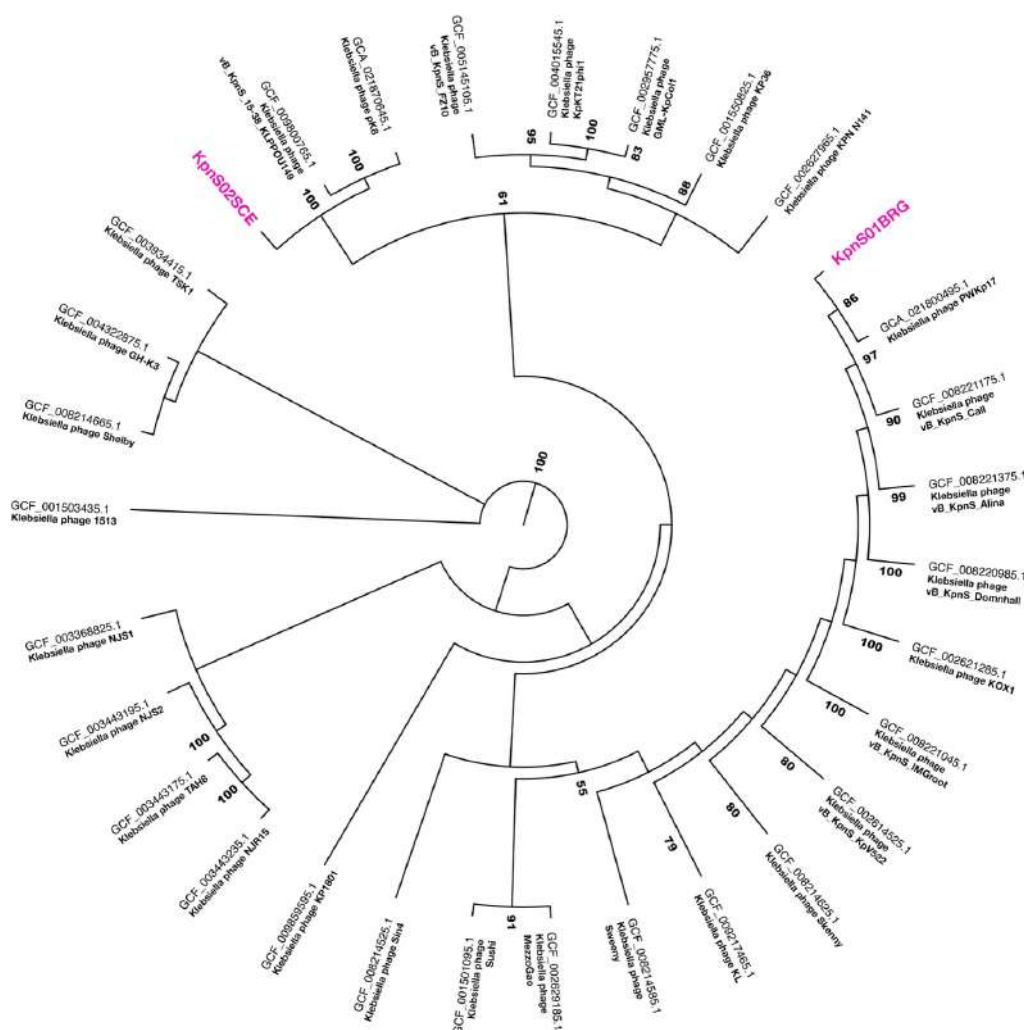


Figure 9 • Phylogenetic tree constructed using soft-core protein clusters from 29 phages, including phages KpnSo1BRG and KpnSo2SCE (highlighted in magenta). Data were sourced from GenBank (GCA) and RefSeq (GCF) genome assemblies. The sequences of these protein clusters were utilized for a Maximum Likelihood (ML) phylogenetic analysis, performed with 999 bootstrap replicates. Only bootstrap values above 50 are represented in the phylogenetic tree.

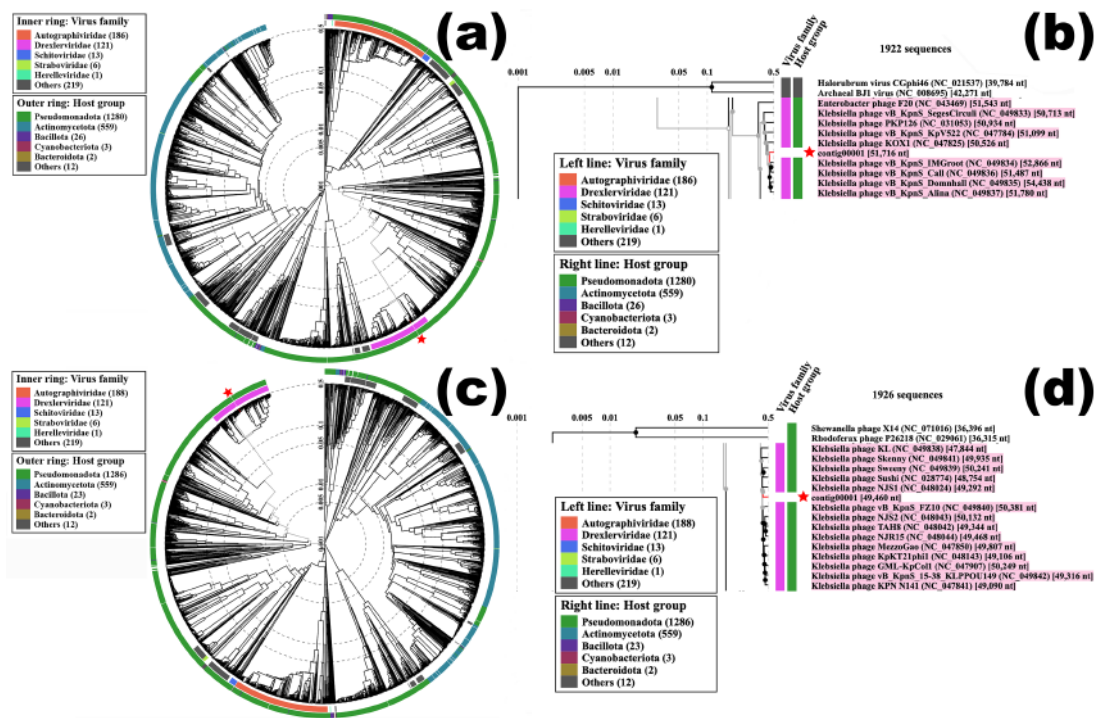


Figure 10 • Viral proteomic trees generated from ViPTree analyses of phages KpnSo1BRG (a,b) and KpnSo2SCE (c,d) alongside related phages. (a,c) Phylogenetic trees were constructed using 5633 phage genomes as reference sequences through ViPTree. In this figure, phages are labeled according to their official ICTV classification, with the inner and outer rings representing their virus family and host group, respectively. (b,d) Expanded views of the tree regions highlighting the most closely related phages. The branches containing phage KpnSo1BRG (b) and phage KpnSo2SCE (d) are highlighted in red. A red star marks the positions of phages KpnSo1BRG (a,b) and KpnSo2SCE (c,d). ICTV: International Committee on Taxonomy of Viruses.

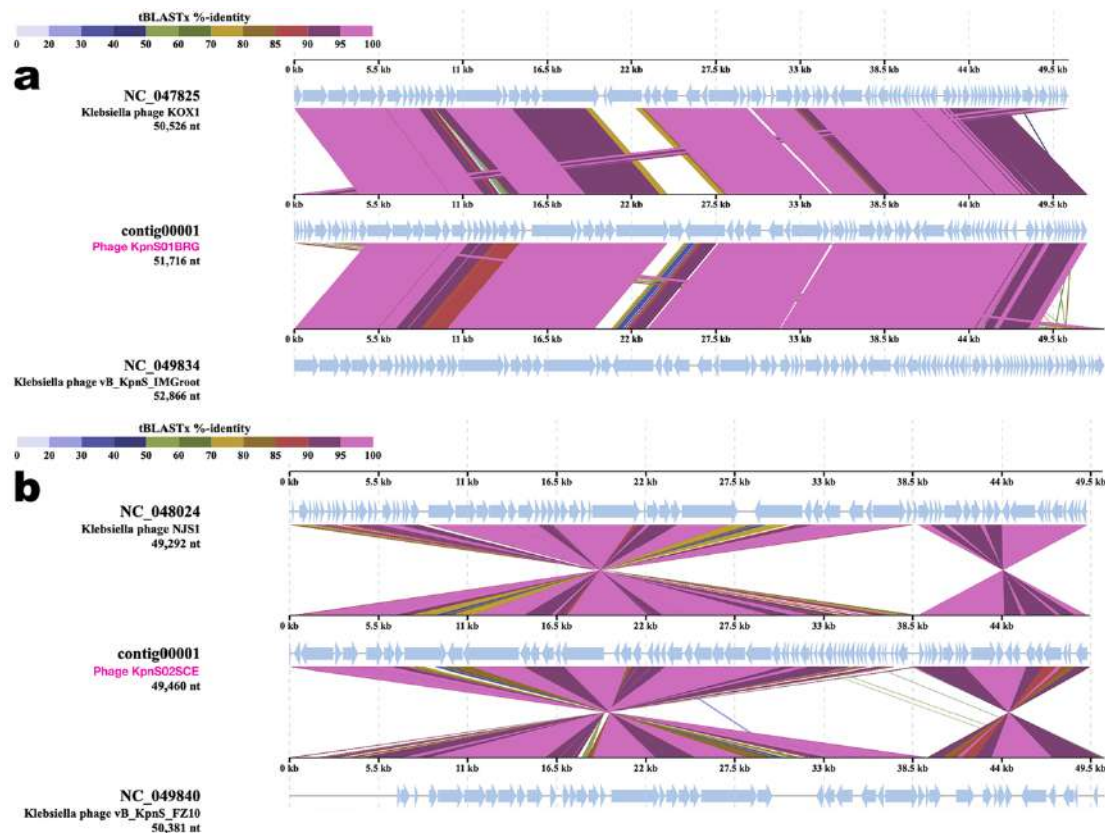


Figure 11 • Genome sequence alignment and comparison of phages KpnSo1BRG (a) and KpnSo2SCE (b) with other *Drexlerviridae* phage genomes, revealing co-linearity identified through tBLASTx analysis using the ViPTree server [78]. Homologous regions detected by tBLASTx are linked by colored segments, with the color intensity representing amino acid identity. The upper color bar indicates the percentage identity from the tBLASTx results.

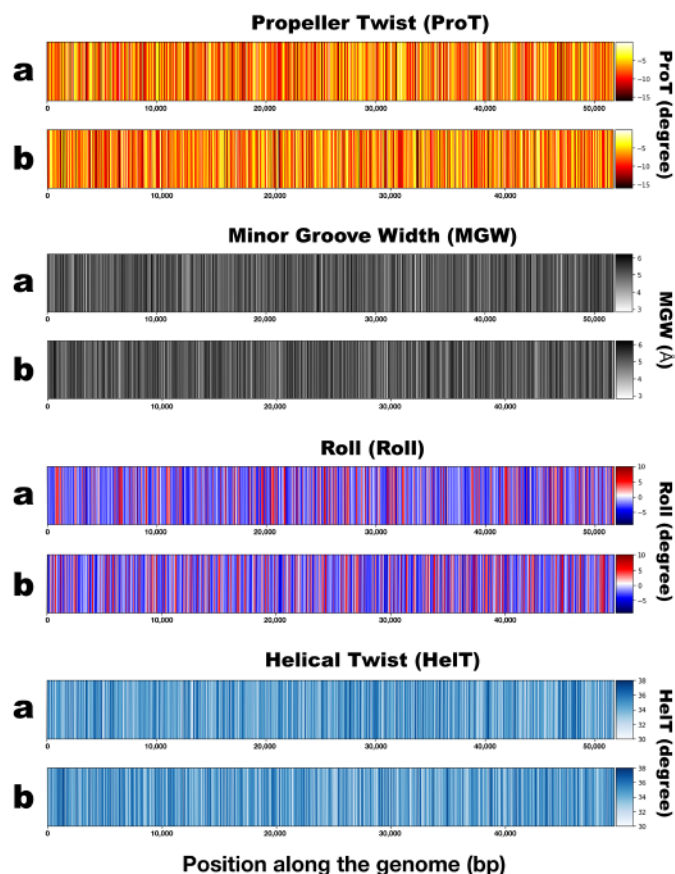


Figure 12 • Heatmap patterns from the results of DNA shape (propeller twist, minor groove width, roll, and helical twist) calculations for the assembled genomes of phage KpnSo1BRG (a) and phage KpnSo2SCE (b).

The fundamental characteristic of DNA is its helicity, a concept pioneered by Watson and Crick [117], with significant experimental contributions from Rosalind Franklin and Maurice Wilkins, also in 1953 [118]. This helical structure is formed by the stacking of individual bases on top of one another, with each base twisting slightly to optimize hydrophobic interactions, ultimately giving rise to the helix [119]. The flexibility of DNA twist is essential for its diverse biological functions. Earlier studies have explored this flexibility using short-base sequences, which are particularly useful for analyzing looping capabilities [115, 120], and have also investigated the bending and twisting deformations required for protein–DNA interactions [121–123]. Together, these results indicate that HelT is a reliable measure of DNA flexibility. Analysis of the HelT heatmaps for both phage genomes showed that the KpnSo2SCE genome (Figure 12b, HelT) exhibits a higher HelT degree, with a mean value of 34.432° , compared to the KpnSo1BRG genome (Figure 12a, HelT), which has a mean value of 34.426° . The HelT heatmaps illustrate the distribution of HelT angles across both genomes, with blue indicating higher angles and white representing lower angles (Figure 12a,b: HelT). The patterns are distinct along the x-axis, in both genomes. For example, a strong blue color pattern at the beginning of the KpnSo2SCE phage genome can be seen, which is absent in the KpnSo1BRG phage genome, suggesting a region with higher HelT angles in the former genome. A Mann–Whitney U test for assessing statistical significance of HelT in both phage genomes yielded a non-significant result, with a p -value of 0.49174, suggesting a no global difference in HelT degree between the genomes. Nonetheless,

variations may exist in the positioning of higher and lower HelT along both genomes.

The roll angle acts as a local angular parameter that describes the bending and twisting between adjacent base pairs [124, 125]. Notably, significant positive roll values in base pair steps suggest weaker stacking interactions [115]. Tolstorukov and colleagues [114] proposed a link between roll and conformational flexibility within the nucleosome. Using molecular dynamic simulations, those researchers demonstrated that setting the roll value to zero at all base pair steps nearly straightens the DNA molecule, highlighting how variations in roll contribute to DNA bending. They also observed that negative roll values cause DNA to bend toward the minor groove, while positive roll values result in bending toward the major groove. These findings emphasize the critical role of roll in determining DNA bendability. The roll angle values for both phage genomes are represented in the heatmaps, with red indicating higher degrees and blue indicating lower degrees of roll (Figure 12a,b: Roll). Notably, differences in the distribution of roll angles are evident between the two phage genomes. The KpnSo1BRG phage genome has an average roll value of -0.691° , while the KpnSo2SCE phage genome shows a slightly more negative mean value of -0.712° . Both genomes exhibit negative roll angles, with a minor variation between them. The more negative value observed in the KpnSo2SCE genome suggests a potential tendency for greater bending toward the minor groove in specific regions, though further studies are required to confirm this hypothesis. A Mann–Whitney U test indicated no statistically significant global difference in the roll angle between the two genomes, with a p -value of 1.0000.

The minor groove width (MGW) varies geometrically depending on the DNA sequence and plays a significant role in how proteins recognize and interact with DNA [126]. A narrower minor groove enhances the accessibility of base edges to proteins, including transcription factors, facilitating their interaction with DNA [115]. This could result from the compression of the groove width, caused by the unusually close spacing between the recognition helices of each helix-turn-helix motif [127]. Conversely, a study by Ogney et al. [128] revealed a connection between flexibility and minor groove width (MGW), suggesting that this relationship is an inherent property of DNA. The MGW can also be visualized in the corresponding heatmaps (Figure 12), where black represents higher MGW values and white indicates lower MGW values (Figure 12a,b: MGW). The distribution of black and white patterns varies across the two phage genome sequences, emphasizing the differing importance of MGW in specific regions of each genome at the local level. The average MGW value for the KpnSo1BRG phage genome is 5.055° , while the KpnSo2SCE genome has a slightly lower mean MGW value of 5.048° . The KpnSo2SCE phage genome displays a narrower and more concentrated MGW distribution compared to the KpnSo1BRG phage genome. The increased narrowness of the minor groove width (MGW) in the KpnSo2SCE genome may imply enhanced protein–DNA interactions and greater bending in specific genomic regions, as proteins such as transcription factors often induce DNA bending. A Mann–Whitney U test unveiled a statistically significant difference in MGW angles between the two phage genomes, with a p -value of 0.02659, highlighting a distinct variation in MGW. This finding suggests a difference in MGW at both the nucleotide and global levels between the KpnSo1BRG and KpnSo2SCE genomes, with the KpnSo2SCE genome exhibiting a narrower MGW.

Following the analysis of the DNA shape of the KpnSo1BRG and KpnSo2SCE phage genomes, a Pearson correlation analysis was performed to examine the linear relationships among the DNA mechanical features of each genome. The results were visualized in two graphs, illustrating the Pearson correlation values for each structural feature (**Figure 13a,b**). The analysis revealed both positive and negative correlations between the features. Notably, the Pearson correlation patterns between the genomes showed similarities. A significant positive correlation was observed between the roll angle and the minor groove width (MGW), with values of 0.51 for the KpnSo1BRG phage genome and 0.50 for the KpnSo2SCE phage genome. This finding aligns with earlier research by Fratini et al. [119], which showed that the roll angle quantifies the extent to which successive base pairs open toward the minor groove. As a result, higher positive roll angles are likely to correspond to wider minor groove widths. On the other hand, a negative correlation was observed between the helical twist (HelT) and the minor groove width (MGW), with values of -0.23° for both the KpnSo1BRG and KpnSo2SCE phage genomes. Liebl et al. [129] demonstrated that DNA overwinding reduces the MGW, increasing the HelT due to greater stacking distances between adjacent base pairs. This supports the likelihood of a negative correlation between HelT and MGW. Additionally, we identified both positive and negative correlations among DNA structural features, such as the negative correlation between roll and HelT, which measured -0.058 for KpnSo1BRG and -0.054 for KpnSo2SCE (**Figure 13**). While the Pearson correlation in this case is close to zero, suggesting a weak association between these features, earlier studies have noted an anti-correlation, likely attributable to the presence of specific pyrimidine/purine steps in the genome [130–132]. Moreover, a negative correlation between ProT and HelT was observed in both phage genomes (**Figure 13**). Propeller twisting serves to optimize stacking interactions. When ProT values approach 36 degrees, they tend toward zero, reflecting optimal stacking. Deviations from this range can lead to reduced stacking efficiency. El Hassan and Calladine [113] demonstrated that more negative ProT angles in double-stranded DNA (ds-DNA) are associated with increased flexibility, while ProT angles between -1 and -3 degrees correspond to greater rigidity in dsDNA. This observation is supported by Basu et al. [81], who also

found that higher ProT values correlate with lower cyclizability, indicating reduced DNA bendability.

From these findings, we can conclude that both the KpnSo1BRG and KpnSo2SCE phage genomes exhibit correlations among DNA structural features, which may influence the mechanical properties of their DNA. These DNA shape characteristics could potentially impact phage biology. For instance, a narrower minor groove width (MGW), distinct helical twist (HelT) conformations, changes in roll, and variations in propeller twist (ProT) might collectively affect biological processes such as DNA translocation. This could occur by enhancing DNA–protein interactions, improving accessibility, and contributing to the stability of the DNA structure during translocation.

While our results suggest differences between the KpnSo2SCE and KpnSo1BRG phage genomes, it is important to emphasize that these analyses were primarily focused on local-level effects rather than global-level implications. Thus, further studies are needed to validate these observations and explore their broader significance. **Figure 13** illustrates the correlations among the four predicted structural features of the KpnSo1BRG and KpnSo2SCE phage genomes.

Figure 14 presents the results of the cross-distant dinucleotide correlation frequency calculations conducted for the two phage genomes, aiming at checking the presence of peak patterns meaning periodicity. The *X* axes in the plots in **Figure 14** represent the number of incremental steps where one can observe such periodicity.

A custom Python script was developed to analyze the frequency of all 16 possible dinucleotides (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT) in the genomes of phages KpnSo1BRG and KpnSo2SCE. The script counted the occurrences of each dinucleotide in both genomes and normalized the counts by dividing them by the respective genome lengths. The resulting dinucleotide frequencies are illustrated in **Figure 15**, which provides a comparative visualization of the dinucleotide distribution in the genomes of phages KpnSo1BRG and KpnSo2SCE.

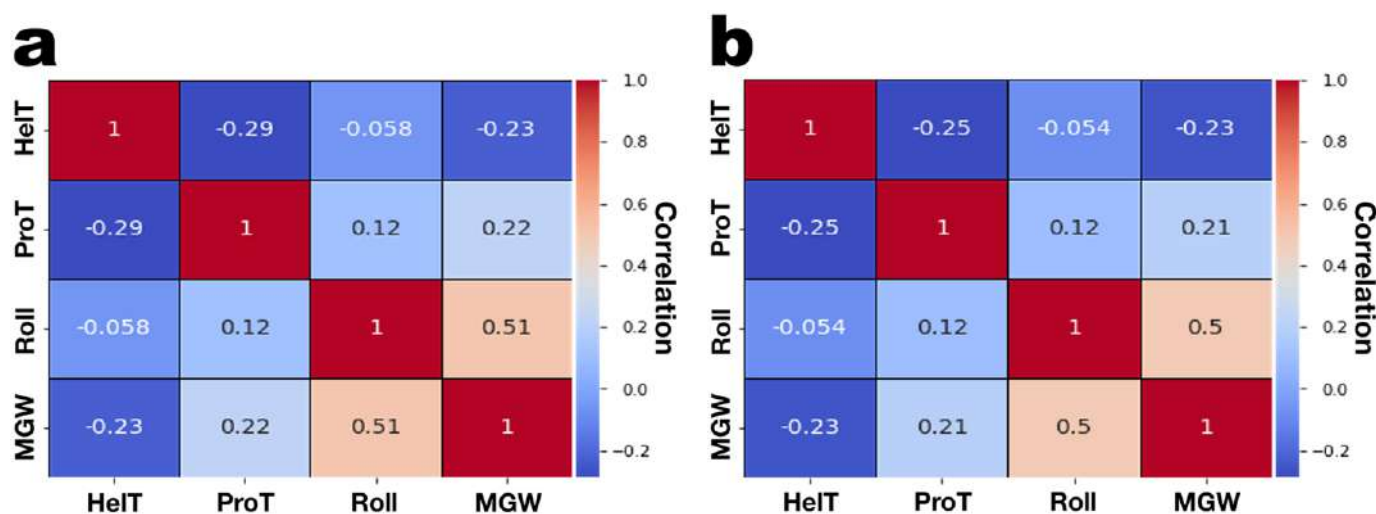


Figure 13 • Correlation between the four predicted structural features of the genomes of phage KpnSo1BRG (a) and phage KpnSo2SCE (b). Dark red represents a high correlation whereas dark blue represents a low correlation.

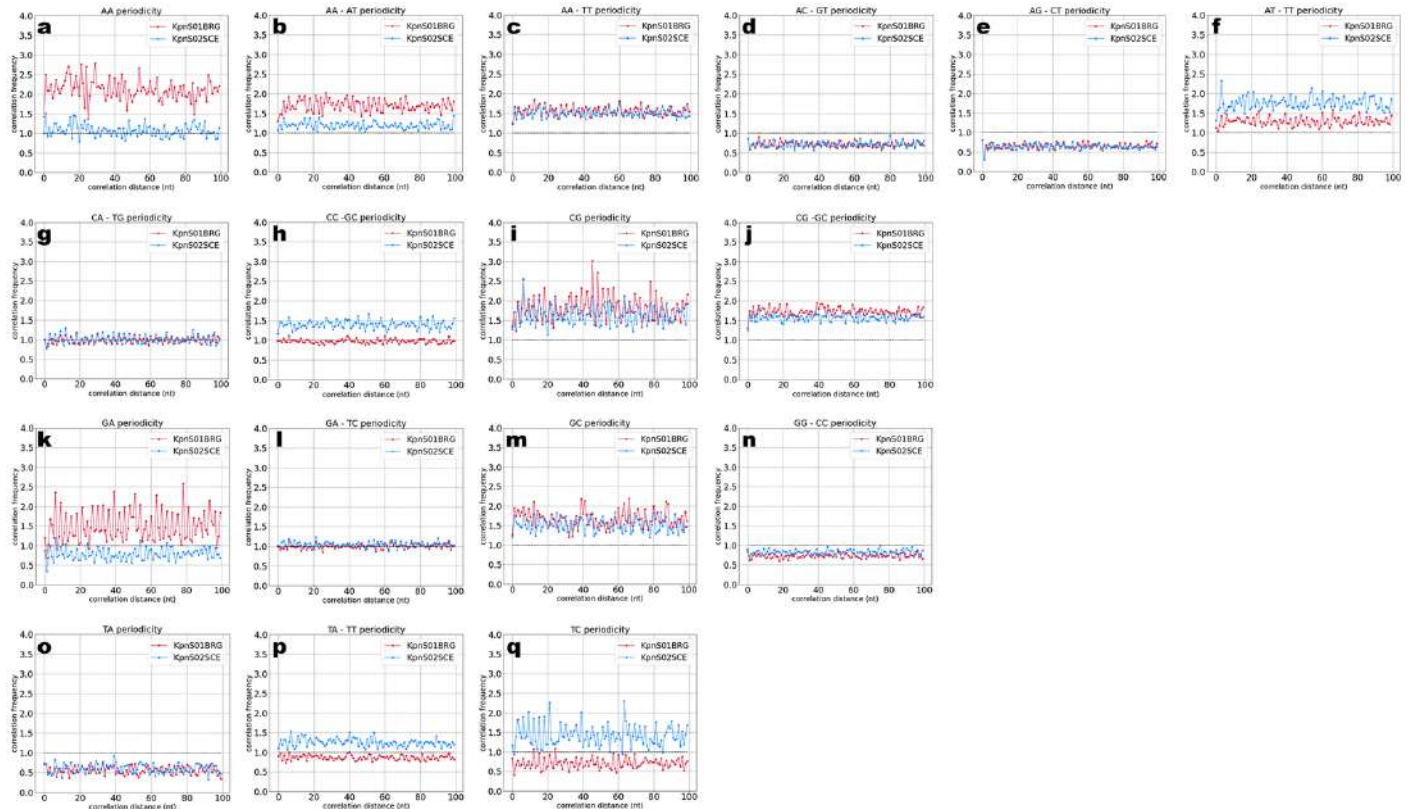


Figure 14 • Cross-distant dinucleotide correlation frequency patterns in the assembled genomes of phages KpnSo1BRG and KpnSo2SCE. (a) AA periodicity; (b) AA-AT periodicity; (c) AA-TT periodicity; (d) AC-GT periodicity; (e) AG-CT periodicity; (f) AT-TT periodicity; (g) CA-TG periodicity; (h) CC-GC periodicity; (i) CG periodicity; (j) CG-GC periodicity; (k) GA periodicity; (l) GA-TC periodicity; (m) GC periodicity; (n) GG-CC periodicity; (o) TA periodicity; (p) TA-TT periodicity; (q) TC periodicity.

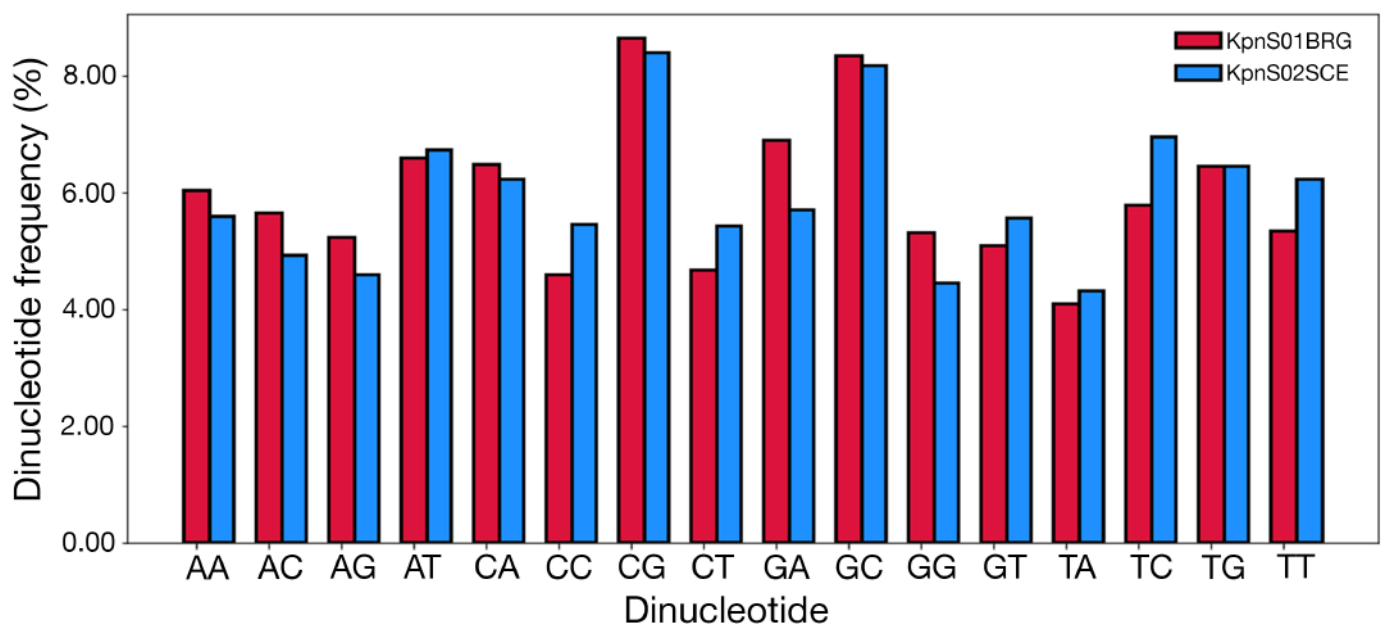


Figure 15 • Dinucleotide frequency in the genomes of phages KpnSo1BRG and KpnSo2SCE.

The occurrence frequencies of all 16 possible dinucleotides (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT) in the genomes of both phages were calculated and visualized in a bar graph (Figure 15). The graph illustrates the percentage representation of each dinucleotide within each genome. Notably, specific dinucleotides showed significantly higher frequencies in

each phage genome, highlighting distinct compositional biases between the two. For the KpnSo1BRG phage genome, dinucleotides AA, AC, AG, CA, CG, GA, GC, and GG demonstrate a higher frequency compared to the KpnSo2SCE phage genome, with percentages of 6.05, 5.66, 5.22, 6.48, 8.64, 6.90, 8.33, and 5.32%, respectively. Conversely, the dinucleotides AT, CC, CT, GT, TA, TC, and TT exhibit a higher frequency in the KpnSo2SCE phage genome

compared to the KpnSo1BRG phage genome, with percentages of 6.72, 5.45, 5.42, 5.56, 4.32, 6.94, and 6.22%, respectively. The CG dinucleotide demonstrates the highest frequency of occurrence in both genomes, surpassing all other dinucleotides, reaching 8.65% for KpnSo1BRG and 8.41% for KpnSo2SCE, followed closely by the GC dinucleotide with the second-highest occurrence frequency at 8.33% for KpnSo1BRG and 8.17% for KpnSo2SCE (**Figure 15**). In contrast, the TA dinucleotide shows the lowest frequency of occurrence in both phage genomes, accounting for 4.08% in KpnSo1BRG and 4.32% in KpnSo2SCE, making it the least common dinucleotide among the 16 possibilities. Initially, the frequencies of all 16 possible dinucleotide combinations were independently calculated for the genomes of phages KpnSo1BRG and KpnSo2SCE (**Figure 15**). To further explore genomic differences, the variations in dinucleotide frequencies between the two phage genomes were analyzed. This involved counting the occurrences of each dinucleotide in each genome, normalizing the counts by dividing them by the respective genome lengths, and then computing the differences in dinucleotide frequencies between the two phages. These differences were visualized as a heatmap (**Figure 16**), where dark red indicates a high differential frequency and dark blue represents a low differential frequency. **Figure 16** illustrates the heatmap of the differential dinucleotide frequencies between the genomes of phages KpnSo1BRG and KpnSo2SCE. The results, comprising 256 frequency values, are visualized in the heatmap of **Figure 16**. In this heatmap, red indicates a high difference, meaning that specific dinucleotides in one phage genome are more represented compared to other dinucleotides from the other phage genome. Conversely, the blue color denotes a low difference, suggesting that these specific dinucleotides are less represented compared to other dinucleotides from the other phage genome (**Figure 16**). We can observe that dinucleotides CG, GA, and GC in the KpnSo1BRG phage genome exhibit a higher differential frequency compared to the majority of dinucleotides in the KpnSo2SCE phage genome. Conversely, CG and GC in the KpnSo2SCE phage genome display a higher differential frequency than most dinucleotides in the KpnSo1BRG phage genome (**Figure 16**).

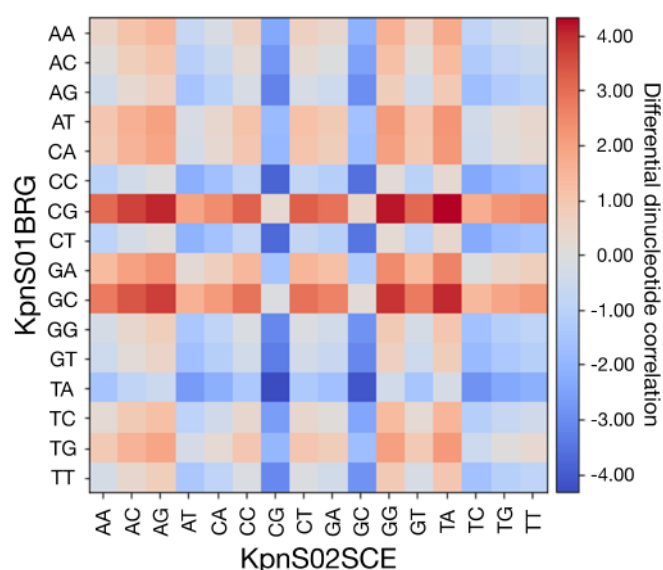


Figure 16 • Heatmap illustrating the differential dinucleotide frequencies between the genomes of phages KpnSo1BRG and KpnSo2SCE. Dark red indicates a high correlation, whereas dark blue indicates a low correlation.

Figure 17 presents the predicted intrinsic cyclizability profiles for the genomes of phages KpnSo1BRG and KpnSo2SCE, displayed at a resolution of seven base pairs. In our study, we have explored the inherent cyclizability of the KpnSo1BRG and KpnSo2SCE phage genomes at a seven base pair resolution, according to the methodology outlined by Basu et al. [81]. Their methodology involved the concurrent assessment of intrinsic cyclizability across a diverse range of 90,000 unique 50 base pair DNA sequences.

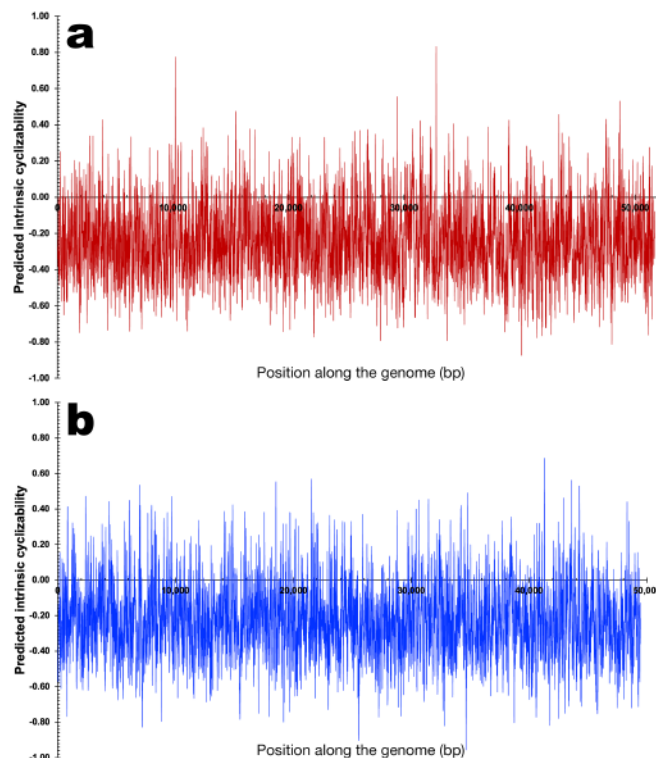


Figure 17 • Predicted intrinsic cyclizability profiles of the phage genomes, analyzed at a seven base pair resolution, for (a) phage KpnSo1BRG and (b) phage KpnSo2SCE.

The cyclizability patterns differ significantly between the two phage genomes (**Figure 17**), with distinct peaks observed in each. In the KpnSo1BRG phage genome, certain peaks reach a maximum cyclizability value of 0.832, while the KpnSo2SCE phage genome shows peaks with a lower maximum value of 0.685. This indicates that specific regions of the KpnSo1BRG genome may exhibit greater cyclizability or flexibility compared to those in the KpnSo2SCE genome. Conversely, the lowest cyclizability peak in the KpnSo1BRG genome is -0.866 , whereas in the KpnSo2SCE genome, it is -0.956 , suggesting regions of reduced cyclizability or flexibility in the latter genome. Despite the higher cyclizability peaks in the KpnSo1BRG genome, its mean cyclizability value is -0.238 , slightly lower than the -0.226 observed for the KpnSo2SCE genome. This implies that, on average, the KpnSo1BRG genome has marginally lower cyclizability compared to the KpnSo2SCE genome. A Mann–Whitney U test confirmed a statistically significant difference between the two genomes, with a p -value of 0.00013799, highlighting distinct cyclizability profiles. Overall, these results might suggest that the KpnSo1BRG phage genome is somewhat more flexible than the KpnSo2SCE phage genome.

Figure 18 presents heatmaps illustrating the predicted intrinsic cyclizability patterns across the genomes of phages KpnSo1BRG and KpnSo2SCE.

Figure 19 displays a box plot (**Figure 19a**) with the statistics of the cyclizabilities of both phage genomes. This figure also provides insight into the distribution of cyclizability values, including their magnitudes and frequency of occurrence, across both phage genomes (**Figure 19b,c**).

A detailed analysis of the data in **Figure 19a** reveals relatively large standard deviations in the mean predicted cyclizabilities. To determine whether the average predicted intrinsic cyclizabilities of the genomes of phages KpnSo1BRG and KpnSo2SCE were statistically distinct, a one-way ANOVA was conducted on the complete dataset of predicted intrinsic cyclizabilities. This test evaluated the null hypothesis that the average predicted intrinsic cyclizabilities of the two phage genomes were similar. The results of this analysis are presented in **Table 4**. Additionally,

Figure 19b,c illustrates the distribution of cyclizability values, including their magnitudes and frequency of occurrence, across both phage genomes. As we can see, the distribution of cyclizability from the KpnSo1BRG genome (**Figure 19b**) goes from ca. -0.85 to 0.80 , and for the KpnSo2SCE genome (**Figure 19c**), it goes from ca. -0.95 to 0.65 . Based on the observed distribution, it can be inferred that the KpnSo1BRG phage genome exhibits a greater number of high cyclizability peaks (albeit limited in quantity) compared to the KpnSo2SCE phage genome. Conversely, the KpnSo2SCE genome displays a higher frequency of negative cyclizability values relative to those found in the KpnSo1BRG genome.

The F -ratio helps determine whether the variance between the means of the predicted intrinsic cyclizability populations for the two phage genomes is significantly different, while the p -value represents the probability of obtaining average predicted intrinsic cyclizabilities as extreme as those observed, assuming the null hypothesis is true.

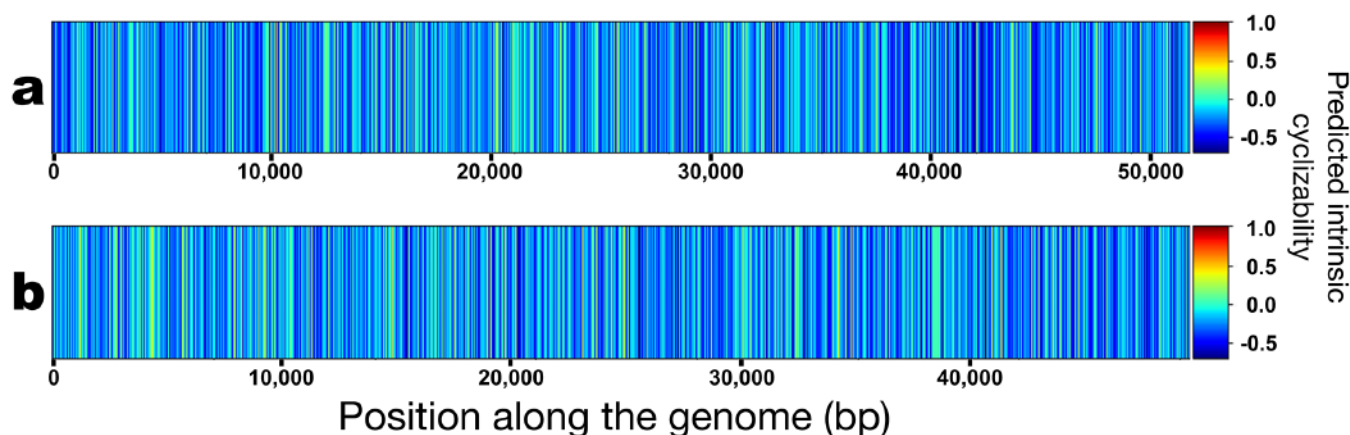


Figure 18 • Heatmap visualization of the cyclizability patterns across the genomes of phages KpnSo1BRG (a) and KpnSo2SCE (b). Dark red indicates a high cyclizability, whereas dark blue indicates a low cyclizability.

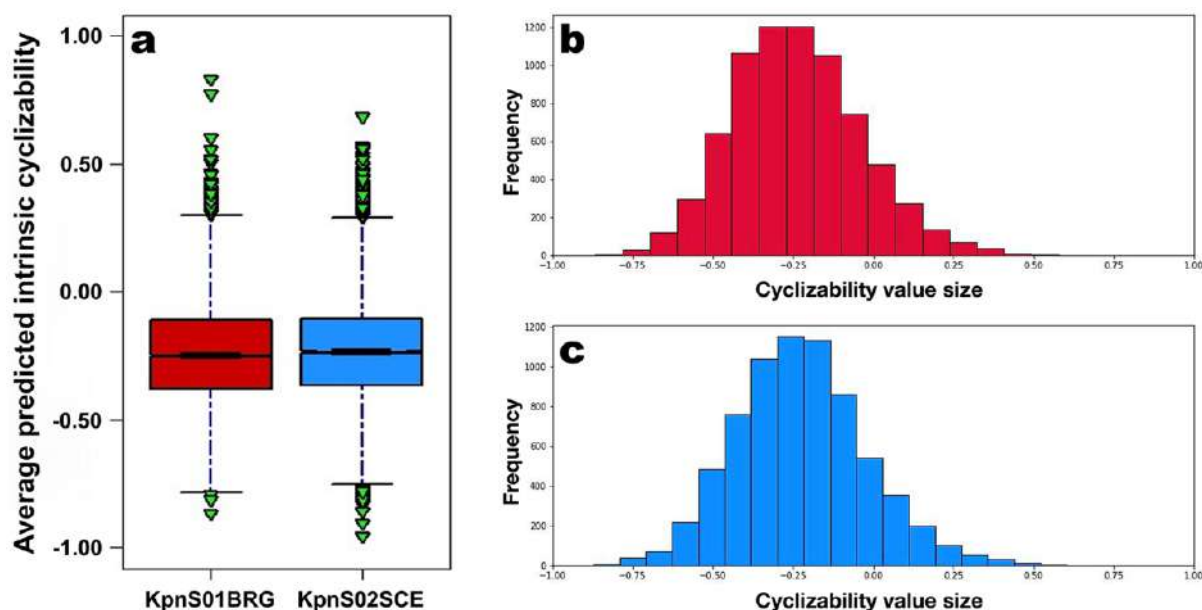


Figure 19 • Distribution of predicted intrinsic cyclizabilities for the genomes of phages KpnSo1BRG and KpnSo2SCE, including statistical summaries (a), alongside the distribution of cyclizability values and their frequency of occurrence in the genomes of phage KpnSo1BRG (b) and phage KpnSo2SCE (c). The black line within each box represents the mean range (a), and the whiskers denote the maximum and minimum cyclizability values. Values outside the majority of the distribution are indicated by green triangles (a).

Table 4 • Results from the one-way ANOVA statistical analysis conducted on the complete set of predicted intrinsic cyclizability data for the phage genomes KpnSo1BRG and KpnSo2SCE.

Source of variation	SS	df	MS	F-ratio	p-value	F-critical
Between groups	0.513	1.000	0.513	12.488	0.00041	3.842
Within groups	593.312	14,438	0.041			
Total	593.825	14,439				

SS: sum of squares; df: degrees of freedom; MS: mean square (variance estimate, calculated as $MS = SS/df$); *F*-ratio: the ratio of MS between groups to MS within groups ($MS_{\text{between}}/MS_{\text{within}}$); *p*-value: the probability that the observed mean is greater than or equal to (or less than or equal to) the expected mean, assuming the null hypothesis is true; *F*-critical: the critical *F*-value (1; 14,438; 0.05).

A closer examination of the one-way ANOVA results in **Table 4** reveals that the *p*-value is less than $\alpha = 0.05$ (or 5%) and the *F*-ratio exceeds the *F*-critical value. This indicates a significant difference between the average intrinsic cyclizabilities of the two phage genomes. In other words, there is at least a 95% probability that a difference in cyclizability will be observed between the genomes of phages KpnSo1BRG and KpnSo2SCE. The critical *F*-value at a 95% confidence level is 3.842, which is much lower than the observed *F*-value of 12.488, confirming that the null hypothesis is false. Therefore, the intrinsic cyclizability data suggest that the differences between the average cyclizability values of the two phage genomes are statistically significant.

4. Discussion

In the research work described herein, deeper molecular studies of two novel isolated lytic bacteriophages for *K. pneumoniae* were conducted, encompassing genomic mechanical features. Phages KpnSo1BRG and KpnSo2SCE formed distinct (clear) lysis plaques on the host lawn, with phage KpnSo2SCE also producing a secondary halo, with diameters of ≈ 1.5 mm (phage KpnSo1BRG, **Figure 2a**) and ca. 2 mm (phage KpnSo2SCE, **Figure 2c**). Both phages were identified as belonging to the class *Caudoviricetes* with siphovirus-like morphotypes (**Figure 2b,d**).

Phage host range was evaluated via spot testing in a number of different bacterial strains, to determine if their virions were able to productively infect the target bacterial strains and produce progeny (hence forming clear plaques of lysis). Beyond the original host, newly discovered phages are expected to infect other bacterial cells that possess similar surface receptors [133].

Phage KpnSo1BRG was only able to inactivate its bacterial host (Table S1), whereas phage KpnSo2SCE was able to infect KpnSo1BRG's host beyond its own isolation host, in the spot tests performed. The lack of bacterial cell destruction in most tested strains could be due to the phage's ability to identify only certain surface receptors on the bacterial outer membrane or the bacteria's defense mechanisms, such as CRISPR, which help them to evade phage infection [134]. This suggests that the two isolated phages have a limited host range. To enhance their effectiveness against *K. pneumoniae*, future research should focus on identifying additional lytic phages to develop a more comprehensive phage cocktail capable of targeting a wider variety of strains.

The extremely narrow host range observed for phages KpnSo1BRG and KpnSo2SCE (1/66 strains) is a hallmark of the *Webervirus* genus. Members of this genus are characterized as "specialist" phages that exhibit high specificity for particular *K. pneumoniae* capsular (K) types, driven by the specialized

depolymerases they encode. *Klebsiella* has over 77 serotypes; if the phage depolymerase does not match the specific K-type, infection cannot occur. This is primarily because *Webervirus* phages rely on highly specific tail fiber-associated depolymerases to penetrate the diverse capsular polysaccharide layer of *Klebsiella*. Given the vast diversity of *Klebsiella* capsules, such high specificity is typical for the genus and serves as a natural bottleneck for phage monotherapy. While this specificity minimizes collateral damage to the host's commensal microbiota (a significant advantage over broad-spectrum antibiotics) it simultaneously underscores that these phages are best suited for inclusion in diverse phage cocktails or as part of a personalized "phage-hunting" therapeutic approach to address the high variability of clinical *K. pneumoniae* infections.

Infection of a susceptible bacterial cell by a phage virion starts with its anchoring onto suitable receptors on the bacterial cell surface [31–33, 41, 135–137] because, due to the unpredictable nature of Brownian motion, free-floating phage particles occasionally come into contact with bacterial cells. Additionally, changes in the structural arrangement of receptor proteins can influence these interactions [9, 31–33, 42, 138]. According to Bull and Gill [139], for phages to be used in the treatment of bacterial infections, 70% or more of the phage virions should be adsorbed onto the target host cells during the first 10 min, and the adsorption rate constant should lie in the range 10^{-8} – 10^{-9} ml/min. A close look to the data in **Figure 3** allows to conclude that approximately 75% of phage KpnSo1BRG virions and approximately 80% of phage KpnSo2SCE virions adsorbed to Kpn NCTC-13439 cells after 10 min, and 100% of the phage particles adsorbed after 35 min, which confirms their potential for the treatment and prevention of bacterial infections by *K. pneumoniae*. The adsorption rates of phages KpnSo1BRG virions ($\delta_{\text{KpnSo1BRG}} = 1.371 \times 10^{-9}$ CFU $^{-1}$ ml min $^{-1}$) and KpnSo2SCE virions ($\delta_{\text{KpnSo2SCE}} = 1.585 \times 10^{-9}$ CFU $^{-1}$ ml min $^{-1}$) (**Figure 3**) onto Kpn NCTC-13439 cells are of the same order of magnitude as the results published by Zurabov and Zhilenkov [140] for *K. pneumoniae* phages (phages vB_KpnS_FZ10 (4.3×10^{-9} ml/min), vB_KpnS_FZ41 (3.1×10^{-9} ml/min), vB_KpnP_FZ12 (5.3×10^{-9} ml/min), and vB_KpnM_FZ14 (4.0×10^{-9} ml/min)), and as the results published by Balcão et al. [31] for phage Kpn31 adsorption onto *K. pneumoniae* CCCD-K001 cells (1.700×10^{-9} CFU $^{-1}$ ml min $^{-1}$), although smaller.

The (first order) detachment rate of the KpnSo2SCE phage virions from reversible phage–bacteria complexes ($\phi_{\text{KpnSo2SCE}} = 6.005 \times 10^{-11}$ min $^{-1}$) was significantly lower, though within the same magnitude, compared to that of phage KpnSo1BRG ($\phi_{\text{KpnSo1BRG}} = 1.200 \times 10^{-7}$ min $^{-1}$). This indicates that fewer KpnSo1BRG phage particles remained bound to their bacterial hosts. Since successful

infection relies on stable attachment to specific surface receptors, followed by viral replication and bacterial cell destruction [80], the reduced desorption rate of phage KpnSo2SCE virions aligns with its greater effectiveness in bacterial inactivation over longer treatment periods (**Figure 5a,b**). Adsorption dynamics revealed that approximately 50% of KpnSo1BRG virions adhered to bacterial cells within 5 min, reaching over 95% adsorption by 25 min (**Figure 3a**). In contrast, around 55% of phage KpnSo2SCE virions bound to host cells within 5 min, with adsorption exceeding 98% after 25 min (**Figure 3b**).

The growth parameters of KpnSo1BRG phage virions (**Figure 4a**) showed a burst size of 17.7 virions/host cell, whereas KpnSo2SCE phage virions showed a burst size of 18.6 virions/host cell (**Figure 4b**), allowing it to be concluded that the two phages replicate well in Kpn NCTC-13439 cells with latency periods of 60 min (phage KpnSo1BRG) and 5.8 min (phage KpnSo2SCE). The latent period of phage KpnSo1BRG correlates well with published data on *K. pneumoniae* phages [26, 140, 141], being of the same order of magnitude, whereas the latent period of phage KpnSo2SCE was one order of magnitude lower than those latent periods published for *K. pneumoniae* phages. The latent period of phage KpnSo2SCE (5.8 min) is exceptionally short compared to that of phage KpnSo1BRG (60 min) and other reported *Klebsiella* phages. While the high metabolic state of the *K. pneumoniae* host during exponential growth provided a robust supply of ribosomes and nucleotides for viral synthesis, the drastic disparity between the two phages suggests an intrinsic evolutionary optimization in KpnSo2SCE. Beyond potentially possessing high-efficiency replication and lysis machinery (e.g., rapid-acting holins), the mechanical analysis of the KpnSo2SCE phage genome revealed significantly higher mean cyclizability and flexibility. Increased DNA flexibility can reduce the energetic cost of genome packaging and facilitate faster access for transcription factors and polymerases by lowering the energy required for local DNA bending and looping and hence correlates with infection rates and virion production by aiding DNA–protein interactions and translocation. These mechanical features likely enable phage KpnSo2SCE to complete its intracellular virion assembly and morphogenesis phase at a significantly accelerated rate, maximizing its fitness in competitive environments. The burst size of phages KpnSo1BRG and KpnSo2SCE were similar to those of phages vB_KpnM_KP15 and vB_KpnM_KP27 (10–15 PFU/host cell) [26], but lower than those of phages vB_KpnS_FZ10 (80 ± 2 PFU/host cell), vB_KpnP_FZ12 (80 ± 7 PFU/host cell), vB_KpnS_FZ41 (118 ± 3 PFU/host cell), and vB_KpnM_FZ14 (120 ± 5 PFU/host cell) [140]. Despite the relatively low burst sizes of phages KpnSo1BRG and KpnSo2SCE, these phages promoted a significant decrease in the growth of *K. pneumoniae* during the whole timeframe of phage–bacteria inactivation assays. *K. pneumoniae* was effectively inactivated by phages KpnSo1BRG and KpnSo2SCE and their cocktail, reaching a maximum inactivation of ca. 94.2% and 84.7%, respectively, after 8 h of incubation with the host at MOI 0.0001, respectively. However, at this MOI, both phages failed at preventing bacterial regrowth after 6 h of contact with their host (**Figure 5a**). At MOI 0.01, phage KpnSo2SCE was quite successful in achieving a high bacterial inactivation, with ca. 94.1% of bacterial reduction being attained after 8 h, which more or less was maintained up to the end of the assay timeframe (**Figure 5a,b**). At this MOI, however, phage KpnSo1BRG failed at preventing bacterial regrowth after 6 h of contact with its host (**Figure 5a**). At MOI 10, both phages were

quite successful in achieving a high bacterial inactivation, with ca. 98.8% of bacterial reduction being attained after 8 h, with only a slight reduction in bacterial inactivation up to the end of the assay timeframe (**Figure 5a,b**; ANOVA, $p > 0.05$). At MOI 10,000, however, the story was quite different. At this MOI, both phages succeeded in achieving a high bacterial inactivation, with ca. 98.8% and 99.9% of bacterial reduction being attained after 8 h for phages KpnSo1BRG and KpnSo2SCE, respectively, which slightly reduced to 95.9% and 99.8%, respectively, by the end of the experiment (**Figure 5a,b**). Hence, MOI 10,000 for either independent phage proved to be quite successful in controlling the bacterial levels during a 12 h timeframe. Therefore, MOI 10,000 was selected for preparing a cocktail with both phages and as expected, the lytic phage cocktail was very effective in promoting a high bacterial inactivation, viz. 98.8% after 8 h, which only slightly reduced to 97.2% after 12 h (**Figure 5a,b**). The bacterial inactivation with both independent phages at MOI 10,000 was much better than that at MOI 0.0001 for both independent phages and then that at MOI 0.01 for phage KpnSo1BRG after 6 h, 8 h, 10 h, and 12 h of incubation (ANOVA, $p < 0.05$). After 8 h of phage–bacteria contact, the difference in bacterial reduction using both independent phages at MOI 10,000 was not statistically different from that using the phage cocktail at MOI 10,000 (ANOVA, $p > 0.05$). Increasing the MOI from 0.0001 to 0.01 to 10 g 10,000 did significantly increase the inactivation factor after 12 h of incubation with both independent phages (ANOVA, $p < 0.05$) (**Figure 5a,b**). When phages KpnSo1BRG and KpnSo2SCE were separately introduced to free-floating (planktonic) bacterial cells, noticeable bacterial regrowth occurred after six hours of incubation at MOI 0.0001 (ANOVA, $p < 0.05$) (**Figure 5a**). However, by the end of the incubation period, bacterial populations in all phage-treated samples were significantly lower across all MOI values compared to the untreated bacterial control (BC, **Figure 5a**).

A key challenge in phage-based bacterial control is the emergence of phage-resistant bacterial variants, an issue particularly prevalent in *K. pneumoniae* [10, 91, 142]. In this study, neither phage KpnSo1BRG nor KpnSo2SCE was able to completely suppress bacterial regrowth throughout the treatment period. Increasing the MOI from 0.0001 to 10,000 for phages KpnSo1BRG and KpnSo2SCE significantly increased the treatment efficiency, an observation in line with the kinetic theory of antibacterial phage treatment, which states that the inactivation of pathogenic bacterial host increases side by side with the value of MOI or starts sooner at higher values of MOI.

The frequency of spontaneous resistant mutants observed for both phages (ranging from 1.164×10^{-3} (0.116%) for phage KpnSo1BRG to 1.724×10^{-3} (0.172%) for phage KpnSo2SCE) is relatively high compared to the 10^{-6} to 10^{-8} range often reported for other therapeutic phages. Such a high frequency (approx 10^{-3}) indicates that resistant variants are pre-existing or rapidly arise within the bacterial population, likely through mutations in the capsular polysaccharide synthesis pathways which serve as the primary phage receptors. This high rate of resistance emergence represents a significant bottleneck that would likely preclude the use of these phages as monotherapies. Consequently, for these phages to be therapeutically viable, they must be deployed as part of a synergistic cocktail. As demonstrated in our in vitro inactivation assays, combining these phages or applying them at a

high MOI is essential to overcome the rapid selection of resistant phenotypes and ensure sustained bacterial clearance.

The observed bacterial regrowth at lower MOI values strongly points to the rapid emergence of spontaneous phage-resistant mutants. A frequent and highly effective bacterial defense mechanism against phage attack is the alteration, masking, or down-regulation of cell surface receptors (such as the *Klebsiella* capsular polysaccharide), thereby preventing initial phage adsorption. Our adsorption kinetic data provides unique insight into this dynamic. While both phages efficiently bound to the host, phage KpnSo2SCE exhibited a drastically lower desorption rate ($6.005 \times 10^{-11} \text{ min}^{-1}$) compared to KpnSo1BRG ($1.200 \times 10^{-7} \text{ min}^{-1}$). This indicates that once KpnSo2SCE binds to the host cell, it remains almost irreversibly attached. From a biological performance standpoint, this ultra-low desorption rate likely provides KpnSo2SCE with a sustained kinetic advantage in mixed populations; even as bacteria attempt to modify their surface structures to evade infection, the phage virions that successfully collide with, and bind to, remaining viable receptors do not detach, maintaining a highly productive infection cycle that contributes to its sustained inactivation efficacy over longer treatment periods.

The high bacterial inactivation efficiency of phages KpnSo1BRG and KpnSo2SCE, combined with their safety, paves the way for new studies, especially in vivo studies, aiming at the development of strategies to control infections caused by *K. pneumoniae*.

Phages KpnSo1BRG and KpnSo2SCE were very stable at 4 °C and 25 °C, without loss of activity at these temperatures during a timeframe of 72 h of incubation (**Figure 7a,b**; ANOVA, $p > 0.05$). At 37 °C, phage KpnSo1BRG lost 2.15 log PFU/ml, whereas phage KpnSo2SCE lost only 0.18 log PFU/ml of lytic activity over a 72 h period of incubation at that temperature. At 41 °C, however, phage KpnSo1BRG lost 3.53 log PFU/ml, whereas phage KpnSo2SCE lost 1.13 log PFU/ml of lytic activity at this temperature. Hence, both phages revealed to be quite resistant to the effects of high(er) temperatures upon their lytic viability, with the reduction in the concentration of viable phage virions being much higher at 41 °C for phage KpnSo1BRG than that for phage KpnSo2SCE (**Figure 7a,b**; ANOVA, $p < 0.05$).

The two isolated phages also revealed an unusual resistance to alkaline conditions (**Figure 7c,d**). While at acidic pH values (viz. 3 and 5) both phages lost a few log PFU/ml of lytic activity over a timeframe of 72 h, phage KpnSo1BRG (reduction of 5.27 log PFU/ml at pH 3.0 and reduction of 2.45 log PFU/ml at pH 5.0) was revealed to be more sensitive than phage KpnSo2SCE (reduction of 2.39 log PFU/ml at pH 3.0 and reduction of 1.29 log PFU/ml at pH 5.0) to lower pH values, with both phages being relatively insensitive to higher pH values (viz. pH 9 and 11), losing only 0.72 log PFU/ml of lytic activity over a 72 h period (**Figure 7c,d**). At neutral pH, both phages remained with nearly all their lytic activity, with phage KpnSo1BRG losing only 0.08 log PFU/ml and phage KpnSo2SCE losing only 0.49 log PFU/ml of lytic activity.

The presence of solar radiation had minimal influence on the persistence of active lytic functionality, which remained stable over the exposure period for both phages. Only a slight variation was observed between the control and experimental groups for phage KpnSo1BRG (**Figure 7e**; ANOVA, $p > 0.05$). Nonetheless, the number of functional lytic particles (SR) dropped by approximately 2.06 log PFU/ml (**Figure 7e**; ANOVA, $p < 0.05$) following

7 h of direct sunlight exposure compared to the control (SR-C), indicating a notable level of resistance in phage KpnSo1BRG when subjected to direct sunlight. Phage KpnSo2SCE (**Figure 7f**), on the other hand, showed no decrease in the concentration of lytic viability (SR) during the whole exposure timeframe when compared to the phage control (SR-C), hence revealing a high resilience to the nefarious effects of solar radiation.

Genomic sequencing analysis revealed that both KpnSo1BRG and KpnSo2SCE phages are siphoviruses (phage morphotype with perfect icosahedral capsids and long, apparently flexible, non-contractile tails) (**Figure 2b,d**; **Table 1**). Both phages belong to the class *Caudoviricetes* (**Figure 2b,d**), family *Drexlerviridae*, genus *Webevirus* (**Table 3**), according to recent ICTV bacterial virus taxonomy [143]. The taxonomic lineage of phages KpnSo1BRG and KpnSo2SCE was therefore established as *Viruses > Duplodnaviria > Heunggongvirae > Uroviricota > Caudoviricetes > Drexlerviridae > Webevirus* [144, 145]. Phages KpnSo1BRG and KpnSo2SCE possessed double-stranded DNA (dsDNA) genomes composed of 51,716 bp and 49,460 bp, respectively, with a GC content of 51.69% and 51.08%, respectively. The phage KpnSo1BRG genome encodes 87 open reading frames, of which 50 are hypothetical proteins (with unknown function) and 37 had assigned putative functions, whereas the phage KpnSo2SCE genome encodes 85 open reading frames, of which 49 are hypothetical proteins (with unknown function) and 36 had assigned putative functions. No tRNA, virulence related, or antibiotic resistance genes were found in either phage genome. Analysis of the genomes of phages KpnSo1BRG and KpnSo2SCE showed no evidence of known integrase genes [146], which are typically associated with lysogenic behavior. However, due to the high proportion of protein-coding genes with unidentified functions in both phages, it remains uncertain whether phages KpnSo1BRG and KpnSo2SCE exclusively follow a strictly lytic lifestyle [31, 32, 96, 147].

Host lysis proteins, viz. holin, endolysin, and spanin, were annotated in the genomes of both phages (**Figure 8**). Holin (or pinholin) accumulates harmlessly in the cytoplasmic membrane until a critical concentration is reached, which triggers the formation of nanopores (pinholes) and causes both depolarization of the host cell membrane and endolysin refolding and release into the periplasmic space [148]. Once the pinholin has permeabilized the host cell membrane, SAR-endolysins are released into the periplasm and breaks down the peptidoglycan layer. This process determines the precise timing of host cell lysis. Participation of the SAR-endolysin and spanin proteins in these sequential events lead to the programmed host cell lysis with concomitant release of the mature phage virions from the host cell.

Webevirus siphoviruses, a genus within the *Drexlerviridae* family, exhibits distinct traits, including a non-enveloped virion with an icosahedral capsid measuring approximately 60 nm in diameter and a lengthy, flexible, non-contractile tail around 151 nm in length and 8 nm in width. The tail terminates in rosette-like formations with four short, bent fibers [149] that play a role in host cell recognition and binding [150]. These phages possess a linear double-stranded DNA genome ranging from 40 to 50 kilobases, encoding roughly 80 genes [150–153]. Known for their lytic activity, *Webevirus*es primarily target *K. pneumoniae* and other members of the Enterobacteriaceae family [29, 140, 154–159]. They are classified under the class *Caudoviricetes* and family *Drexlerviridae* [149], which predominantly infect Gram-negative

bacteria such as Enterobacteriaceae and *Pseudomonas* [150]. The structural and genomic characteristics of these phages, summarized in **Table 3**, are clearly illustrated in the TEM images shown in **Figure 2b,d**.

Genome circular maps were constructed for both the KpnSo1BRG (**Figure 8a**) and KpnSo2SCE (**Figure 8b**) phage genomes to illustrate their similarities and differences. These maps include representations of various genomic features such as GC content (outward indicating higher than average GC content compared to the whole genome, and inward indicating the opposite), GC skew (outward indicating values >0 and inward indicating the opposite), cyclizability, helical twist values, minor groove width, roll values, propeller twist values, and coding sequences/regions (CDSs).

The KpnSo1BRG phage genome comprises 87 coding regions, zero tRNAs, and zero rRNAs, while the KpnSo2SCE phage genome includes 85 coding regions, zero tRNAs, and zero rRNAs. Additionally, the KpnSo1BRG genome contains 50 predicted hypothetical proteins, whereas the KpnSo2SCE genome contains 49 (**Table 3**). The absence of tRNAs suggests a dependency on the host's translational machinery for both KpnSo1BRG and KpnSo2SCE phage genomes.

In the KpnSo1BRG phage genome, various proteins were identified, including tail proteins, a modifier of suppressor of tRNAs, capsid proteins, and minor and major tail proteins. The presence of the modifier of suppressor of tRNAs might explain the absence of tRNAs in this genome, although further research is required for confirmation.

In contrast, the KpnSo2SCE genome features capsid proteins, minor and major tail proteins, phage proteins, transcriptional regulators, helicases, and others.

Phage KpnSo1BRG was classified alongside 30 other phages at the family level and clustered with 13 at the genus level. In contrast, phage KpnSo2SCE shared a family-level grouping with 30 phages but was associated with only seven at the genus level (**Figure 9**). Both phages, KpnSo1BRG and KpnSo2SCE, occupy distinct branches on the phylogenetic tree, indicating limited evolutionary relatedness to other members of their respective clusters (**Figure 9**). The phage closest to the KpnSo1BRG phage is *Klebsiella* phage PWKp17 (GCA_021800495.1), whereas the phage closest to the KpnSo2SCE phage is *Klebsiella* phage vB_KpnS_15-38_KLPPOU149 (GCF_009800765.1) (**Figure 9**), exhibiting the smallest distance to the KpnSo1BRG and KpnSo2SCE phage genomes (genomic distance metrics calculated with the program Mash, that calculates distances between genomes using kmer sketches).

The connections and similarities between phages KpnSo1BRG, KpnSo2SCE, and other double-stranded DNA viruses infecting prokaryotes were examined using the online ViPTree tool [18, 78, 105]. This analysis generated a viral proteomic tree, illustrated in **Figure 10**. The findings revealed that phage KpnSo1BRG formed a distinct cluster with nine other phages (**Figure 10a,b**), while phage KpnSo2SCE was grouped with 14 additional phages (**Figure 10c,d**). All these phages are classified under the *Drexelviridae* family, as per the latest ICTV taxonomy. Here we can see that phage KpnSo1BRG and phages KOX1 (NC_047825) and vB_KpnS_IMGroot (NC_049834) (the phages closest to the KpnSo1BRG phage) are alone on their

branch, showing high phylogenetic similarity (**Figure 10b**). Based on the results obtained from the ViPTree tool and tBLASTx genomic sequence alignment, phages KpnSo1BRG (**Figure 11a**) and KpnSo2SCE (**Figure 11b**) were compared to other phage genomes available in the Virus-Host Database [106], and the phages closest to the KpnSo1BRG phage are phages KOX1 (NC_047825) and vB_KpnS_IMGroot (NC_049834), with 95.0% and 94.8% similarity, respectively, whereas the phages closest to phage KpnSo2SCE are phages NJS1 (NC_048024) and vB_KpnS_FZ10 (NC_049840), with 93.1% and 93.0% similarity, respectively.

Cyclizability, pairwise distant distribution function, and dinucleotide frequency. The structural mechanics of genomic DNA are crucial for its stability, function, and interactions with proteins and other molecules inside the cell. Its flexibility is influenced by multiple factors, including the rigidity of the sugar-phosphate backbone, the specific nucleotide sequence, and the forces that govern base stacking [160].

DNA flexibility can be studied at various levels, from the movement of individual base pair steps to the overall conformation of the double helix. This property plays a key role in genome organization and gene expression (i.e., transcription) regulation. A clear example is seen in viral genomes, where the DNA undergoes substantial bending to fit within the confined space of the viral capsid [161]. In these processes, transcription factors interact with DNA, causing it to bend and form specialized structures. These structural changes help recruit proteins that play a crucial role in regulating gene expression [162]. Increasing evidence indicates that differences in DNA sequences can impact its structural flexibility by modifying how adjacent base pairs interact and stack together [163]. A method to measure DNA flexibility in relation to its biological role involves evaluating how effectively double-stranded DNA forms loops or circular structures (cyclization).

DNA cyclization is influenced by the length of the DNA molecule. Shore and Baldwin [164] were the first to explore this phenomenon, revealing the helical nature of DNA through the cyclic variation in cyclization efficiency [165]. Following their work, various experimental and low-throughput methods have been developed to study the tendency of DNA to form circular structures (i.e., its cyclization propensity).

In recent years, Basu et al. [166] developed a high-throughput method called loop-seq to assess the inherent ability of DNA to form loops. This technique quantifies cyclization by analyzing the logarithmic ratio of sequence occurrences in a looped set compared to a control group. Additionally, intrinsic cyclizability was determined as the average value derived from multiple iterations of the loop-seq process.

Our research investigated the natural looping tendency of the KpnSo1BRG and KpnSo2SCE phage genomes at a fine-scale resolution of seven base pairs. This analysis was conducted using an approach inspired by the techniques developed by Basu et al. [81]. Their approach evaluated the natural ability of DNA to form circular structures by analyzing 90,000 distinct 50 base pair sequences. In the genome of phage KpnSo1BRG, some regions showed cyclizability peaks as high as 0.832, whereas in the KpnSo2SCE genome, the highest peaks reached 0.685. This indicates that certain areas of the KpnSo1BRG genome may have a greater tendency to circularize (i.e., to bend) compared to those in the KpnSo2SCE genome. Conversely, the lowest cyclizability value

observed in the KpnSo1BRG genome was -0.866 (**Figure 17a**), while the KpnSo2SCE genome recorded a minimum of -0.956 (**Figure 17b**), suggesting that the latter genome may contain more regions with limited cyclizability (i.e., bending) potential.

The average cyclizability score for the KpnSo1BRG genome is -0.238 , whereas the KpnSo2SCE genome has a slightly higher average of -0.226 . This difference becomes clearer when analyzing the distribution and frequency of cyclizability values across both genomes (**Figure 19**). The skewness of these values varies significantly between the two phages (**Figure 19b**: KpnSo1BRG; **Figure 19c**: KpnSo2SCE), highlighting their distinct cyclizability patterns. A Mann–Whitney U test confirmed this distinction, revealing a statistically significant difference with a p -value of 1.3799×10^{-4} .

By heuristically comparing the mechanical properties of these genomes to their biological performance, a clear correlation emerges between genome flexibility and viral morphogenesis efficiency. On average, the KpnSo2SCE genome exhibits a higher overall intrinsic cyclizability (-0.226) than the KpnSo1BRG genome (-0.238), coupled with a higher frequency of the highly flexible TA dinucleotide (4.32% vs. 4.08%). This increased global flexibility in KpnSo2SCE translates directly to its superior biological performance during the infection cycle. While both phages produce similar virion morphogenesis yields, the more flexible KpnSo2SCE genome is associated with a slightly higher burst size (18.6 vs. 17.7 virions/host cell) and, most notably, a drastically shorter latent period (5.8 min vs. 60 min for KpnSo1BRG). These data strongly suggest that a more pliable, flexible genome sequence offers less thermodynamic resistance during the extreme physical constraints of infection. Consequently, the highly flexible KpnSo2SCE genome likely facilitates a much faster translocation into the bacterial cytoplasm, easier negotiation with the host's RNA polymerase for rapid transcription, and more efficient spooling into the newly assembled viral capsids, ultimately accelerating the entire lytic cycle.

It is important to recognize that DNA looping predominantly occurs in short sequences of base pairs, as these segments inherently possess the ability to form loops. Our study identified localized cyclizability values within the genomes of phages KpnSo1BRG and KpnSo2SCE, pinpointing specific areas with either elevated or reduced cyclizability (**Figure 18**). While this analysis provides insight into cyclizability at a localized scale, it does not reflect the overall mechanical properties of the entire phage genome.

From these findings, we deduce that the KpnSo1BRG genome contains certain regions with greater cyclizability compared to the KpnSo2SCE genome. However, most regions in the KpnSo1BRG genome display cyclizability values close to -0.238 , whereas the KpnSo2SCE genome averages -0.226 . This implies that, on average, the KpnSo2SCE genome has more flexible regions than the KpnSo1BRG genome. These flexible regions are likely shaped by DNA structural features such as ProT, MGW, HelT, and roll, as discussed earlier. These results align closely with prior findings from our research group [18, 33, 43, 80], indicating that greater genome flexibility correlates with higher virion production during morphogenesis. In this study, phage KpnSo1BRG generated 17.7 virions per host cell, while phage KpnSo2SCE produced 18.6 virions per host cell (**Figure 4**). Although the difference is modest, the morphogenesis yields of the two phages within their host cells are indeed distinct.

Multiple studies suggest that DNA flexibility and looping potential are influenced by specific pairs of adjacent nucleotides. These dinucleotide (pairing of two nucleotides along the DNA sequence) combinations play a role in shaping the structural and geometric properties of the DNA molecule, affecting its overall adaptability [167].

The distribution pattern of specific dinucleotide pairs within a DNA sequence is often used as a unique genetic identifier for different microorganisms [168]. The expected frequency of a dinucleotide is calculated by multiplying the occurrences of its individual nucleotides. In a sequence without bias, the observed frequency of dinucleotides matches the expected frequency, producing a ratio (observed divided by expected) of 1 [169]. As a result, a ratio greater than 1 suggests that a particular dinucleotide appears more frequently than expected, whereas a ratio below 1 indicates its occurrence is lower than anticipated [170].

The repeated occurrence of a dinucleotide at regular intervals within a DNA sequence suggests a pattern of periodicity. Wu et al. [171] observed that certain dinucleotides display periodicity, which correlates with DNA flexibility. Their findings indicate that this pattern often emerges around every 10 base pairs, aligning with the helical pitch of DNA, though the exact spacing can vary based on sequence composition. In general, periodicity can manifest at any consistent interval within a sequence.

Research has shown that the arrangement of dinucleotides within the genome plays a significant role in determining DNA flexibility [160, 167, 171, 172]. Basu et al. [81] explored how the spatial distribution of specific dinucleotides influences local cyclizability. Using high-throughput analysis, they calculated the pairwise distance distribution function to assess how frequently certain dinucleotide pairs appear at specific intervals. This approach helped reveal their connection to DNA mechanical properties, such as looping potential, across a span of 100 base pairs.

In their research, Basu et al. [81] investigated how 136 different dinucleotide combinations (NN-NN) influence the local cyclizability of DNA. They found that dinucleotide pairs made exclusively of adenines (A) or thymines (T) enhanced DNA cyclizability, while pairs composed solely of guanines (G) or cytosines (C) also exhibited a similar positive effect. Based on this insight and aiming to identify which dinucleotide pairs are associated with cyclizability or flexibility in the KpnSo1BRG and KpnSo2SCE phage genomes, we performed a pairwise distance correlation analysis of 256 dinucleotide pairs steps. Subsequently, we examined the similarities and differences between both genomes and compared our results to those reported by Basu et al. [81].

Basu et al. [81] found that dinucleotides consisting solely of As, Ts, Gs, and Cs contribute positively to local DNA flexibility. In our analysis, we specifically examined these dinucleotides and discovered that the AA pair appears more frequently in the KpnSo1BRG phage genome than in the KpnSo2SCE phage genome (**Figure 14a**), with a distant correlation frequency of approximately 2.3. The distant correlation frequency is defined as the frequency of finding a specific dinucleotide step at a specific distance, which has been normalized by the frequency of finding the same dinucleotide step in a random expected sequence.

The distant correlation frequency of the AA dinucleotide in the KpnSo2SCE phage genome is similar to the random expected sequence (**Figure 14a**).

The AA-AT step exhibits a higher distant correlation frequency in both KpnSo1BRG and KpnSo2SCE phage genomes (**Figure 14b**), compared to a random expected sequence. However, the distant correlation frequency in the KpnSo1BRG genome is approximately 1.8, while in the KpnSo2SCE genome, it is ca. 1.3, indicating a greater over-representation of this step in the KpnSo1BRG phage genome (**Figure 14b**). The AA-TT step is over-represented in both phage genomes (**Figure 14c**), with a distant correlation frequency of approximately 1.5. Previously, the AA-TT dinucleotide step has been related to a narrow MGW in both free and bound DNA [128]. Since this dinucleotide step is over-represented in both phage genomes, it might play an important role in the MGW in both KpnSo1BRG and KpnSo2SCE phage genomes.

Similarly, the AT-TT step occurs more frequently than in the random expected sequence (**Figure 14f**). The distant correlation frequency of this dinucleotide in the KpnSo2SCE phage genome is higher than in the KpnSo1BRG phage genome, approximately 1.8 compared to 1.3, respectively (**Figure 14f**). The TA dinucleotide appears less frequently in both phage genomes compared to the random expected sequence (**Figure 14o**). The TA-TT step is over-represented in the KpnSo2SCE phage genome compared to the KpnSo1BRG phage genome (**Figure 14p**), with a distant correlation frequency of approximately 1.3 and 0.8, respectively.

Both the CG (**Figure 14i**) and GC (**Figure 14m**) dinucleotides have a higher occurrence in both phage genomes compared to the random expected sequence. The CG-GC step (**Figure 14j**) is observed with higher occurrence in both phage genomes compared to the random expected sequence, with both phage genomes showing a similar distant correlation frequency of approximately 1.8. However, the CC-GC step (**Figure 14h**) is over-represented in the KpnSo2SCE phage genome, with a distant correlation frequency of approximately 1.3, while in the KpnSo1BRG phage genome, the correlation is similar to that of the random expected sequence (**Figure 14h**).

We can observe that the majority of dinucleotide steps identified by Basu and colleagues [81] as positively correlated with intrinsic cyclizability are over-represented in one or both phage genomes. This might suggest that these particular dinucleotide steps are making a positive contribution to the cyclizability of both KpnSo1BRG and KpnSo2SCE phage genomes to some extent.

On the other hand, Wu et al. [171] have also examined the association between specific dinucleotide steps and DNA flexibility, finding other dinucleotide steps associated with flexibility. These researchers employed three models to assess the flexibility of various DNA sequences and dinucleotide patterns at the local level, and observed strong positive correlations between CA-TG, CG, GC, GG-CC, AG-CT, AC-GT, and GA-TC steps and DNA flexibility, while the AA-TT step exhibited the lowest correlation with DNA flexibility.

Based on these observations, we decided to examine if those dinucleotide steps previously reported by Wu et al. [171] are also playing an important role in DNA bendability in both KpnSo1BRG and KpnSo2SCE phage genomes.

In our results, we observed that the CA-TG (**Figure 14g**) and GA-TC (**Figure 14l**) dinucleotide steps exhibited distant correlation frequencies similar to those of the random expected sequence in both phage genomes, while there was an under-representation

of the GG-CC (**Figure 14n**), AG-CT (**Figure 14e**), and AC-GT (**Figure 14d**) steps in both phage genomes, with distant correlation frequencies of approximately 0.8. The CG (**Figure 14i**) and GC (**Figure 14m**) dinucleotides were previously discussed, and they showed an over-representation in both phage genomes.

Wu et al. [171] reported that the AA-TT step exhibits the lowest correlation with DNA flexibility. Conversely, Basu et al. [81] reported a positive correlation between this dinucleotide step and DNA flexibility. This discrepancy could be attributed to the context-dependent nature of certain dinucleotide steps' influence on DNA flexibility and rigidity. Their impact is not universally defined and can vary among different organisms.

In conclusion, we analyzed 256 dinucleotide steps within the KpnSo1BRG and KpnSo2SCE phage genomes aiming at unveiling similarities and differences between them. Variations in the occurrence of specific dinucleotides within 100 steps were observed in both phage genomes. However, no discernible oscillations or periodic patterns were evident in any of the dinucleotide steps, contrary to suggestions made by Wu et al. [171] who mentioned that certain dinucleotides exhibit periodicity observed at repetitive distances within a sequence. One potential reason for not observing these oscillations could be due to differences in the organisms studied. While Wu et al. [171] conducted their research in yeast genomes, we are investigating phage genomes, which are entirely distinct entities.

Nevertheless, a close inspection of the pairwise distant distribution function of the GC dinucleotide (**Figure 14m**) might reveal a periodic pattern of nearly every 20 base pairs but further research is required to validate this finding.

Furthermore, our analysis hinted at a potential intricate relationship between the occurrence of specific dinucleotide steps in both KpnSo1BRG and KpnSo2SCE phage genomes and DNA bendability to some extent, as certain dinucleotides were over-represented in both phage genomes, correlating with findings from the study by Basu et al. [81]. Nonetheless, further investigations are necessary to substantiate this assertion.

Based on the findings from the pairwise distance distribution function, we decided to analyze the overall frequency of each dinucleotide in the phage genomes, disregarding their spatial arrangement. **Figure 15** presents a graphical representation of the percentage distribution of all 16 possible dinucleotide combinations (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT) across the two phage genomes.

Basu et al. [81] identified TA as the dinucleotide with the strongest positive effect on DNA flexibility, while CG was found to be the least flexible in terms of intrinsic cyclizability. However, in both phage genomes analyzed, TA appears less frequently at specific distances (**Figure 15**). Previous research by Packer et al. [131] highlighted TA's role in influencing DNA mechanics, particularly in enhancing flexibility. More recent studies by Back and Walther [173] further reinforce this idea, showing a clear link between TA content and DNA bendability.

These findings suggest that a higher occurrence of TA dinucleotides may contribute to increased DNA flexibility, while a greater abundance of CG pairs could make the DNA structure more rigid.

Our analysis showed that the TA dinucleotide appeared at a frequency of 4.32% and 4.08% in the KpnSo2SCE and KpnSo1BRG phage genomes, respectively, and a CG frequency of 8.41% and 8.65% in the KpnSo2SCE and KpnSo1BRG phage genomes, respectively (**Figure 15**). The TA dinucleotide is under-represented in the pairwise distant distribution function (**Figure 14o**); however, despite TA appearing less frequently than expected in a randomly generated sequence, its role in enhancing DNA flexibility remains significant.

This finding implies that the KpnSo2SCE phage genome may exhibit greater flexibility than the KpnSo1BRG genome in regions where this specific dinucleotide is abundant. Conversely, the CG dinucleotide, which appears most frequently in both genomes (**Figures 14i** and **15**), could suggest increased rigidity in regions where CG dinucleotides are dominant. Rao et al. [174] discovered that cytosine methylation predominantly occurs at CG dinucleotides, altering DNA structure and local shape parameters, particularly roll and ProT. Their findings demonstrated that methylation significantly impacts DNA conformation, increasing the roll angle while reducing the ProT angle in CpG steps, with effects ranging from 50% to 100% compared to unmethylated sequences. Additionally, Basu et al. [81] reported that CpG methylation enhances DNA rigidity within a single sequence.

In our analysis, CG was the most frequent dinucleotide (**Figure 15**) and was over-represented in pairwise distance correlations spanning 100 steps (**Figure 14i**). This suggests that methylation could potentially occur at these sites in the KpnSo1BRG and KpnSo2SCE phage genomes. However, this remains speculative and requires further investigation.

After calculating the dinucleotide frequencies in each genome, we computed the differential dinucleotide frequency (**Figure 16**) to obtain a better understanding of the dinucleotide genomic differences. In our results, the dinucleotides CG, GA, and GC in the KpnSo1BRG phage genome demonstrate a higher differential frequency than most dinucleotides in the KpnSo2SCE phage genome (**Figure 16**). Conversely, CG, GC, and TC in the KpnSo2SCE phage genome exhibit a higher differential frequency than most dinucleotides in the KpnSo1BRG phage genome.

This finding aligns with the dinucleotide frequency (**Figure 15**) and pairwise distance distribution function of these specific dinucleotides (**Figure 14i,k,m,q**). The elevated occurrence of CG and GC dinucleotides in both the KpnSo1BRG and KpnSo2SCE phage genomes suggests a potential role for these dinucleotides in shaping DNA structure and enhancing or decreasing DNA flexibility in both genomes.

The transition from simply reading a bacteriophage's genetic code (nucleotide sequence) to understanding its mechanical and physical properties represents a fascinating paradigm shift in virology. In phage biology, analyzing the mechanical features of a viral genome provides crucial insights into how efficiently a phage can infect a host, replicate, and assemble new virions. In our opinion, here is why these mechanical analyses are so significant to understanding phage biology and optimizing phage therapy: [i] Decoding the “mechanical code” of DNA: DNA is not just an abstract string of information; it is a physical, three-dimensional polymer. The unique sequence of base pairs dictates localized structural variations (the DNashape). Parameters like roll (the angle between adjacent base pairs) and minor groove width (MGW) determine how easily the DNA helix can bend and twist

(i.e., flexibility vs. rigidity). Cyclizability measures the intrinsic probability of a DNA sequence to loop back on itself and form a closed ring. A genome with high cyclizability is highly flexible, whereas a low cyclizability score indicates a rigid molecule. [ii] The physics of DNA packaging (virion morphogenesis): One of the most extreme biomechanical events in nature is the packaging of a phage genome. During viral assembly (morphogenesis), the double-stranded DNA must be tightly spooled into a rigid, pre-formed protein shell (the capsid), by a molecular motor. The DNA is packed to near-crystalline density, creating internal pressures of up to 60 atm. If a phage genome possesses high intrinsic cyclizability (i.e., mechanical flexibility), it offers less physical resistance to being bent and compressed. This “pliable” DNA molecule conforms more easily to the confined space of the capsid, leading to a much more energetically favorable and efficient assembly process. [iii] “Negotiating” with the host’s transcription machinery: Once a phage injects its DNA into a susceptible bacterium, it must immediately hijack the host’s cellular machinery to survive. To transcribe the early viral genes, the bacterium’s RNA polymerase (RNAP) must bind to the phage DNA and physically bend it to open the transcription bubble. A mechanically flexible phage genome acts as a “highly cooperative” template. It “negotiates” easily with the host’s RNAP because it requires less thermodynamic energy to achieve the necessary structural distortions. This results in rapid transcription of the phage genome and production of phage proteins in large amounts, allowing the phage to take over the cell at blinding speed. [iv] Direct correlation with infection kinetics: The biological significance of genome mechanics ultimately manifests in the phage’s infection kinetics (the very metrics that determine if a phage is a good candidate for clinical therapy). Recent studies from our research group [18, 33, 43, 80] have established that highly flexible genomes correlate strongly with shorter eclipse periods (because the DNA is packaged and transcribed so efficiently, the time from initial infection to the assembly of the first mature virion is significantly reduced) and larger burst sizes (the ease of packaging flexible DNA means the host cell’s resources can be utilized to assemble a much higher number of mature virions before the cell undergoes lysis (i.e., bursts)). [v] Predictive power for phage therapy: Finding a highly lytic, efficient phage for biocontrol or human therapy relied heavily on tedious, trial-and-error in vitro assays. By performing mechanical analyses on newly sequenced genomes, one can use bioinformatics as a “predictive biomarker”, screening vast libraries of sequenced phages in silico and prioritize those with optimal mechanical signatures (high cyclizability, optimal roll values) as the most promising “super-phages” for treating multidrug-resistant infections.

Ultimately, it is important to highlight that our analyses—focusing on DNA shape, pairwise distance correlations, and intrinsic cyclizability—primarily impact DNA structure at the local or nucleotide level. However, these factors are less influential when considering the overall global architecture of the genome, since other major properties influence the DNA at this level. Those properties have not been addressed in this study. Our findings offer a detailed characterization of the KpnSo1BRG and KpnSo2SCE phage genomes at the nucleotide level, shedding light on their flexibility and cyclization potential. While these analyses provide valuable insights, we recognize that a broader, genome-wide study would enhance our understanding. This perspective will be considered in future research.

Limitations and translational perspectives. Both phages show potential for combating Kpn-related bacterial infections, but in order to enhance their effectiveness, additional lytic phages should be isolated and used to create a broader-spectrum phage cocktail capable of targeting a wide range of Kpn strains. While the comprehensive molecular and mechanical characterization of phages KpnSo1BRG and KpnSo2SCE provides valuable insights into their lytic machinery, several key limitations must be addressed before considering clinical translation. First, both phages exhibited a highly narrow host range, effectively lysing only their isolation host out of the 66 strains tested. This high specificity is a double-edged sword; while it ensures minimal disruption to the host microbiome, it renders these phages unsuitable as standalone monotherapies. For clinical viability, KpnSo1BRG and KpnSo2SCE must be integrated into carefully curated, broad-spectrum phage cocktails targeting a diverse array of *K. pneumoniae* capsule types and surface receptors. Second, our in vitro inactivation assays demonstrated that an exceptionally high initial viral dose (MOI 10,000) was required to achieve maximal efficacy and prevent bacterial regrowth. This directly correlates with the observed frequency of spontaneous resistant mutants (0.116% to 0.172%). Because *K. pneumoniae* can rapidly alter its surface receptors or up-regulate capsular polysaccharide production to evade infection, a massive initial phage titer is necessary to overwhelm the bacterial population and achieve bacterial clearance before these resistant mutant subpopulations can out-compete the phages. In addition, the frequency of emergence of *K. pneumoniae* NCTC-13439 spontaneous resistant mutants to the phage cocktail was not evaluated. Finally, although our data demonstrate that these phages possess robust stability against abiotic environmental factors (such as extreme pH, temperature, and UV radiation), the present study relies exclusively on in vitro models. It should be noted, however, that lab conditions may not reflect real-world storage. The current lack of in vivo data represents a significant limitation. Systemic application in a clinical setting introduces complex biological variables, including rapid clearance by the reticuloendothelial system, potential neutralization by the human innate immune response, and reduced phage penetrability into deep-tissue infection sites. Therefore, future studies utilizing in vivo models (e.g., murine models or *Galleria mellonella*) are strictly required to assess the pharmacokinetics, immunogenicity, and true therapeutic efficacy of these phages in a living organism.

5. Conclusions

Both phages show potential for combating Kpn-related bacterial infections. However, to enhance their effectiveness, additional lytic phages should be identified and combined with these two to create a broader-spectrum phage cocktail capable of targeting diverse Kpn strains.

Genome mechanics analysis revealed that the KpnSo1BRG genome contains specific regions with greater cyclizability compared to KpnSo2SCE. This suggests that, on average, the KpnSo2SCE genome has more flexible regions, likely influenced by DNA shape characteristics. These findings align with previous research from our group, indicating that increased genome flexibility correlates with higher virion production in infected host cells.

In conclusion, we have isolated and thoroughly characterized two novel virulent *Webervirus* siphoviruses, KpnSo1BRG and KpnSo2SCE, which target multidrug-resistant *K. pneumoniae*. While both phages exhibit potent antibacterial activity in vitro, their markedly different replication kinetics (specifically the exceptionally short 5.8 min latent period of phage KpnSo2SCE) underscore the diverse evolutionary strategies within this genus. Our molecular analysis reveals that these differences in replicative performance correlate significantly with the mechanical properties of their DNA, such as genomic flexibility and cyclizability. Consequently, this study demonstrates that genome mechanical features may predict phage replicative efficiency, offering a new and valuable criterion for the rational selection and optimization of phages for therapeutic applications.

Funding

This research was sponsored by the São Paulo Research Foundation (FAPESP), under grant numbers 2022/10775-9 and 2023/03797-9. V.M.B. was a visiting researcher fellow from FAPESP (Project PsaPhageKill), under grant number 2018/05522-9. V.M.B. received a research fellowship award from the National Council for Scientific and Technological Development (CNPq), under grant number 301978/2022-0.

Author contributions

Conceptualization, V.M.B.; methodology, V.M.B., S.C.E., B.R.G., and M.M.D.C.V.; Python code writing, A.B., C.A.Q., and B.C.; validation, V.M.B. and B.C.; formal analysis, V.M.B., B.C., C.A.Q., S.S., A.A.A., E.B.O., and M.M.D.C.V.; investigation, S.C.E. and B.R.G.; resources, V.M.B.; data curation, V.M.B.; writing—original draft preparation, V.M.B.; writing—review and editing, V.M.B., B.C., C.A.Q., S.S., A.A.A., E.B.O., and M.M.D.C.V.; visualization, V.M.B.; supervision, V.M.B.; project administration, V.M.B.; funding acquisition, V.M.B. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare that they have no competing interests. The funding sponsors played no role in the study's conception, execution, or interpretation, or in the preparation of this article.

Data availability statement

The genomic data generated in this study has been made available within the NCBI GenBank database, under accession numbers PV081216 (phage KpnSo1BRG), and PV081217 (phage KpnSo2SCE).

Supplementary materials

The supplementary materials are available at <https://doi.org/10.20935/AcadMolBioGen8364>.

Additional information

Received: 26 March 2026

Accepted: 5 June 2026

Published: 17 June 2026

Academia Molecular Biology and Genomics papers should be cited as *Academia Molecular Biology and Genomics* 2026, ISSN 3064-9765, <https://doi.org/10.20935/AcadMolBioGen8364>. The journal's official abbreviation is *Acad. Mol. Biol. Genom.*

Publisher's note

Academia.edu Journals stays neutral with regard to jurisdictional claims in published maps and institutional affiliations. All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

Copyright

© 2026 copyright by the authors. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

References

- Kim Y, Lee SM, Nong LK, Kim J, Kim SB, Kim D, et al. Characterization of *Klebsiella pneumoniae* bacteriophages, KP1 and KP12, with deep learning-based structure prediction. *Front Microbiol.* 2023; 13: 990910. doi: 10.3389/fmicb.2022.990910
- Ochońska D, Ścibik Ł, Brzywczy-Włoch M. Biofilm formation of clinical *Klebsiella pneumoniae* strains isolated from tracheostomy tubes and their association with antimicrobial resistance, virulence and genetic diversity. *Pathogens.* 2021; 10 (10): 1345. doi: 10.3390/pathogens10101345
- Chang D, Sharma L, Dela Cruz CS, Zhang D. Clinical epidemiology, risk factors, and control strategies of *Klebsiella pneumoniae* infection. *Front Microbiol.* 2021; 12: 750662. doi: 10.3389/fmicb.2021.750662
- Pacios O, Blasco L, Ortiz Cartagena C, Bleriot I, Fernández-García L, López M, et al. Molecular studies of phages-*Klebsiella pneumoniae* in mucoid environment: innovative use of mucolytic agents prior to the administration of lytic phages. *Front Microbiol.* 2023; 14: 1286046. doi: 10.3389/fmicb.2023.1286046
- Kamruzzaman M, Iredell JR. CRISPR-cas system in antibiotic resistance plasmids in *Klebsiella pneumoniae*. *Front Microbiol.* 2020; 10: 2934. doi: 10.3389/fmicb.2019.02934
- Mackow NA, Shen J, Adnan M, Khan AS, Fries BC, Diagonavarró E. CRISPR-CAS influences the acquisition of antibiotic resistance in *Klebsiella pneumoniae*. *PLoS ONE.* 2019; 14 (11): e0225131. doi: 10.1371/journal.pone.0225131
- Mohd Asri NA, Ahmad S, Mohamud R, Mohd Hanafi N, Mohd Zaidi NF, Irekeola AA, et al. Global prevalence of nosocomial multidrug-resistant *Klebsiella pneumoniae*: a systematic review and meta-analysis. *Antibiotics.* 2021; 10 (12): 1508. doi: 10.3390/antibiotics10121508
- Gan L, Fu H, Tian Z, Cui J, Yan C, Xue G, et al. Bacteriophage effectively rescues pneumonia caused by prevalent multidrug-resistant *Klebsiella pneumoniae* in the early stage. *Microbiol Spectr.* 2022; 10 (4): e0235822. doi: 10.1128/spectrum.02358-22
- Harada LK, Silva EC, Campos WF, Del Fiol FS, Vila MMDC, Dąbrowska K, et al. Biotechnological applications of bacteriophages: state of the art. *Microbiol Res.* 2018; 212–3: 38–58. doi: 10.1016/j.micres.2018.04.007
- Rios AC, Moutinho CG, Pinto FC, Del Fiol FS, Jozala AF, Chaud MV, et al. Alternatives to overcoming bacterial resistances: state-of-the-art. *Microbiol Res.* 2016; 191: 51–80. doi: 10.1016/j.micres.2016.04.008
- Samir S, El-Far A, Okasha H, Mahdy R, Samir F, Nasr S. Isolation and characterization of lytic bacteriophages from sewage at an egyptian tertiary care hospital against methicillin-resistant staphylococcus aureus clinical isolates. *Saudi J Biol Sci.* 2022; 29 (5): 3097–106. doi: 10.1016/j.sjbs.2022.03.019
- Samir S. Phages for treatment of staphylococcus aureus infection. *Prog Mol Biol Transl Sci.* 2023; 200: 275–302. doi: 10.1016/bs.pmbts.2023.03.027
- Lin DM, Koskella B, Lin HC. Phage therapy: an alternative to antibiotics in the age of multi-drug resistance. *World J Gastrointest Pharmacol Ther.* 2017; 8 (3): 162–73. doi: 10.4292/wjgpt.v8.i3.162
- Samir S. Bacteriophages as therapeutic agents: alternatives to antibiotics. *Recent Pat Biotechnol.* 2021; 15 (1): 25–33. doi: 10.2174/1872208315666210121094311
- Ferry T, Kolenda C, Laurent F, Leboucher G, Merabischvilli M, Djebara S, et al. Personalized bacteriophage therapy to treat pandrug-resistant spinal pseudomonas aeruginosa infection. *Nat Commun.* 2022; 13 (1): 4239. doi: 10.1038/s41467-022-31837-9
- Zalewska-Piątek B. Phage therapy-challenges, opportunities and future prospects. *Pharmaceuticals.* 2023; 16 (12): 1638. doi: 10.3390/ph16121638
- Samir S. Basic guidelines for bacteriophage isolation and characterization. *Recent Pat Biotechnol.* 2023; 17 (4): 312–31. doi: 10.2174/1872208317666221017094715
- Silva EC, Quinde CA, Cieza B, Basu A, Vila MMDC, Balcão VM. Molecular characterization and genome mechanical features of two newly isolated polyvalent bacteriophages infecting *Pseudomonas syringae* pv. *garcae*. *Genes.* 2024; 15 (1): 113. doi: 10.3390/genes15010113
- Gao M, Yi L, Wang Y, Gao J, Liu H, Zhang X, et al. Characterization and genomic analysis of bacteriophage VB_KpnM_IME346 targeting clinical *Klebsiella pneumoniae* strain of the K63 capsular type. *Curr Microbiol.* 2022; 79 (5): 160. doi: 10.1007/s00284-022-02834-4

20. Dennehy JJ, Abedon ST. Phage Infection and Lysis. In: Harper DR, Abedon ST, Burrowes BH, McConville ML, editors. *Bacteriophages*. Cham, Switzerland: Springer; 2021. p. 341–83.
21. Fister S, Robben C, Witte AK, Schoder D, Wagner M, Rossmanith P. Influence of environmental factors on phage-bacteria interaction and on the efficacy and infectivity of phage P100. *Front Microbiol.* 2016; 7: 1152. doi: 10.3389/fmicb.2016.01152
22. Naureen Z, Dautaj A, Anpilogov K, Camilleri G, Dhuli K, Tanzi B, et al. Bacteriophages presence in nature and their role in the natural selection of bacterial populations. *Acta Biomed.* 2020; 91 (13): e2020024. doi: 10.23750/ABM.V91I13-S.10819
23. Harper DR, Abedon ST, Burrowes BH, McConville ML, editors. *Bacteriophages: biology, technology, therapy*. 1st ed. Cham, Switzerland: Springer; 2021.
24. Obradović M, Malešević M, Di Luca M, Kekić D, Gajić I, McAuliffe O, et al. Isolation, characterization, genome analysis and host resistance development of two novel lastavirus phages active against pandrug-resistant *Klebsiella pneumoniae*. *Viruses.* 2023; 15 (3): 628. doi: 10.3390/v15030628
25. Dunstan RA, Bamert RS, Belousoff MJ, Short FL, Barlow CK, Pickard DJ, et al. Mechanistic Insights into the capsule-targeting depolymerase from a *Klebsiella pneumoniae* bacteriophage. *Microbiol Spectr.* 2021; 9 (2): e0102321. doi: 10.1128/spectrum.01023-21
26. Kęsik-Szeloch A, Drulis-Kawa Z, Weber-Dąbrowska B, Kassner J, Majkowska-Skrobek G, Augustyniak D, et al. Characterising the biology of novel lytic bacteriophages infecting multidrug resistant *Klebsiella pneumoniae*. *Virol J.* 2013; 10: 100. doi: 10.1186/1743-422X-10-100
27. De Koster S, Rodriguez Ruiz JP, Rajakani SG, Lammens C, Glupczynski Y, Goossens H, et al. Diversity in the characteristics of *Klebsiella pneumoniae* ST101 of human, environmental, and animal origin. *Front Microbiol.* 2022; 13: 838207. doi: 10.3389/fmicb.2022.838207
28. Majkowska-Skrobek G, Latka A, Berisio R, Squeglia F, Maciejewska B, Briers Y, et al. Phage-borne depolymerases decrease *Klebsiella pneumoniae* resistance to innate defense mechanisms. *Front Microbiol.* 2018; 9: 2517. doi: 10.3389/fmicb.2018.02517
29. Pertics BZ, Kovács T, Schneider G. Characterization of a lytic bacteriophage and demonstration of its combined lytic effect with a K2 depolymerase on the hypervirulent *Klebsiella pneumoniae* strain 52145. *Microorganisms.* 2023; 11 (3): 669. doi: 10.3390/microorganisms11030669
30. Silva EC, Rodrigues LMR, Vila MMDC, Balcão VM. Newly isolated phages preying on *Pseudomonas syringae* pv. garcae: in vitro and ex vivo inactivation studies in coffee plant leaf. *Enzyme Microb Technol.* 2023; 171: 110325. doi: 10.1016/j.enzymtec.2023.110325
31. Balcão VM, Moreli FC, Silva EC, Belline BG, Martins LF, Rossi FPN, et al. Isolation and molecular characterization of a novel lytic bacteriophage that inactivates mdr *Klebsiella pneumoniae* strains. *Pharmaceutics.* 2022; 14 (7): 1421. doi: 10.3390/pharmaceutics14071421
32. Balcão VM, Belline BG, Silva EC, Almeida PFFB, Baldo DÂ, Amorim LRP, et al. Isolation and molecular characterization of two novel lytic bacteriophages for the biocontrol of *Escherichia coli* in uterine infections: in vitro and ex vivo preliminary studies in veterinary medicine. *Pharmaceutics.* 2022; 14 (11): 2344. doi: 10.3390/pharmaceutics14112344
33. Harada LK, Silva EC, Rossi FPN, Cieza B, Oliveira TJ, Pereira C, et al. Characterization and in vitro testing of newly isolated lytic bacteriophages for biocontrol of *Pseudomonas aeruginosa*. *Future Microbiol.* 2022; 17: 111–41. doi: 10.2217/fmb-2021-0027
34. Kharina AV, Zaika SA, Yumyna YM, Zelena PP, Kornienko NO, Kosenko YA, et al. Detection of *Proteus mirabilis* and *Enterobacter cloacae* in tomato and pepper fruits and isolation of their bacteriophages. *Naukovi Dopovidi Natsionalnoho Universytetu Bioresursiv i Pryrodokorystuvannia Ukrainy.* 2015; 13: 80–93.
35. Kropinski AM, Mazzocco A, Waddell TE, Lingohr E, Johnson RP. Enumeration of bacteriophages by double agar overlay plaque assay. In: Clokie MRJ, Kropinski AM, editors. *Bacteriophages: methods and protocols*, volume 1: isolation, characterization, and interactions. Totowa, NJ, USA: Humana Press; 2009. p. 69–76.
36. Pinheiro LAM, Pereira C, Barreal ME, Gallego PP, Balcão VM, Almeida A. Use of phage Φ6 to inactivate *Pseudomonas syringae* pv. actinidiae in kiwifruit plants: in vitro and ex vivo experiments. *Appl Microbiol Biotechnol.* 2019; 104 (3): 1319–30. doi: 10.1007/s00253-019-10301-7
37. Pinheiro LAM, Pereira C, Frazão C, Balcão VM, Almeida A. Efficiency of phage Φ6 for biocontrol of *Pseudomonas syringae* Pv. syringae: an in vitro preliminary study. *Microorganisms.* 2019; 7 (9): 286. doi: 10.3390/microorganisms7090286
38. Khan Mirzaei M, Nilsson AS. Isolation of phages for phage therapy: a comparison of spot tests and efficiency of plating analyses for determination of host range and efficacy. *PLoS ONE.* 2015; 10 (3): e0118557. doi: 10.1371/journal.pone.0118557
39. Stuer-Lauridsen B, Janzen T, Schnabl J, Johansen E. Identification of the host determinant of two prolate-headed phages infecting *Lactococcus lactis*. *Virology.* 2003; 309 (1): 10–7. doi: 10.1016/S0042-6822(03)00012-6
40. Watanabe K, Takesue S, Ishibashi K, Nakahara S. A computer simulation of the adsorption of *Lactobacillus* phage PL-1 to host cells: some factors affecting the process. *Agric Biol Chem.* 1982; 46 (3): 697–702. doi: 10.1271/bbb1961.46.697
41. Moldovan R, Chapman-McQuiston E, Wu XL. On kinetics of phage adsorption. *Biophys J.* 2007; 93 (1): 303–15. doi: 10.1529/biophysj.106.102962
42. Storms ZJ, Sauvageau D. Modeling tailed bacteriophage adsorption: insight into mechanisms. *Virology.* 2015; 485: 355–62. doi: 10.1016/j.virol.2015.08.007

43. Mota LC, Silva EC, Quinde CA, Cieza B, Basu A, Rodrigues LMR, et al. Potential of a newly isolated lytic bacteriophage to control *pseudomonas coronafaciens* pv. *garcae* in coffee plants: molecular characterization with in vitro and ex vivo experiments. *Enzyme Microb Technol.* 2025; 184: 110573. doi: 10.1016/j.enzmictec.2024.110573
44. Pereira AO, Barros NMA, Guerrero BR, Emencheta SC, Baldo DÂ, Oliveira JM Jr, et al. An edible biopolymeric microcapsular wrapping integrating lytic bacteriophage particles for salmonella enterica: potential for integration into poultry feed. *Antibiotics.* 2023; 12 (6): 988. doi: 10.3390/antibiotics12060988
45. Filippov AA, Sergueev KV, He Y, Huang XZ, Gnade BT, Mueller AJ, et al. Bacteriophage-resistant mutants in *yersinia pestis*: identification of phage receptors and attenuation for mice. *PLoS ONE.* 2011; 6 (9): e25486. doi: 10.1371/journal.pone.0025486
46. Ewing B, Hillier L, Wendl MC, Green P. Base-calling of automated sequencer traces using phred. I. accuracy assessment. *Genome Res.* 1998; 8 (3): 175–85. doi: 10.1101/gr.8.3.175
47. Andrews S. FastQC: a quality control tool for high throughput sequence data. Cambridge, UK: Babraham Bioinformatics; 2010. Available from: <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>
48. Ewels P, Magnusson M, Lundin S, Käller M. MultiQC: summarize analysis results for multiple tools and samples in a single report. *Bioinformatics.* 2016; 32 (19): 3047–8. doi: 10.1093/bioinformatics/btw354
49. Kokot M, Dlugosz M, Deorowicz S. KMC 3: counting and manipulating k-mer statistics. *Bioinformatics.* 2017; 33 (17): 2759–61. doi: 10.1093/bioinformatics/btx304
50. Bolger AM, Lohse M, Usadel B. Trimmomatic: a flexible trimmer for illumina sequence data. *Bioinformatics.* 2014; 30 (15): 2114–20. doi: 10.1093/bioinformatics/btu170
51. Song L, Florea L, Langmead B. Lighter: fast and memory-efficient sequencing error correction without counting. *Genome Biol.* 2014; 15 (11): 509. doi: 10.1186/s13059-014-0509-9
52. Magoč T, Salzberg SL. FLASH: fast length adjustment of short reads to improve genome assemblies. *Bioinformatics.* 2011; 27 (21): 2957–63. doi: 10.1093/bioinformatics/btr507
53. Bankevich A, Nurk S, Antipov D, Gurevich AA, Dvorkin M, Kulikov AS, et al. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. *J Comput Biol.* 2012; 19 (5): 455–77. doi: 10.1089/cmb.2012.0021
54. Li H, Durbin R. Fast and accurate short read alignment with burrows-wheeler transform. *Bioinformatics.* 2009; 25 (14): 1754–60. doi: 10.1093/bioinformatics/btp324
55. Walker BJ, Abeel T, Shea T, Priest M, Abouelliel A, Sakthikumar S, et al. Pilon: an integrated tool for comprehensive microbial variant detection and genome assembly improvement. *PLoS ONE.* 2014; 9 (11): e112963. doi: 10.1371/journal.pone.0112963
56. Wood DE, Lu J, Langmead B. Improved metagenomic analysis with kraken 2. *Genome Biol.* 2019; 20 (1): 257. doi: 10.1186/s13059-019-1891-0
57. Lu J, Rincon N, Wood DE, Breitwieser FP, Pockrandt C, Langmead B, et al. Metagenome analysis using the kraken software suite. *Nat Protoc.* 2022; 17 (12): 2815–39. doi: 10.1038/s41596-022-00738-y
58. Li H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics.* 2018; 34 (18): 3094–100. doi: 10.1093/bioinformatics/bty191
59. Nishimura Y, Watai H, Honda T, Mihara T, Omae K, Roux S, et al. Environmental viral genomes shed new light on virus-host interactions in the ocean. *mSphere.* 2017; 2 (2): e00359-16. doi: 10.1128/msphere.00359-16
60. Bouras G, Nepal R, Houtak G, Psaltis AJ, Wormald PJ, Vreugde S. Pharokka: a fast scalable bacteriophage annotation tool. *Bioinformatics.* 2023; 39 (1): btac776. doi: 10.1093/bioinformatics/btac776
61. Chan PP, Lin BY, Mak AJ, Lowe TM. tRNAscan-SE 2.0: improved detection and functional classification of transfer RNA genes. *Nucleic Acids Res.* 2021; 49 (16): 9077–96. doi: 10.1093/nar/gkab688
62. Laslett D, Canback B. ARAGORN, a program to detect tRNA genes and tmRNA genes in nucleotide sequences. *Nucleic Acids Res.* 2004; 32 (1): 11–6. doi: 10.1093/nar/gkh152
63. Bland C, Ramsey TL, Sabree F, Lowe M, Brown K, Kyrpides NC, et al. CRISPR recognition tool (CRT): a tool for automatic detection of clustered regularly interspaced palindromic repeats. *BMC Bioinform.* 2007; 8: 209. doi: 10.1186/1471-2105-8-209
64. McNair K, Zhou C, Dinsdale EA, Souza B, Edwards RA. PHANOTATE: a novel approach to gene identification in phage genomes. *Bioinformatics.* 2019; 35 (22): 4537–42. doi: 10.1093/bioinformatics/btz265
65. Mirdita M, Steinegger M, Breitwieser F, Söding J, Levy Karin E. Fast and sensitive taxonomic assignment to metagenomic contigs. *Bioinformatics.* 2021; 37 (18): 3029–31. doi: 10.1093/bioinformatics/btab184
66. Terzian P, Olo Ndela E, Galiez C, Lossouarn J, Pérez Bucio RE, Mom R, et al. PHROG: families of prokaryotic virus proteins clustered using remote homology. *NAR Genom Bioinform.* 2021; 3 (3): lqab067. doi: 10.1093/nargab/lqab067
67. Alcock BP, Huynh W, Chalil R, Smith KW, Raphenya AR, Wlodarski MA, et al. CARD 2023: expanded curation, support for machine learning, and resistome prediction at the comprehensive antibiotic resistance database. *Nucleic Acids Res.* 2023; 51 (D1): D690–9. doi: 10.1093/nar/gkac920
68. Liu B, Zheng D, Zhou S, Chen L, Yang J. VFDB 2022: a general classification scheme for bacterial virulence factors. *Nucleic Acids Res.* 2022; 50 (D1): D912–7. doi: 10.1093/nar/gkab1107

69. Cook R, Brown N, Redgwell T, Rihtman B, Barnes M, Clokie M, et al. Infrastructure for a phage reference database: identification of large-scale biases in the current collection of cultured phage genomes. *Phage*. 2021; 2 (4): 214–23. doi: 10.1089/phage.2021.0007
70. Ondov BD, Treangen TJ, Melsted P, Mallonee AB, Bergman NH, Koren S, et al. Mash: fast genome and metagenome distance estimation using minhash. *Genome Biol*. 2016; 17 (1): 132. doi: 10.1186/s13059-016-0997-x
71. Camacho C, Coulouris G, Avagyan V, Ma N, Papadopoulos J, Bealer K, et al. BLAST+: architecture and applications. *BMC Bioinformatics*. 2009; 10: 421. doi: 10.1186/1471-2105-10-421
72. Sayers EW, Bolton EE, Brister JR, Canese K, Chan J, Comeau DC, et al. Database resources of the National Center for Biotechnology Information. *Nucleic Acids Res*. 2022; 50 (D1): D20–6. doi: 10.1093/nar/gkab1112
73. Schwengers O, Hain T, Chakraborty T, Goesmann A. ReferenceSeeker: rapid determination of appropriate reference genomes. *J Open Source Softw*. 2020; 5 (46): 1994. doi: 10.21105/joss.01994
74. Marçais G, Delcher AL, Phillippy AM, Coston R, Salzberg SL, Zimin A. MUMmer4: a fast and versatile genome alignment system. *PLoS Comput Biol*. 2018; 14 (1): e1005944. doi: 10.1371/journal.pcbi.1005944
75. O’Leary NA, Wright MW, Brister JR, Ciuffo S, Haddad D, McVeigh R, et al. Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation. *Nucleic Acids Res*. 2016; 44 (D1): D733–45. doi: 10.1093/nar/gkv1189
76. Katz LS, Griswold T, Morrison SS, Caravas JA, Zhang S, den Bakker HC, et al. Mashtree: a rapid comparison of whole genome sequence files. *J Open Source Softw*. 2019; 4 (34): 1762. doi: 10.21105/joss.01762
77. Howe K, Bateman A, Durbin R. QuickTree: building huge neighbour-joining trees of protein sequences. *Bioinformatics*. 2002; 18 (11): 1546–7. doi: 10.1093/bioinformatics/18.11.1546
78. Nishimura Y, Yoshida T, Kuronishi M, Uehara H, Ogata H, Goto S. ViPTree: the viral proteomic tree server. *Bioinformatics*. 2017; 33 (15): 2379–80. doi: 10.1093/bioinformatics/btx157
79. Zhou T, Yang L, Lu Y, Dror I, Machado ACD, Ghane T, et al. DNASHape: a method for the high-throughput prediction of DNA structural features on a genomic scale. *Nucleic Acids Res*. 2013; 41 (W1): W56–62. doi: 10.1093/nar/gkt437
80. Balcão VM, Basu A, Cieza B, Rossi FN, Pereira C, Vila MMDC, et al. Pseudomonas-tailed lytic phages: genome mechanical analysis and putative correlation with virion morphogenesis yield. *Future Microbiol*. 2022; 17 (13): 1009–26. doi: 10.2217/fmb-2021-0293
81. Basu A, Bobrovnikov DG, Cieza B, Arcon JP, Qureshi Z, Orozco M, et al. Deciphering the mechanical code of the genome and epigenome. *Nat Struct Mol Biol*. 2022; 29 (12): 1178–87. doi: 10.1038/s41594-022-00877-6
82. Basu A, Bobrovnikov DG, Ha T. DNA mechanics and its biological impact. *J Mol Biol*. 2021; 433 (18): 166861. doi: 10.1016/j.jmb.2021.166861
83. Zhang Y, Basu A, Ha T, Bialek W. Searching for sequence features that control DNA flexibility. *arXiv*. 2020. doi: 10.48550/arXiv.2012.06127
84. Bates DM, Watts DG. Nonlinear regression analysis and its applications. 1st ed. Hoboken, NJ, USA: Wiley-Blackwell; 1988.
85. Balcão VM, Vieira MC, Malcata FX. Adsorption of protein from several commercial lipase preparations onto a hollow-fiber membrane module. *Biotechnol Prog*. 1996; 12 (2): 164–72. doi: 10.1021/bp950070n
86. Balcão VM, Oliveira TA, Xavier Malcata F. Stability of a commercial lipase from *Mucor javanicus*: kinetic modelling of pH and temperature dependencies. *Biocatal Biotransform*. 1998; 16 (1): 45–66. doi: 10.3109/10242429809040110
87. Hughes KA, Sutherland IW, Clark J, Jones MV. Bacteriophage and associated polysaccharide depolymerases—novel tools for study of bacterial biofilms. *J Appl Microbiol*. 1998; 85 (3): 583–90. doi: 10.1046/j.1365-2672.1998.853541.x
88. Pires DP, Oliveira H, Melo LDR, Sillankorva S, Azeredo J. Bacteriophage-encoded depolymerases: their diversity and biotechnological applications. *Appl Microbiol Biotechnol*. 2016; 100 (5): 2141–51. doi: 10.1007/s00253-015-7247-0
89. Oliveira H, São-José C, Azeredo J. Phage-derived peptidoglycan degrading enzymes: challenges and future prospects for in vivo therapy. *Viruses*. 2018; 10 (6): 292. doi: 10.3390/v10060292
90. Birch EW, Ruggero NA, Covert MW. Determining host metabolic limitations on viral replication via integrated modeling and experimental perturbation. *PLoS Comput Biol*. 2012; 8 (10): e1002746. doi: 10.1371/journal.pcbi.1002746
91. Lima R, Del Fiol FS, Balcão VM. Prospects for the use of new technologies to combat multidrug-resistant bacteria. *Front Pharmacol*. 2019; 10: 692. doi: 10.3389/fphar.2019.00692
92. Pereira C, Moreirinha C, Lewicka M, Almeida P, Clemente C, Cunha Â, et al. Bacteriophages with potential to inactivate salmonella typhimurium: use of single phage suspensions and phage cocktails. *Virus Res*. 2016; 220: 179–92. doi: 10.1016/j.virusres.2016.04.020
93. Pereira C, Moreirinha C, Lewicka M, Almeida P, Clemente C, Romalde JL, et al. Characterization and in vitro evaluation of new bacteriophages for the biocontrol of *Escherichia coli*. *Virus Res*. 2017; 227: 171–82. doi: 10.1016/j.virusres.2016.09.019
94. Lopes A, Pereira C, Almeida A. Sequential combined effect of phages and antibiotics on the inactivation of *E. coli*. *Microorganisms*. 2018; 6 (4): 125. doi: 10.3390/microorganisms6040125
95. Levin BR, Bull JJ. Population and evolutionary dynamics of phage therapy. *Nat Rev Microbiol*. 2004; 2 (2): 166–73. doi: 10.1038/nrmicro822

96. Gill JJ, Hyman P. Phage choice, isolation, and preparation for phage therapy. *Curr Pharm Biotechnol*. 2010; 11 (1): 2–14. doi: 10.2174/138920110790725311
97. Labrie SJ, Samson JE, Moineau S. Bacteriophage resistance mechanisms. *Nat Rev Microbiol*. 2010; 8 (5): 317–27. doi: 10.1038/nrmicro2315
98. Townsend EM, Kelly L, Gannon L, Muscatt G, Dunstan R, Michniewski S, et al. Isolation and characterization of klebsiella phages for phage therapy. *Phage*. 2021; 2 (1): 26–42. doi: 10.1089/phage.2020.0046
99. Yoo S, Lee KM, Kim N, Vu TN, Abadie R, Yong D. Designing phage cocktails to combat the emergence of bacteriophage-resistant mutants in multidrug-resistant *Klebsiella pneumoniae*. *Microbiol Spectr*. 2024; 12 (2): e0125823. doi: 10.1128/spectrum.01258-23
100. Fang Q, Feng Y, McNally A, Zong Z. Characterization of phage resistance and phages capable of intestinal decolonization of carbapenem-resistant *Klebsiella pneumoniae* in mice. *Commun Biol*. 2022; 5 (1): 48. doi: 10.1038/s42003-022-03001-y
101. Peng Q, Ma Z, Han Q, Xiang F, Wang L, Zhang Y, et al. Characterization of bacteriophage VB_KleM_KB2 possessing high control ability to pathogenic *Klebsiella pneumoniae*. *Sci Rep*. 2023; 13 (1): 9815. doi: 10.1038/s41598-023-37065-5
102. Nang SC, Lu J, Yu HH, Wickremasinghe H, Azad MAK, Han M, et al. Diversified repertoire of phage resistance in *Klebsiella pneumoniae* and bidirectional steering effects impacting antibiotic susceptibility. *bioRxiv*. 2022. doi: 10.1101/2022.11.14.516531
103. Ciacci N, D'Andrea MM, Marmo P, Demattè E, Amisano F, Pilato VD, et al. Characterization of vB_Kpn_F48, a newly discovered lytic bacteriophage for *Klebsiella pneumoniae* of sequence type 101. *Viruses*. 2018; 10 (9): 482. doi: 10.3390/v10090482
104. Grant JR, Enns E, Marinier E, Mandal A, Herman EK, Chen CY, et al. Proksee: in-depth characterization and visualization of bacterial genomes. *Nucleic Acids Res*. 2023; 51 (W1): W484–92. doi: 10.1093/nar/gkad326
105. Xuan G, Kong J, Wang Y, Lin H, Wang J. Characterization of the newly isolated pseudomonas phage vB_Pae_LC3I3. *Virus Res*. 2023; 323: 198978. doi: 10.1016/j.virusres.2022.198978
106. Mihara T, Nishimura Y, Shimizu Y, Nishiyama H, Yoshikawa G, Uehara H, et al. Linking virus genomes with host taxonomy. *Viruses*. 2016; 8 (3): 66. doi: 10.3390/v8030066
107. Rohs R, West SM, Sosinsky A, Liu P, Mann RS, Honig B. The role of DNA shape in protein-DNA recognition. *Nature*. 2009; 461 (7268): 1248–53. doi: 10.1038/nature08473
108. Abe N, Dror I, Yang L, Slattery M, Zhou T, Bussemaker HJ, et al. Deconvolving the recognition of DNA shape from sequence. *Cell*. 2015; 161 (2): 307–18. doi: 10.1016/j.cell.2015.02.008
109. Dickerson RE, Klug A. Base sequence and helix structure variation in B and A DNA. *J Mol Biol*. 1983; 166 (4): 419–41. doi: 10.1016/s0022-2836(83)80093-x
110. Kriegel F, Matek C, Dršata T, Kulenkampff K, Tschirpke S, Zacharias M, et al. The temperature dependence of the helical twist of DNA. *Nucleic Acids Res*. 2018; 46 (16): 7998–8009. doi: 10.1093/nar/gky599
111. Laughton C, Luisi B. The mechanics of minor groove width variation in DNA, and its implications for the accommodation of ligands. *J Mol Biol*. 1999; 288 (5): 953–63. doi: 10.1006/jmbi.1999.2733
112. Stella S, Cascio D, Johnson RC. The shape of the DNA minor groove directs binding by the DNA-bending protein FIS. *Genes Dev*. 2010; 24 (8): 814–26. doi: 10.1101/gad.1900610
113. El Hassan MA, Calladine CR. Propeller-twisting of base-pairs and the conformational mobility of dinucleotide steps in DNA. *J Mol Biol*. 1996; 259 (1): 95–103. doi: 10.1006/jmbi.1996.0304
114. Tolstorukov MY, Colasanti AV, McCandlish DM, Olson WK, Zhurkin VB. A novel roll-and-slide mechanism of DNA folding in chromatin: implications for nucleosome positioning. *J Mol Biol*. 2007; 371 (3): 725–38. doi: 10.1016/j.jmb.2007.05.048
115. Zhou T, Shen N, Yang L, Abe N, Horton J, Mann RS, et al. Quantitative modeling of transcription factor binding specificities using DNA shape. *Proc Natl Acad Sci USA*. 2015; 112 (15): 4654–9. doi: 10.1073/pnas.1422023112
116. Ussery DW. DNA structure: A-, B- and Z-DNA helix families. In: *Encyclopedia of life sciences*. Chichester, UK: John Wiley & Sons; 2002. doi: 10.1038/npg.els.0003122
117. Watson JD, Crick FH. Molecular structure of nucleic acids: a structure for deoxyribose nucleic acid. *Nature*. 1953; 171 (4356): 737–8. doi: 10.1038/171737a0
118. Franklin RE, Gosling RG. Molecular configuration in sodium thymonucleate. *Nature*. 1953; 171 (4356): 740–1. doi: 10.1038/171740a0
119. Fratini AV, Kopka ML, Drew HR, Dickerson RE. Reversible bending and helix geometry in a B-DNA dodecamer: CGC-GAATTBrCGCG. *J Biol Chem*. 1982; 257 (24): 14686–707. doi: 10.1016/s0021-9258(18)33335-0
120. Cohen RS, Meselson M. Periodic interactions of heat shock transcriptional elements. *Nature*. 1988; 332 (6167): 856–8. doi: 10.1038/332856a0
121. Zhang Y, Crothers DM. Statistical mechanics of sequence-dependent circular DNA and its application for DNA cyclization. *Biophys J*. 2003; 84 (1): 136–53. doi: 10.1016/s0006-3495(03)74838-3
122. Crothers DM, Drak J, Kahn JD, Levene SD. DNA bending, flexibility, and helical repeat by cyclization kinetics. *Methods Enzymol*. 1992; 212: 3–29. doi: 10.1016/0076-6879(92)12003-9

123. Shore D, Langowski J, Baldwin RL. DNA flexibility studied by covalent closure of short fragments into circles. *Proc Natl Acad Sci USA*. 1981; 78 (8): 4833–7. doi: 10.1073/pnas.78.8.4833
124. Richmond TJ, Davey CA. The structure of DNA in the nucleosome core. *Nature*. 2003; 423 (6936): 145–50. doi: 10.1038/nature01595
125. Zhurkin VB, Lysov YP, Ivanov VI. Anisotropic flexibility of DNA and the nucleosomal structure. *Nucleic Acids Res*. 1979; 6 (3): 1081–96. doi: 10.1093/nar/6.3.1081
126. Wing R, Drew H, Takano T, Broka C, Tanaka S, Itakura K, et al. Crystal structure analysis of a complete turn of B-DNA. *Nature*. 1980; 287 (5784): 755–8. doi: 10.1038/287755a0
127. Kostrewa D, Granzin J, Koch C, Choe HW, Raghunathan S, Wolf W, et al. Three-dimensional structure of the *E. coli* DNA-binding protein FIS. *Nature*. 1991; 349 (6305): 178–80. doi: 10.1038/349178a0
128. Oguey C, Foloppe N, Hartmann B. Understanding the sequence-dependence of DNA groove dimensions: implications for DNA interactions. *PLoS ONE*. 2010; 5 (12): e15931. doi: 10.1371/journal.pone.0015931
129. Liebl K, Drsata T, Lankas F, Lipfert J, Zacharias M. Explaining the striking difference in twist-stretch coupling between DNA and RNA: a comparative molecular dynamics analysis. *Nucleic Acids Res*. 2015; 43 (20): 10143–56. doi: 10.1093/nar/gkv1028
130. Olson WK, Gorin AA, Lu XJ, Hock LM, Zhurkin VB. DNA sequence-dependent deformability deduced from protein-DNA crystal complexes. *Proc Natl Acad Sci USA*. 1998; 95 (19): 11163–8. doi: 10.1073/pnas.95.19.11163
131. Packer MJ, Dauncey MP, Hunter CA. Sequence-dependent DNA structure: dinucleotide conformational maps. *J Mol Biol*. 2000; 295 (1): 71–83. doi: 10.1006/jmbi.1999.3236
132. Kannan S, Kohlhoff K, Zacharias M. B-DNA under stress: over- and untwisting of DNA during molecular dynamics simulations. *Biophys J*. 2006; 91 (8): 2956–65. doi: 10.1529/biophysj.106.087163
133. Hyman P. Phages for phage therapy: isolation, characterization, and host range breadth. *Pharmaceuticals*. 2019; 12 (1): 35. doi: 10.3390/ph12010035
134. Seed KD. Battling Phages: how bacteria defend against viral attack. *PLoS Pathog*. 2015; 11 (6): e1004847. doi: 10.1371/journal.ppat.1004847
135. Jin J, Li ZJ, Wang SW, Wang SM, Huang DH, Li YH, et al. Isolation and characterization of ZZ1, a novel lytic phage that infects *acinetobacter baumannii* clinical isolates. *BMC Microbiol*. 2012; 12: 156. doi: 10.1186/1471-2180-12-156
136. Ceysens PJ. Isolation and characterization of lytic bacteriophages infecting *Pseudomonas aeruginosa* [Ph.D. thesis]. Leuven, Belgium: Katholieke Universiteit Leuven; 2009
137. Hyman P, Abedon ST. Practical methods for determining phage growth parameters. In: Clokie MRJ, Kropinski AM, editors. *Bacteriophages: methods and protocols*, volume 1: isolation, characterization, and interactions. Totowa, NJ, USA: Humana Press; 2009. p. 175–202.
138. Rakhuba DV, Kolomiets EI, Szwajcer Dey E, Novik GI. Bacteriophage receptors, mechanisms of phage adsorption and penetration into host cell. *Pol J Microbiol*. 2010; 59 (3): 145–55. doi: 10.33073/pjm-2010-023
139. Bull JJ, Gill JJ. The habits of highly effective phages: population dynamics as a framework for identifying therapeutic phages. *Front Microbiol*. 2014; 5: 618. doi: 10.3389/fmicb.2014.00618
140. Zurabov F, Zhilenkov E. Characterization of four virulent *Klebsiella pneumoniae* bacteriophages, and evaluation of their potential use in complex phage preparation. *Virology*. 2021; 18 (1): 9. doi: 10.1186/s12985-020-01485-w
141. Jamal M, Hussain T, Rajanna Das C, Andleeb S. Characterization of siphoviridae phage Z and studying its efficacy against multidrug-resistant *Klebsiella pneumoniae* planktonic cells and biofilm. *J Med Microbiol*. 2015; 64 (Pt 4): 454–62. doi: 10.1099/jmm.0.000040
142. Herridge WP, Shibu P, O'Shea J, Brook TC, Hoyle L. Bacteriophages of *Klebsiella* spp., their diversity and potential therapeutic uses. *J Med Microbiol*. 2020; 69 (2): 176–94.
143. Turner D, Shkoporov AN, Lood C, Millard AD, Dutilh BE, Alfenas-Zerbini P, et al. Abolishment of morphology-based taxa and change to binomial species names: 2022 taxonomy update of the ICTV bacterial viruses subcommittee. *Arch Virol*. 2023; 168 (2): 74. doi: 10.1007/s00705-022-05694-2
144. UniProt Consortium. UniProt: The universal protein knowledgebase in 2021. *Nucleic Acids Res*. 2021; 49 (D1): D480–9. doi: 10.1093/nar/gkaa1100
145. Schoch CL, Ciuffo S, Domrachev M, Hottel CL, Kannan S, Khovanskaya R, et al. NCBI taxonomy: a comprehensive update on curation, resources and tools. *Database*. 2020; 2020: baaa062. doi: 10.1093/database/baaa062
146. Kasurinen J, Spruit CM, Wicklund A, Pajunen MI, Skurnik M. Screening of bacteriophage encoded toxic proteins with a next generation sequencing-based assay. *Viruses*. 2021; 13 (5): 750. doi: 10.3390/v13050750
147. Kakasis A, Panitsa G. Bacteriophage therapy as an alternative treatment for human infections. a comprehensive review. *Int J Antimicrob Agents*. 2019; 53 (1): 16–21. doi: 10.1016/j.ijantimicag.2018.09.004
148. Samir S. Molecular machinery of the triad holin, endolysin, and spanin: key players orchestrating bacteriophage-induced cell lysis and their therapeutic applications. *Protein Pept Lett*. 2024; 31 (1): 1–12. doi: 10.2174/0109298665181166231212051621
149. Khan MSI, Gao X, Liang K, Mei S, Zhan J. Virulent drexelviral bacteriophage MSK, morphological and genome resemblance with RTP bacteriophage inhibits the multidrug-resistant bacteria. *Front Microbiol*. 2021; 12: 706700. doi: 10.3389/fmicb.2021.706700
150. Ku H, Kabwe M, Chan HT, Stanton C, Petrovski S, Batinovic S, et al. Novel drexelviral bacteriophage KMI8 with specific lytic activity against *klebsiella michiganensis* and its biofilms. *PLoS ONE*. 2021; 16 (9): e0257102. doi: 10.1371/journal.pone.0257102

151. German GJ, Misra R, Kropinski AM. The T1-like bacteriophages. In: Calendar R, editor. The bacteriophages. 2nd ed. New York, NY, USA: Oxford University Press; 2005. p. 211–24.
152. Adriaenssens EM, Sullivan MB, Knezevic P, van Zyl LJ, Sarkar BL, Dutilh BE, et al. Taxonomy of prokaryotic viruses: 2018–2019 update from the ICTV bacterial and archaeal viruses subcommittee. Arch Virol. 2020; 165 (5): 1253–60. doi: 10.1007/s00705-020-04577-8
153. Koonjan S, Seijsing F, Cooper CJ, Nilsson AS. Infection kinetics and phylogenetic analysis of vB_EcoD_SU57, a virulent T1-like drexlerviridae coliphage. Front Microbiol. 2020; 11: 565556. doi: 10.3389/fmicb.2020.565556
154. Rahimi S, Bakht M, Javadi A, Foroughi F, Marashi SMA, Nikkhahi F. Characterization of novel bacteriophage PSKP16 and its therapeutic potential against β -lactamase and biofilm producer strain of K2-hypervirulent *Klebsiella pneumoniae* pneumonia infection in mice model. BMC Microbiol. 2023; 23 (1): 233. doi: 10.1186/s12866-023-02979-7
155. Pertics BZ, Cox A, Nyúl A, Szamek N, Kovács T, Schneider G. Isolation and characterization of a novel lytic bacteriophage against the K2 capsule-expressing hypervirulent *Klebsiella pneumoniae* strain 52145, and identification of its functional depolymerase. Microorganisms. 2021; 9 (3): 650. doi: 10.3390/microorganisms9030650
156. Gilcrease EB, Casjens SR, Bhattacharjee A, Goel R. A *Klebsiella pneumoniae* NDM-1+ bacteriophage: adaptive polyvalence and disruption of heterogenous biofilms. Front Microbiol. 2023; 14: 1100607. doi: 10.3389/fmicb.2023.1100607
157. Domingo-Calap P, Beamud B, Mora-Quilis L, González-Candelas F, Sanjuán R. Isolation and characterization of two *Klebsiella pneumoniae* phages encoding divergent depolymerases. Int J Mol Sci. 2020; 21 (9): 3160. doi: 10.3390/ijms21093160
158. Khan F, Bhuiyan MSS, Tabassum SN, Hossain SA, Rahman A, Sadique A, et al. Characterization and genome sequencing of a novel lytic bacteriophage infecting hospital-associated carbapenem-resistant *Klebsiella pneumoniae* strain from Dhaka, Bangladesh. medRxiv. 2023. doi: 10.1101/2023.08.13.23294043
159. Majkowska-Skrobek G, Markwitz P, Sosnowska E, Lood C, Lavigne R, Drulis-Kawa Z. The evolutionary trade-offs in phage-resistant *Klebsiella pneumoniae* entail cross-phage sensitization and loss of multidrug resistance. Environ Microbiol. 2021; 23 (12): 7723–40. doi: 10.1111/1462-2920.15476
160. Travers AA. The structural basis of DNA flexibility. Philos Trans A Math Phys Eng Sci. 2004; 362 (1820): 1423–38. doi: 10.1098/rsta.2004.1390
161. Garcia HG, Grayson P, Han L, Inamdar M, Kondev J, Nelson PC, et al. Biological consequences of tightly bent DNA: the other life of a macromolecular celebrity. Biopolymers. 2007; 85 (2): 115–30. doi: 10.1002/bip.20627
162. Johnson S, Lindén M, Phillips R. Sequence dependence of transcription factor-mediated DNA looping. Nucleic Acids Res. 2012; 40 (16): 7728–38. doi: 10.1093/nar/gks473
163. Mitchell JS, Glowacki J, Grandchamp AE, Manning RS, Maddocks JH. Sequence-dependent persistence lengths of DNA. J Chem Theory Comput. 2017; 13 (4): 1539–55. doi: 10.1021/acs.jctc.6b00904
164. Shore D, Baldwin RL. Energetics of DNA twisting. I. Relation between twist and cyclization probability. J Mol Biol. 1983; 170 (4): 957–81. doi: 10.1016/s0022-2836(83)80198-3
165. Podtelezhnikov AA, Mao C, Seeman NC, Vologodskii A. Multimerization-cyclization of DNA fragments as a method of conformational analysis. Biophys J. 2000; 79 (5): 2692–704. doi: 10.1016/s0006-3495(00)76507-6
166. Basu A, Bobrovnikov DG, Qureshi Z, Kayikcioglu T, Ngo TTM, Ranjan A, et al. Measuring DNA mechanics on the genome scale. Nature. 2021; 589 (7842): 462–7. doi: 10.1038/s41586-020-03052-3
167. Wang DD, Yan H. The relationship between periodic dinucleotides and the nucleosomal DNA deformation revealed by normal mode analysis. Phys Biol. 2011; 8 (6): 066004. doi: 10.1088/1478-3975/8/6/066004
168. Gu H, Fan RLY, Wang D, Poon LLM. Dinucleotide evolutionary dynamics in influenza A virus. Virus Evol. 2019; 5 (2): vez038. doi: 10.1093/ve/vez038
169. Karlin S, Mrázek J. Compositional differences within and between eukaryotic genomes. Proc Natl Acad Sci USA. 1997; 94 (19): 10227–32. doi: 10.1073/pnas.94.19.10227
170. Di Giallonardo F, Schlub TE, Shi M, Holmes EC. Dinucleotide composition in animal RNA viruses is shaped more by virus family than by host species. J Virol. 2017; 91 (8): e02381-16. doi: 10.1128/jvi.02381-16
171. Wu Q, Zhou W, Wang J, Yan H. Correlation between the flexibility and periodic dinucleotide patterns in yeast nucleosomal DNA sequences. J Theor Biol. 2011; 284 (1): 92–8. doi: 10.1016/j.jtbi.2011.06.026
172. El Hassan MA, Calladine CR. Conformational characteristics of DNA: empirical classifications and a hypothesis for the conformational behavior of dinucleotide steps. Philos Trans A Math Phys Eng Sci. 1997; 355 (1722): 43–100. doi: 10.1098/rsta.1997.0002
173. Back G, Walther D. Predictions of DNA mechanical properties at a genomic scale reveal potentially new functional roles of DNA flexibility. NAR Genom Bioinform. 2023; 5 (4): lqad097. doi: 10.1093/nargab/lqad097
174. Rao S, Chiu TP, Kribelbauer JF, Mann RS, Bussemaker HJ, Rohs R. Systematic prediction of DNA shape changes due to CpG methylation explains epigenetic effects on protein-DNA binding. Epigenetics Chromatin. 2018; 11 (1): 6. doi: 10.1186/s13072-018-0174-4